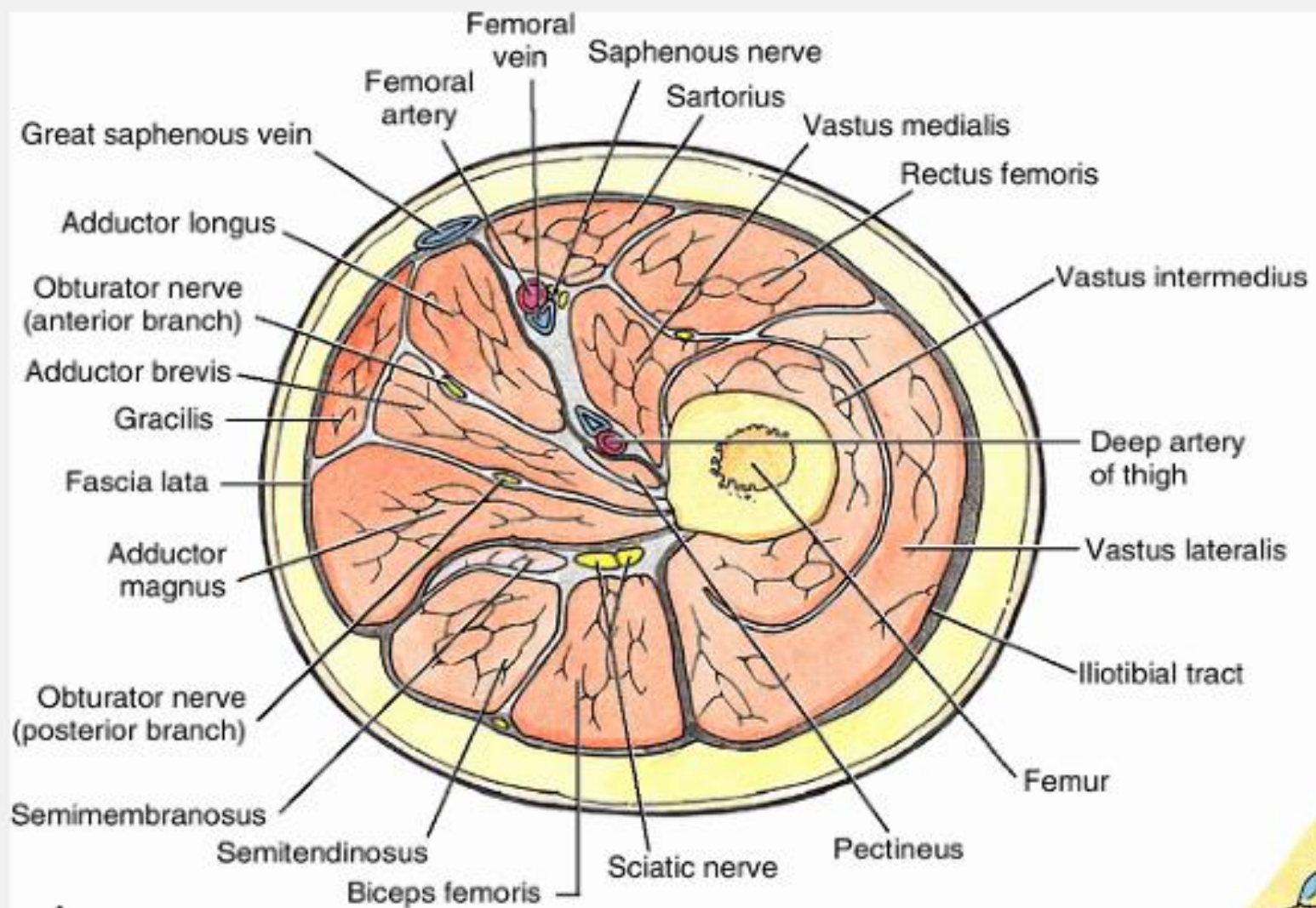


Thigh and gluteal region

Thigh

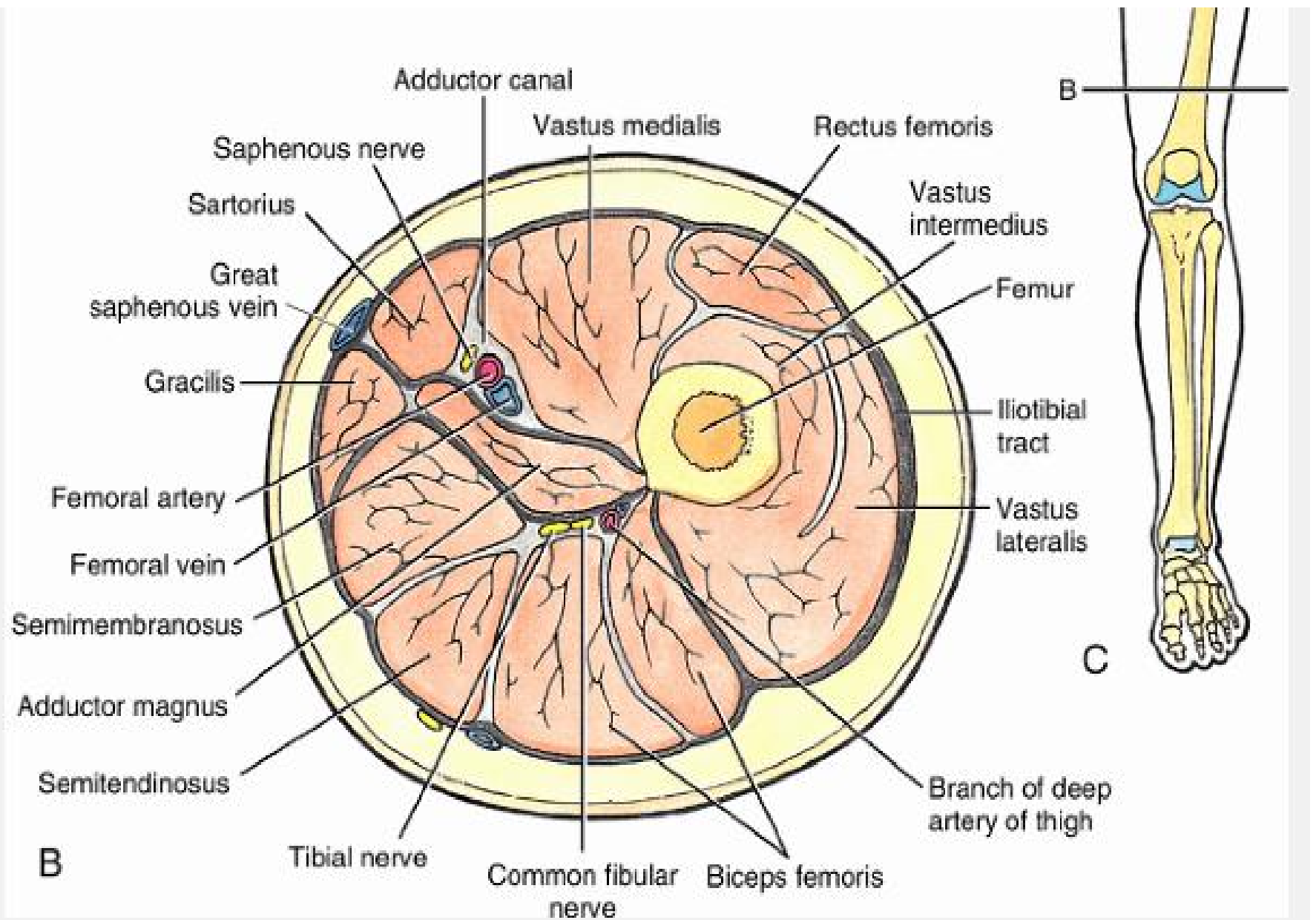
Thigh muscles are organized into 3 compartments by intermuscular septa

- ✓ Anterior or extensor
- ✓ Medial or adductor
- ✓ Posterior or flexor



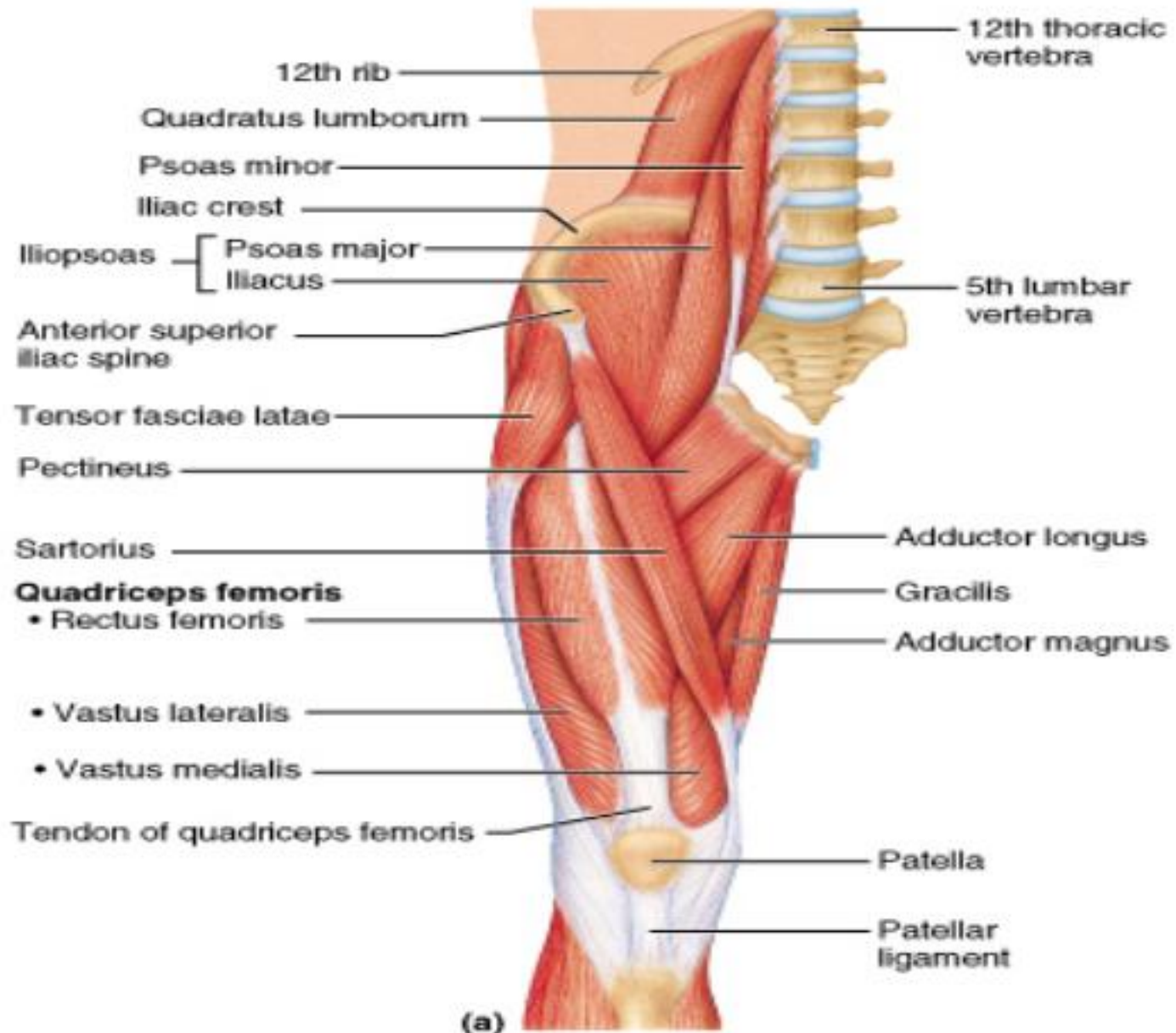
A





Anterior thigh muscles

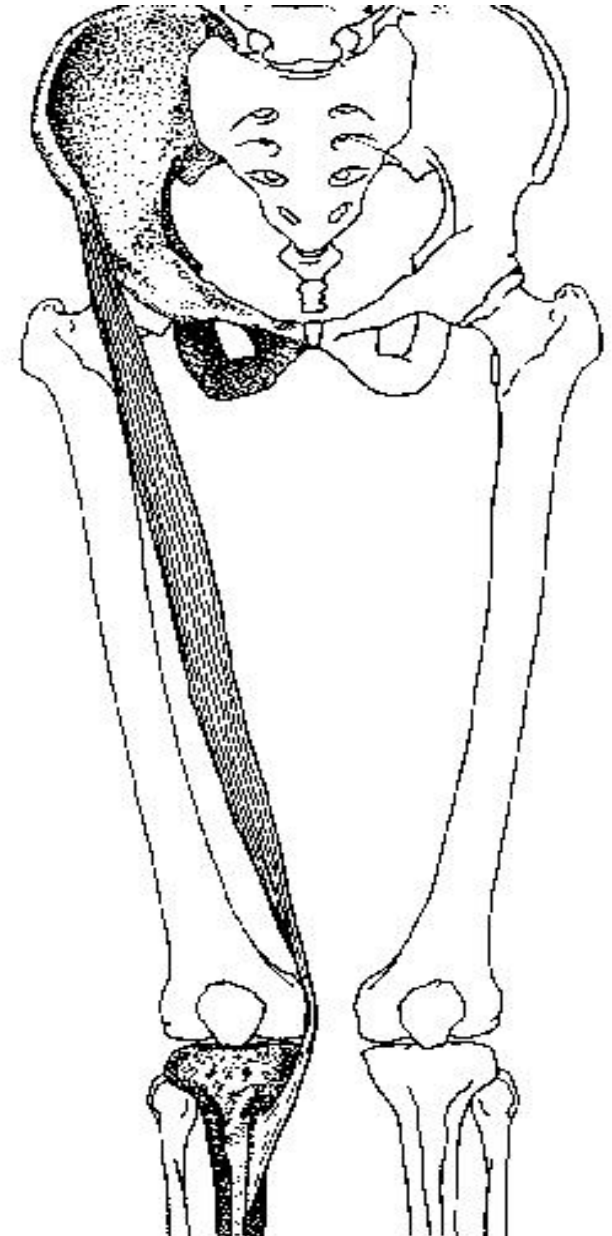
- Found in anterior compartment of thigh
- Flexors of hip and extensors of knee



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Sartorius

- Long, ribbon-like muscle
- The most superficial muscle in the anterior thigh
- Passes obliquely (lateral to medial) across the superoanterior part of the thigh
- Origin
 - anterior superior iliac spine
- Insertion
 - medial aspect of proximal tibia (inferior to the tuberosity)
- Innervation
 - femoral nerve
- Action
 - flexes, abducts and laterally rotates thigh at hip joint
 - flexes leg at knee joint
 - when acting bilaterally, the muscles bring the lower limbs into the cross-legged sitting position; "tailor's position"



Quadriceps femoris

- Four headed muscle
- Forms the main bulk of the anterior thigh
- The four parts
 - Rectus femoris: anterior
 - Vastus lateralis: lateral
 - Vastus medialis: medial
 - Vastus intermedius: deep to rectus femoris between lateralis and medialis
- All insert into tibial tuberosity by a common distal tendon - quadriceps tendon

- The patellar ligament, attached to the tibial tuberosity, is the continuation of the quadriceps tendon in which the patella is embedded
- The vastus medialis and lateralis also attach independently to the patella and form aponeuroses, the medial and lateral patellar retinacula, which reinforce the joint capsule of the knee on each side of the patella en route to attachment to the anterior border of the tibial plateau
- The patella provides additional leverage for the quadriceps in placing the tendon more anteriorly, farther from the joint's axis, causing it to approach the tibia from a position of greater mechanical advantage

Quadriceps femoris

Extends the knee.

Rectus femoris

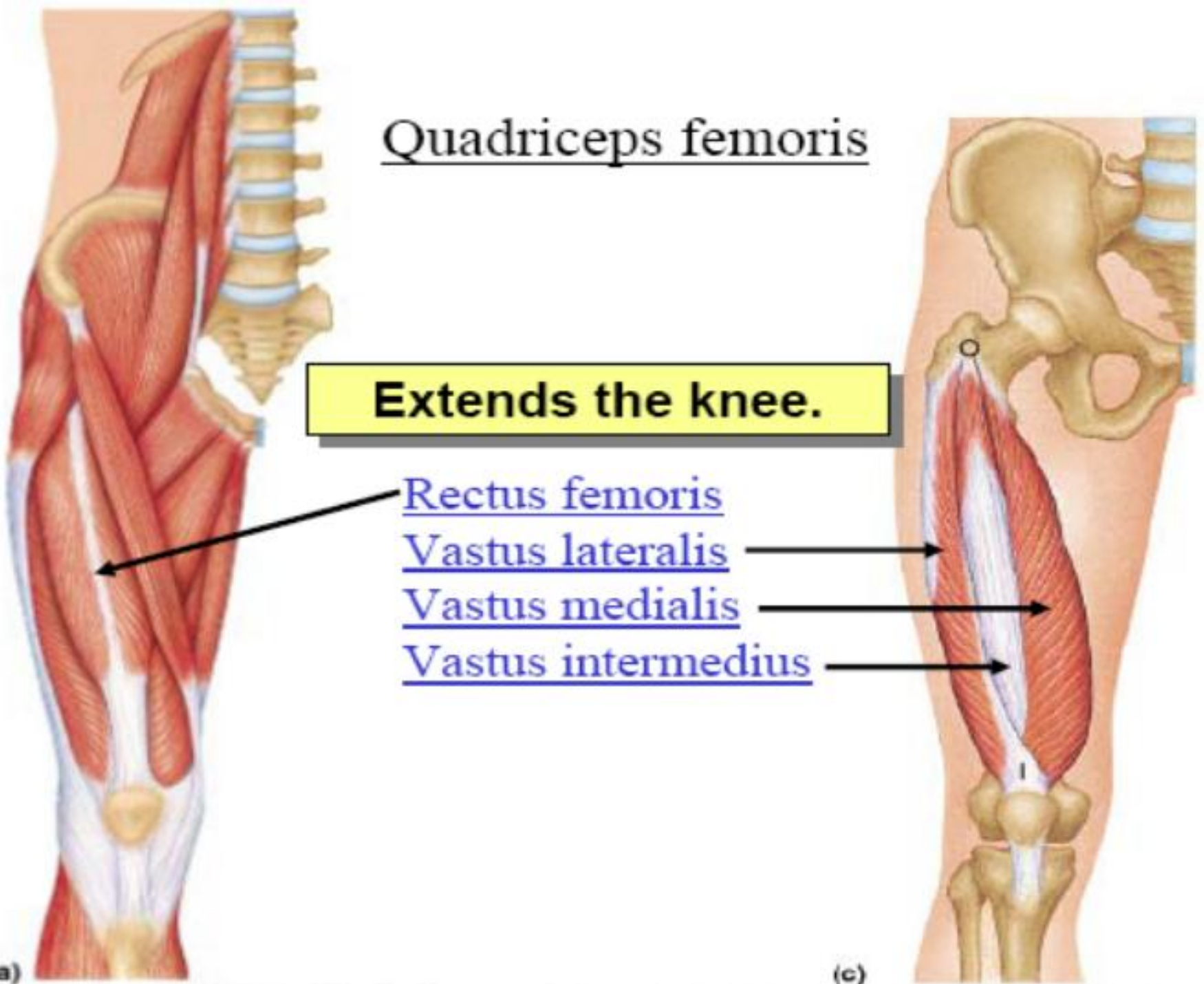
Vastus lateralis

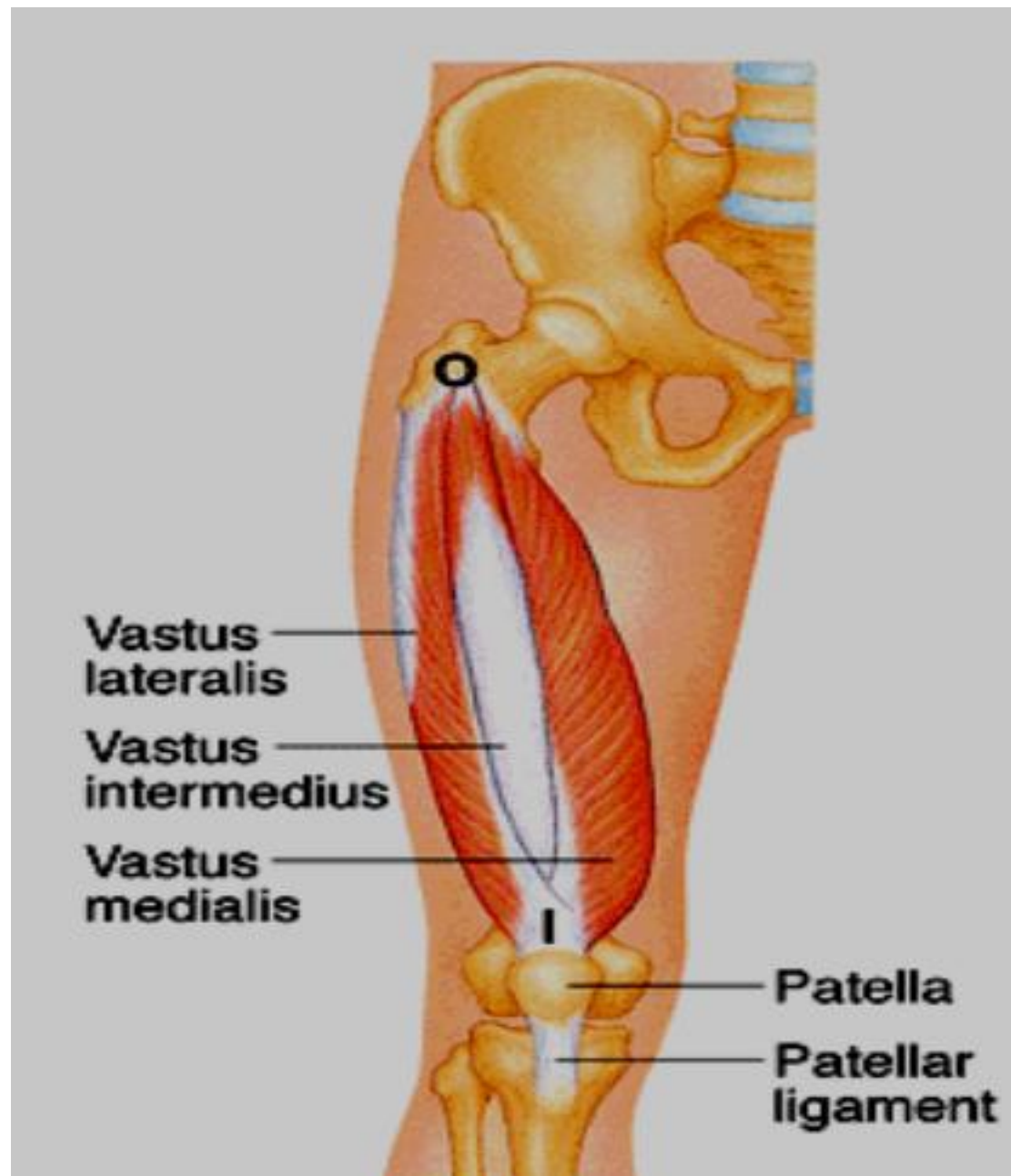
Vastus medialis

Vastus intermedius

(a)

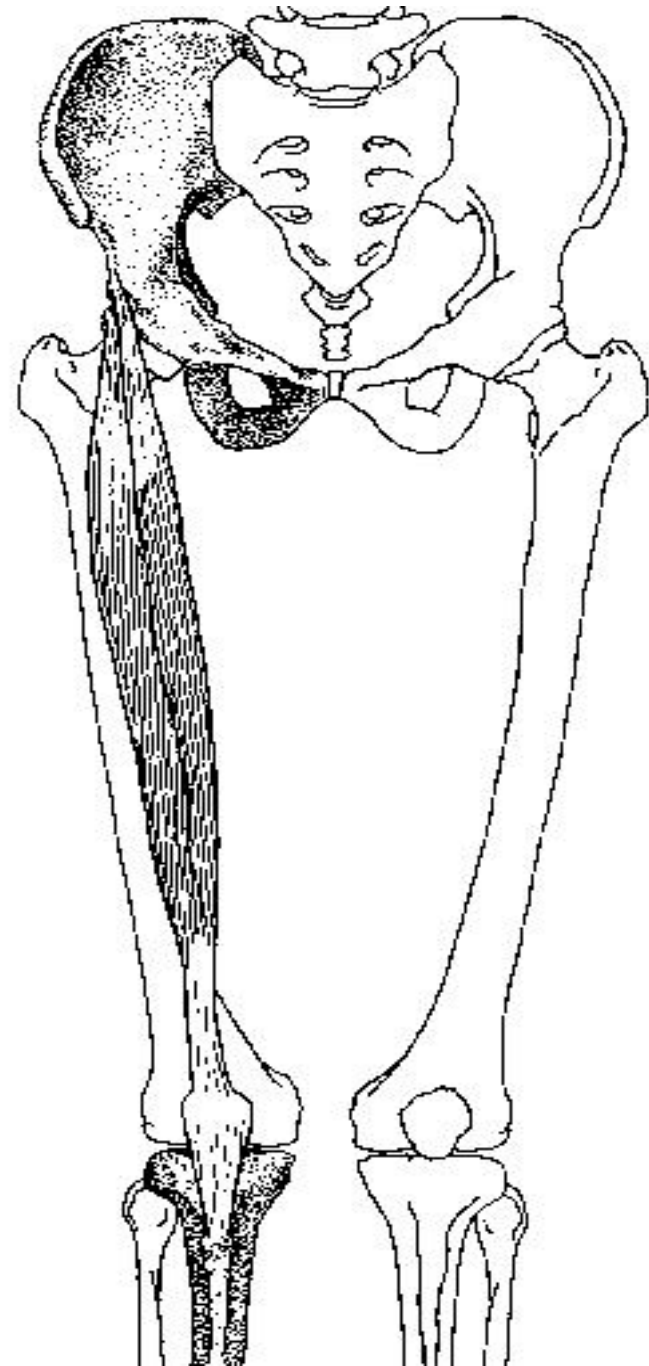
(c)





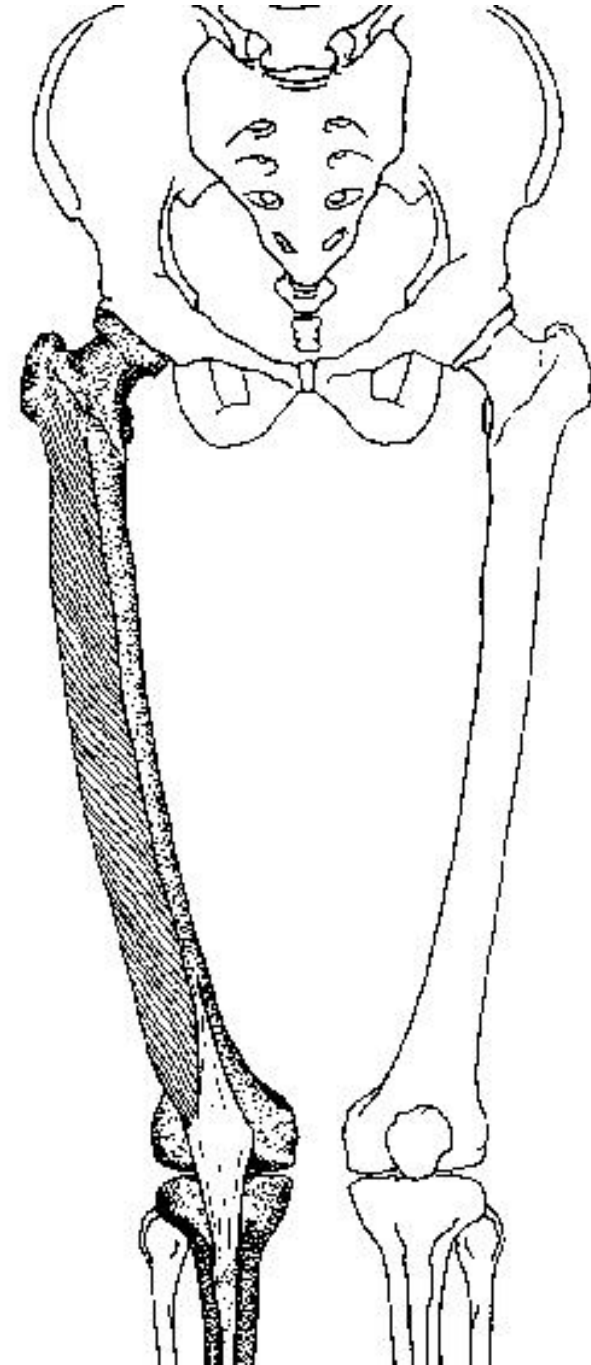
Rectus femoris

- Origin - anterior inferior iliac spine (by its straight head) and acetabular margin (by its reflected head)
- Thus, it can also flex the hip joint
- Insertion - patella and tibial tuberosity
- Action - extends knee and flexes thigh at hip
- Innervation - femoral nerve



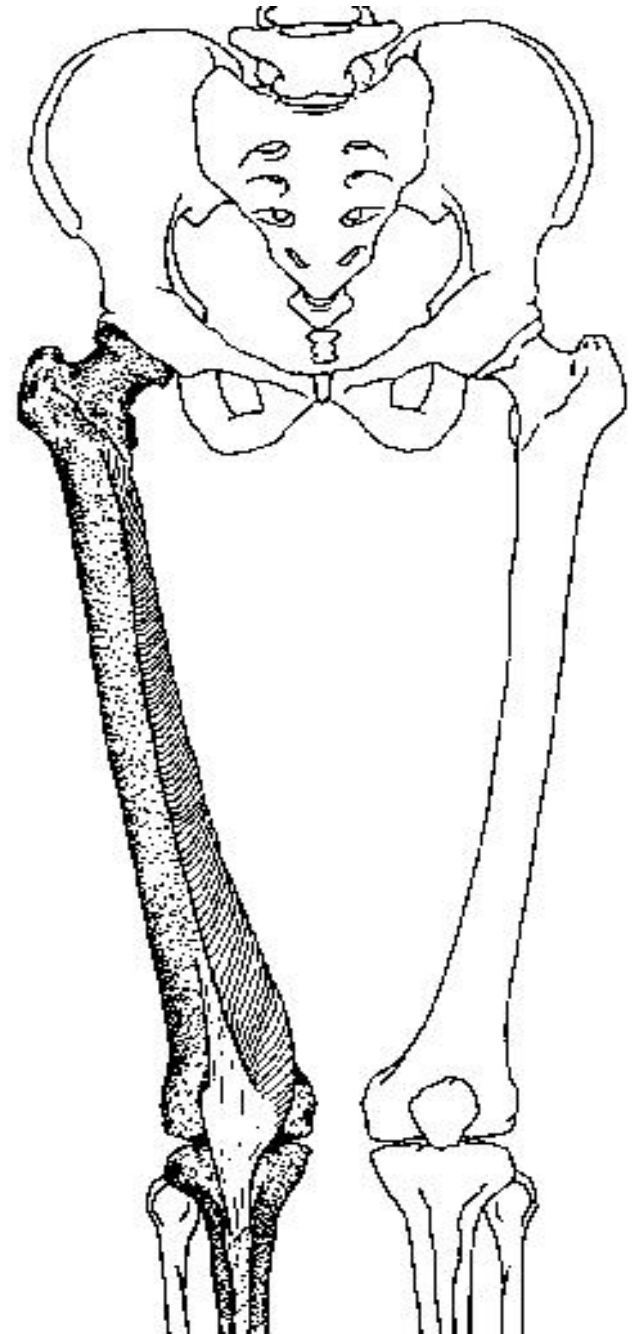
Vastus lateralis

- Origin - lateral lip of the linea aspera; greater trochanter, intertrochanteric line
- Insertion - patella and tibial tuberosity
- Action - extends knee
- Innervation - femoral nerve



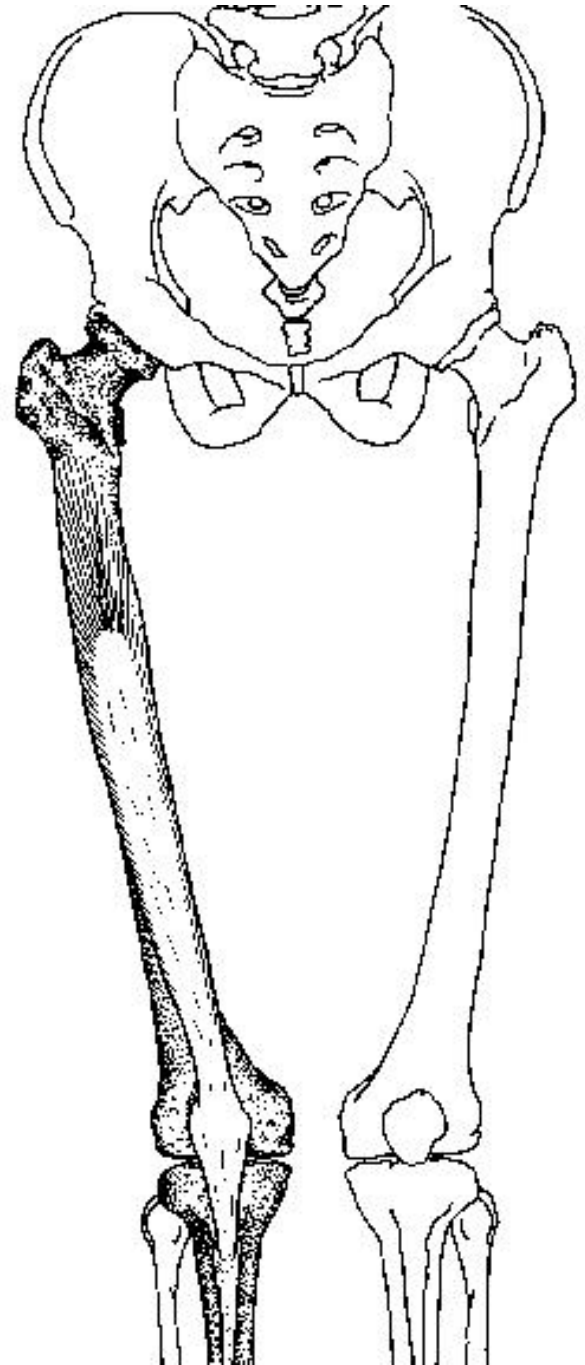
Vastus medialis

- Origin - medial lip of the linea aspera; intertrochanteric line
- Insertion - patella and tibial tuberosity
- Action - extends knee
- Innervation - femoral nerve



Vastus intermedius

- Origin - anterior and lateral surfaces of proximal shaft of femur
 - Its most distal fibers attach to the synovial capsule at the knee and retracts the capsule during extension of knee, thereby preventing folds of the membrane from being compressed between the femur and the patella within the knee joint. This portion is called the articularis genu
- Insertion - patella and tibial tuberosity
- Action - extend knee
- Innervation - femoral nerve



Paralysis of quadriceps

- Tested with a person in supine position with the knee partly flexed then extend the knee against resistance
- If the muscle is paralyzed, the person cannot extend

Patellar tendon reflex

- Tapping the patellar ligament with a reflex hammer normally elicits patellar tendon reflex (knee jerk)
- Causes leg to extend
- It tests L2-L4 nerves
- Absence of patellar tendon reflex may result from any lesion that interrupts innervation of quadriceps muscle

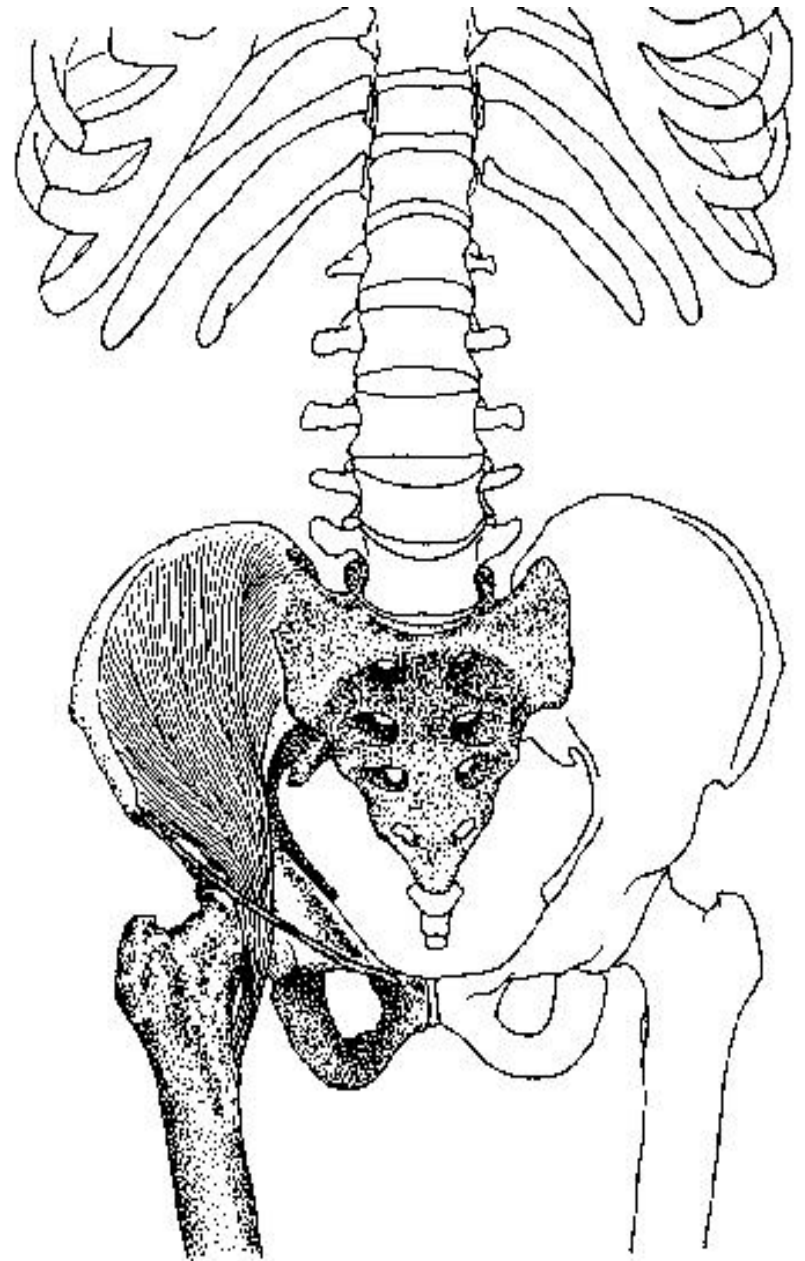
Muscles of the Hip

Medial muscles of the hip

- Psoas major and minor
- Iliacus

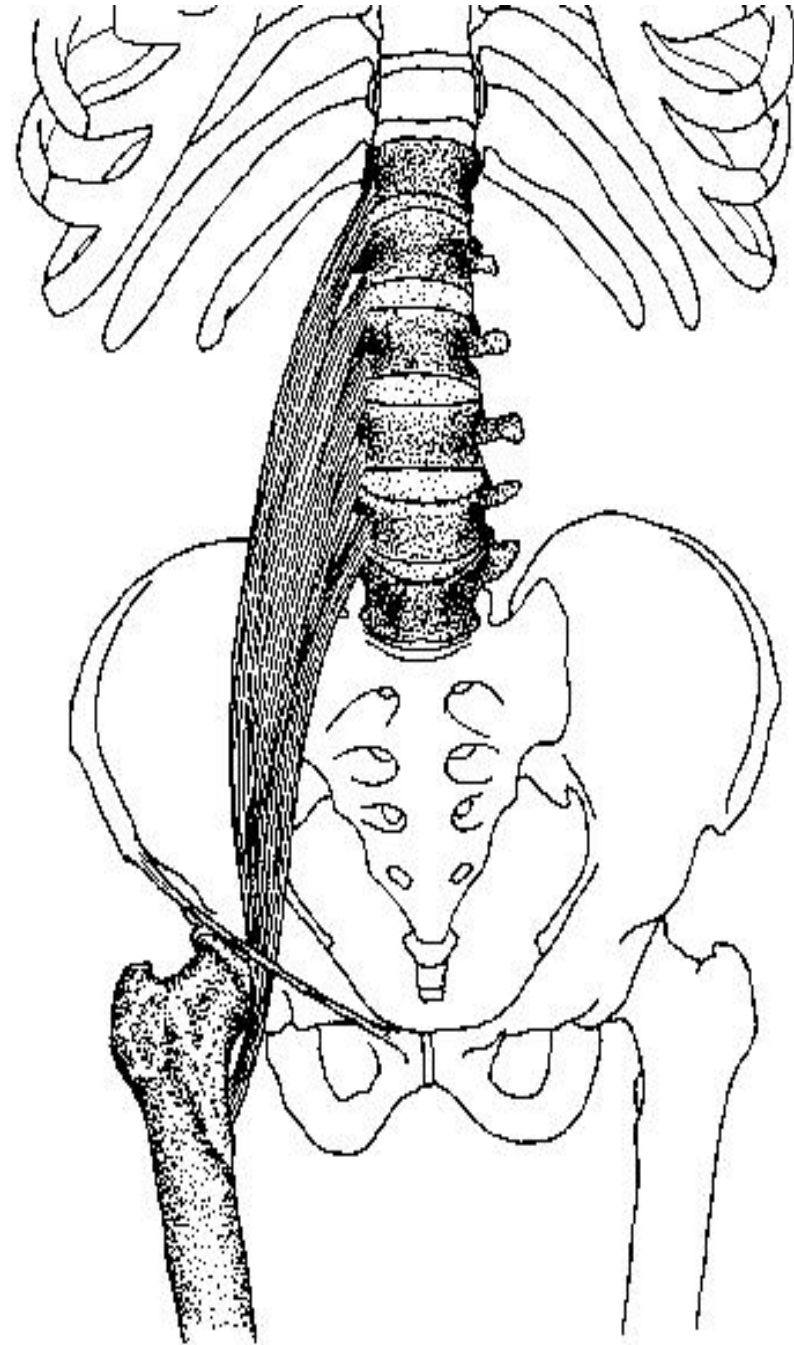
Iliacus

- Origin - iliac fossa
- Insertion - lesser trochanter of femur
- Action - hip flexion
- Innervation - femoral branch in the abdomen



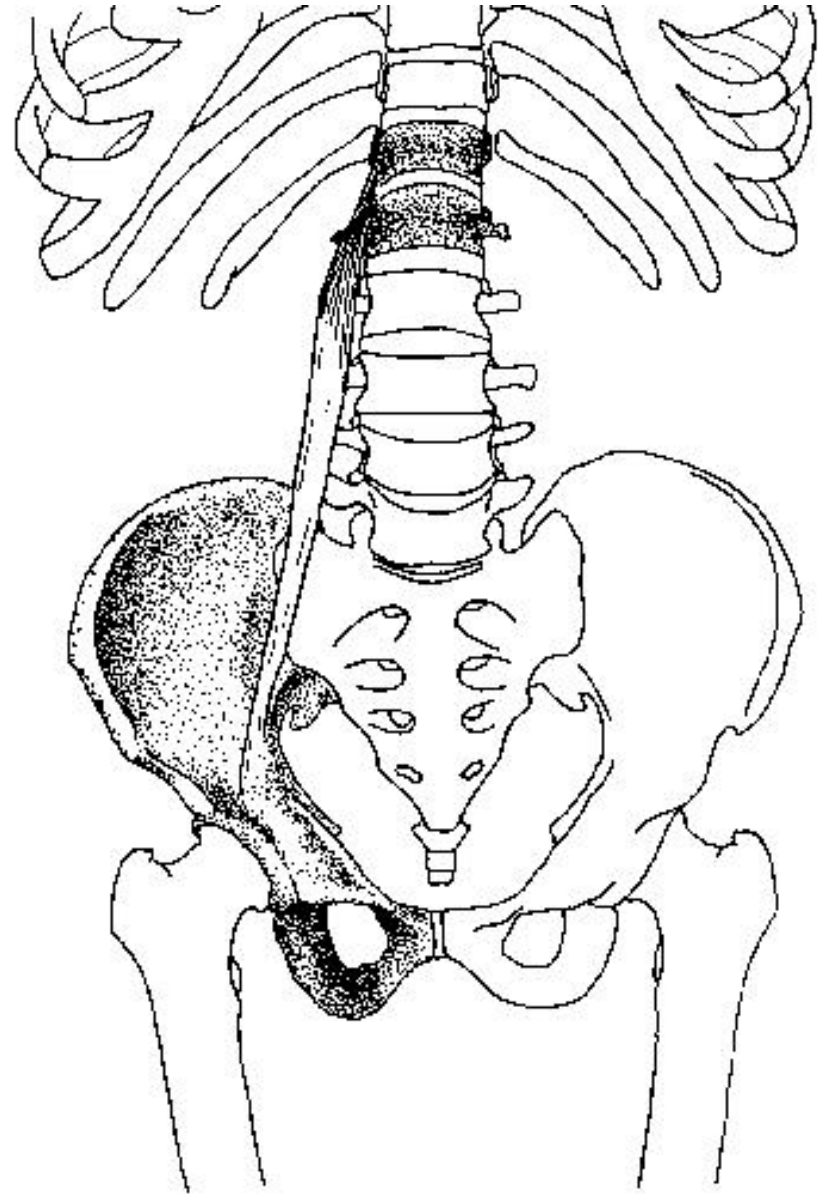
Psoas major

- Origin - transverse processes and bodies of T12 and all lumbar vertebrae
- Insertion - lesser trochanter of femur
- Action - hip flexion
- Innervation - branches from lumbar plexus



Psoas minor

- Origin - transverse processes and bodies of T12 and L1
- Insertion - rim of acetabulum
- Action - flexes lumbar vertebrae



Medial thigh muscles

Collectively called adductor group and found in medial compartment.

Attach from the pubis and ischium to the linea aspera

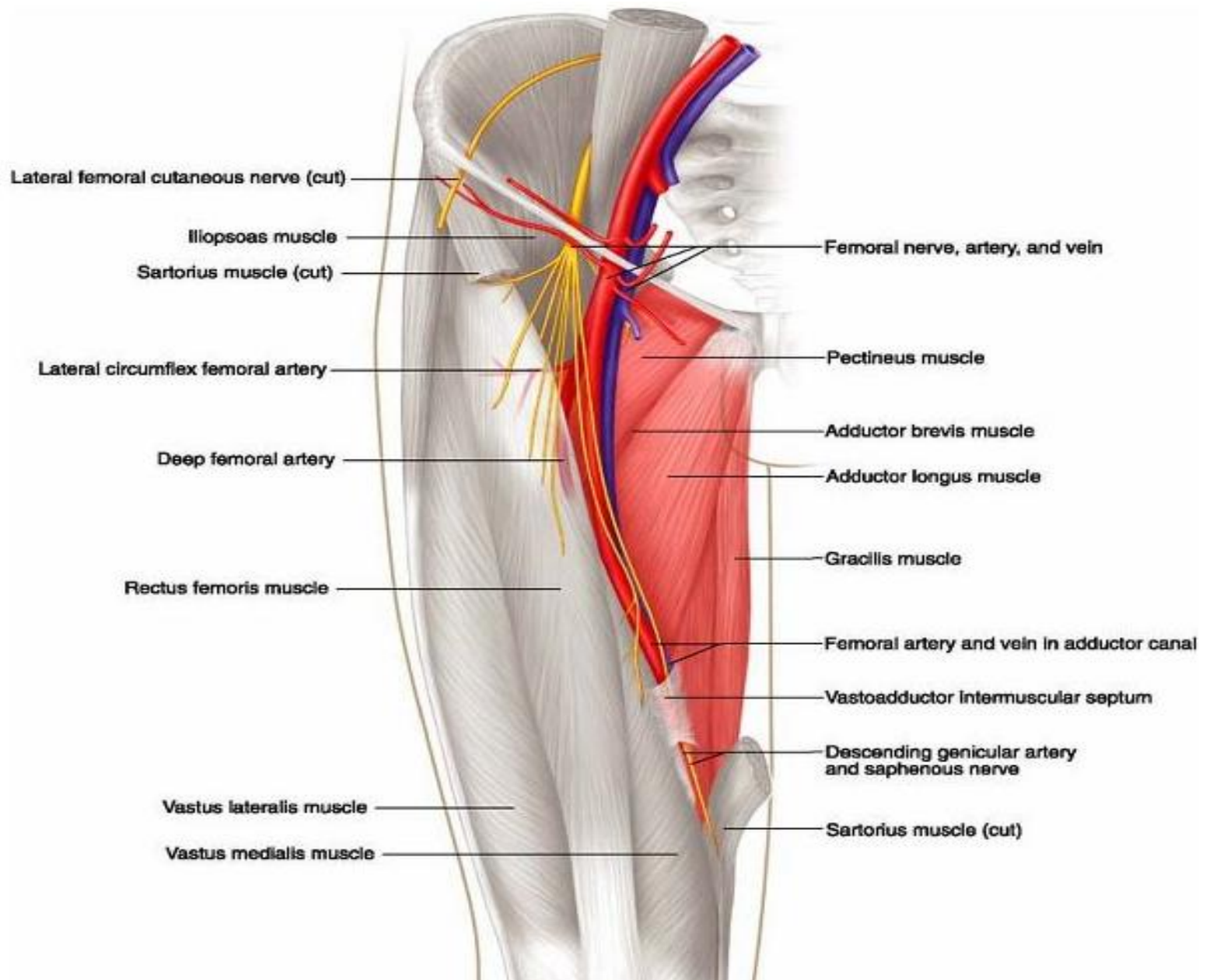
Adductor muscles are:

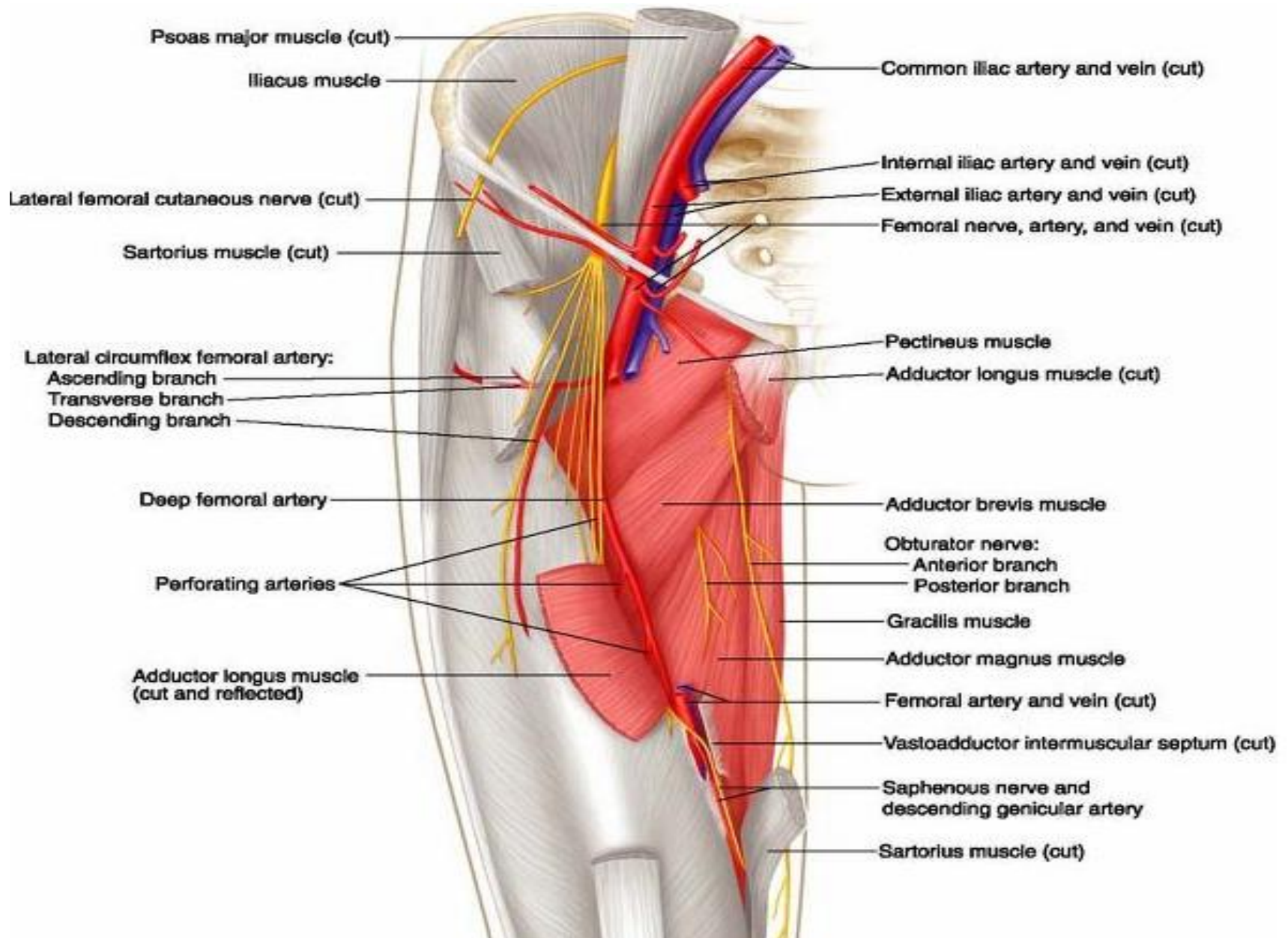
- ✓ Adductor Magnus
- ✓ Adductor longus
- ✓ Adductor brevis
- ✓ Pectineus
- ✓ Gracilis
- ✓ Obturator externus

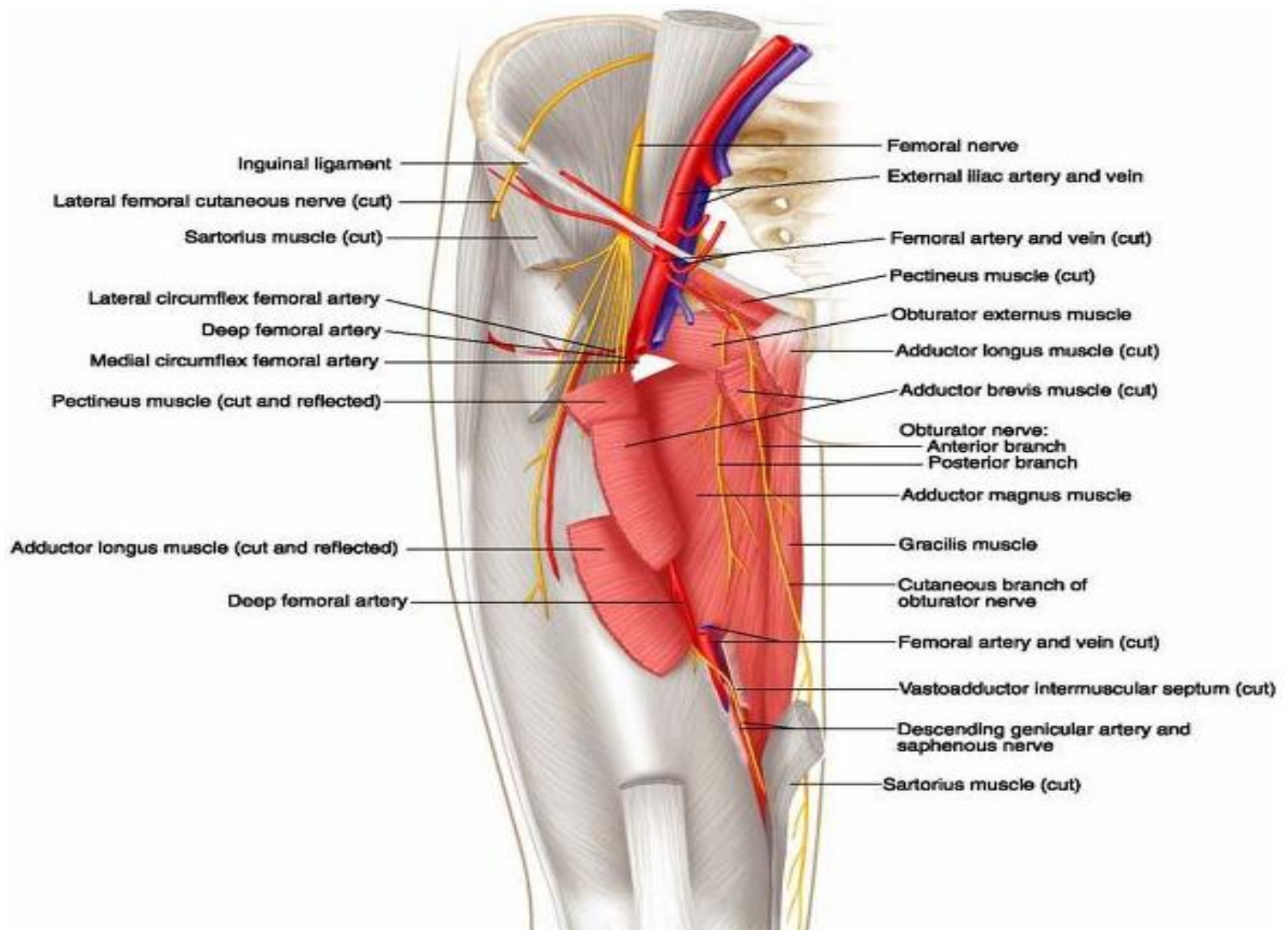
All are supplied by obturator nerve except pectineus and hamstring part of adductor magnus

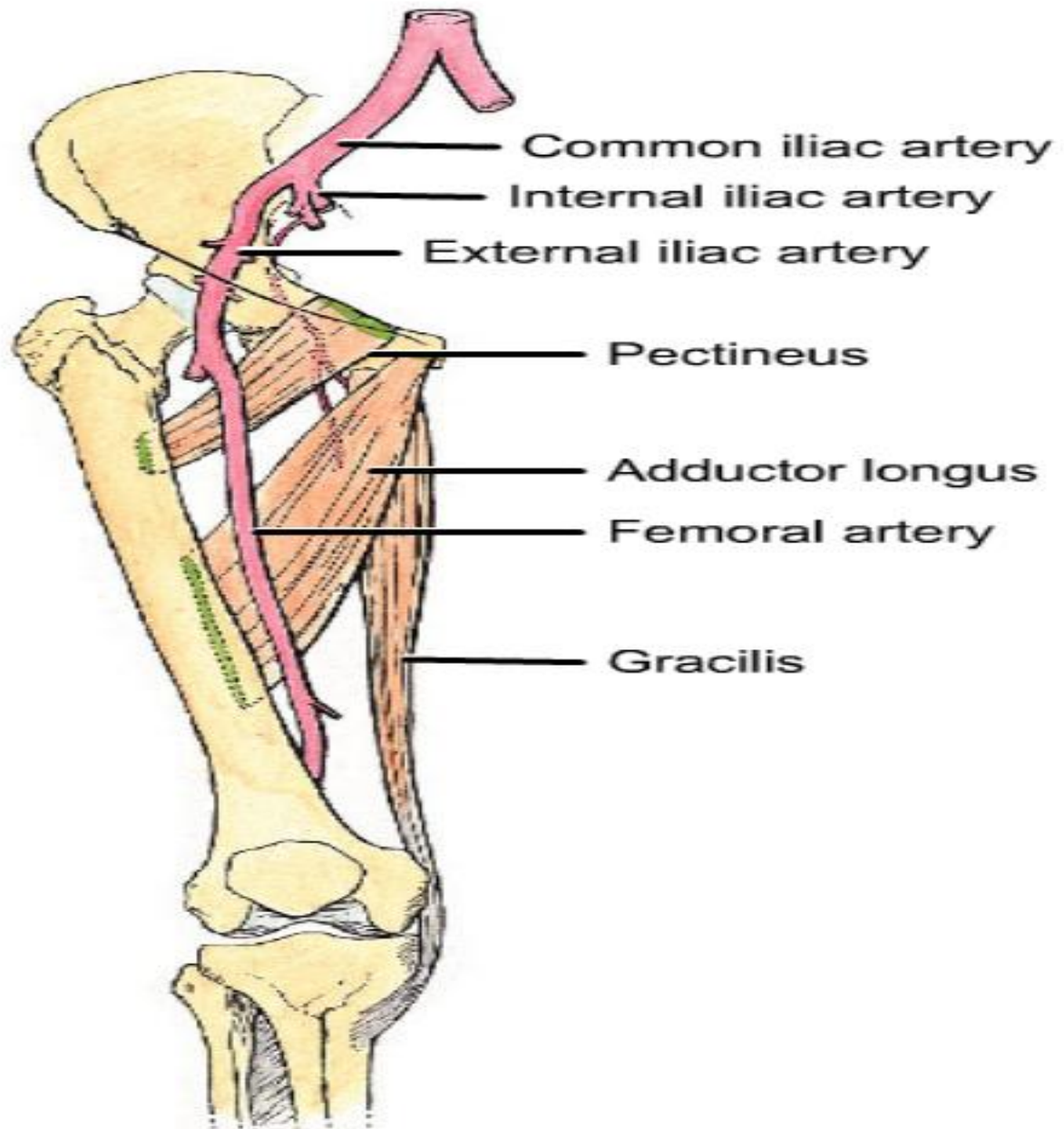
Three layers

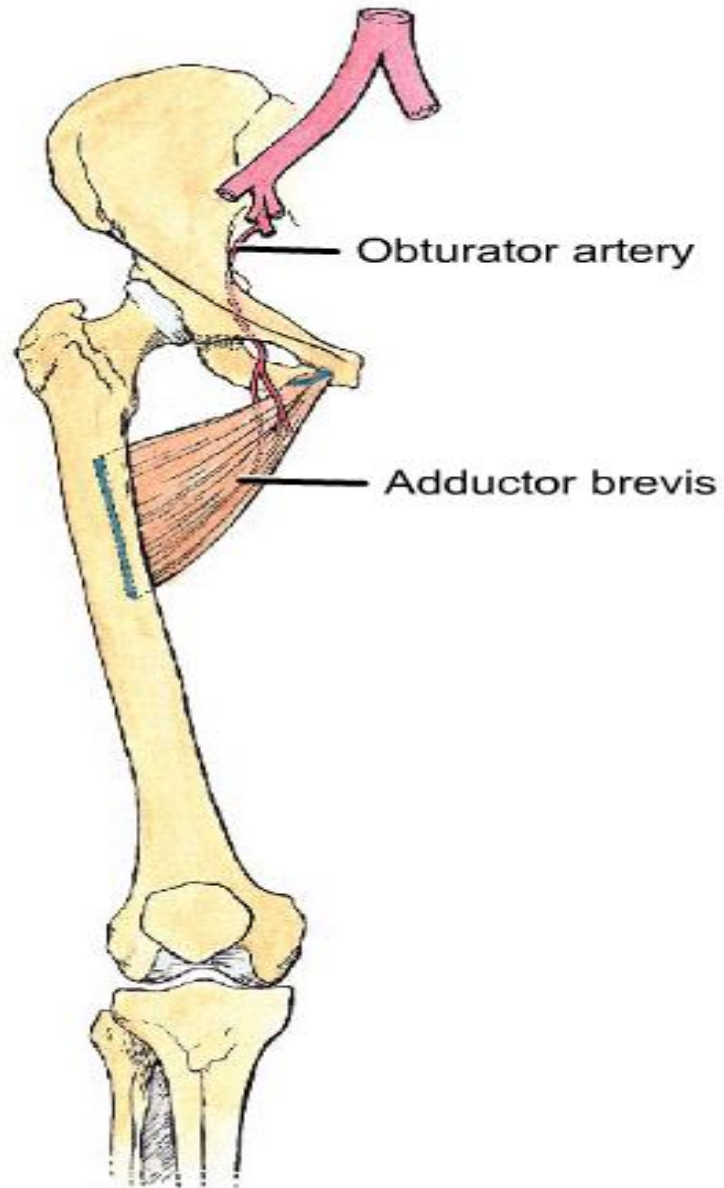
- ✓ Superficial - pectineus, gracilis and adductor longus
- ✓ Middle - adductor brevis
- ✓ Deep - adductor magnus & obturator externus

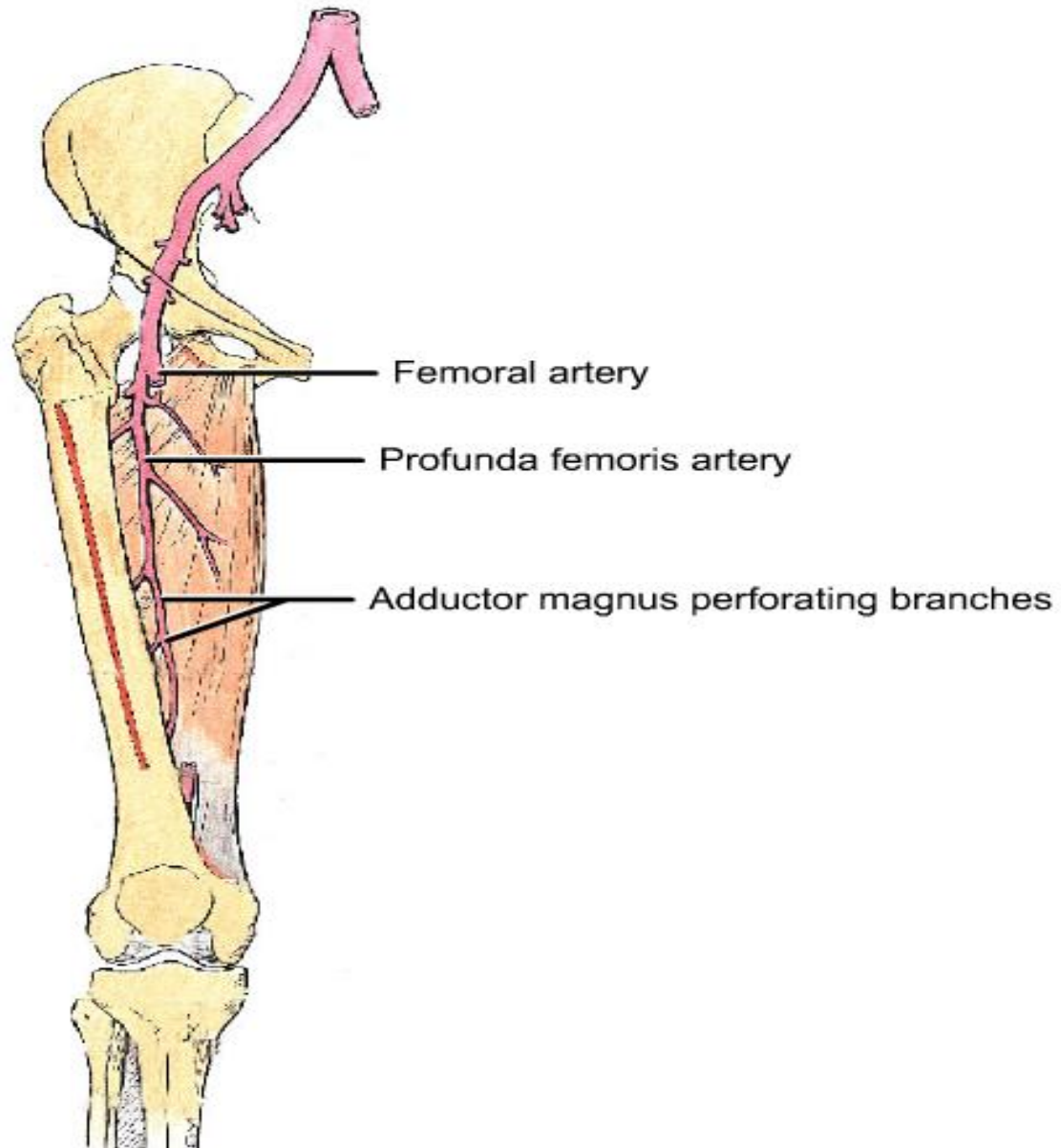


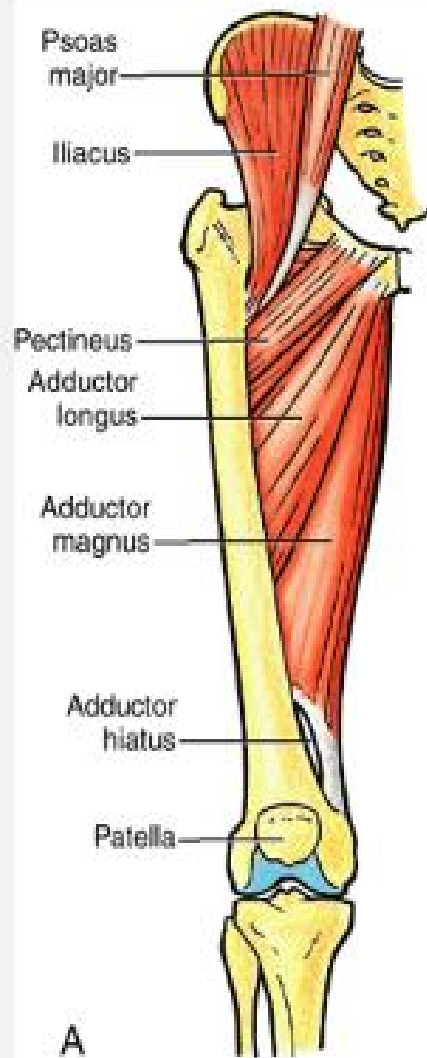






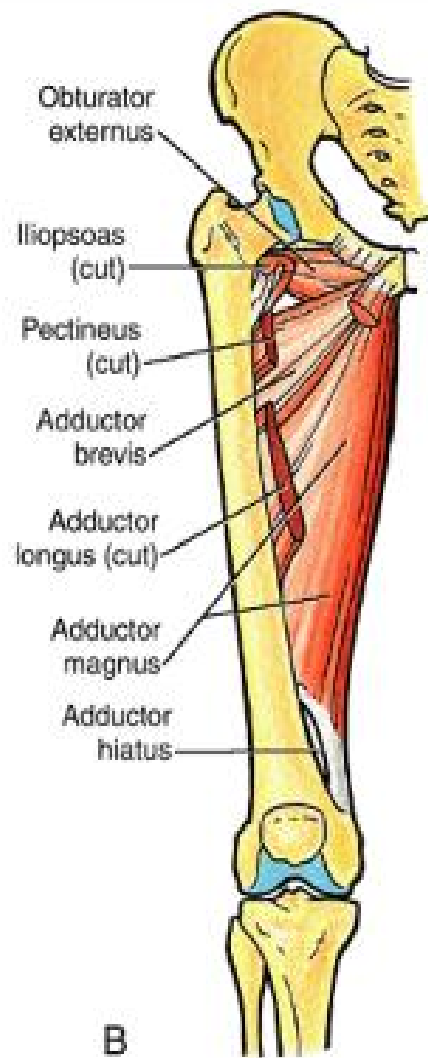




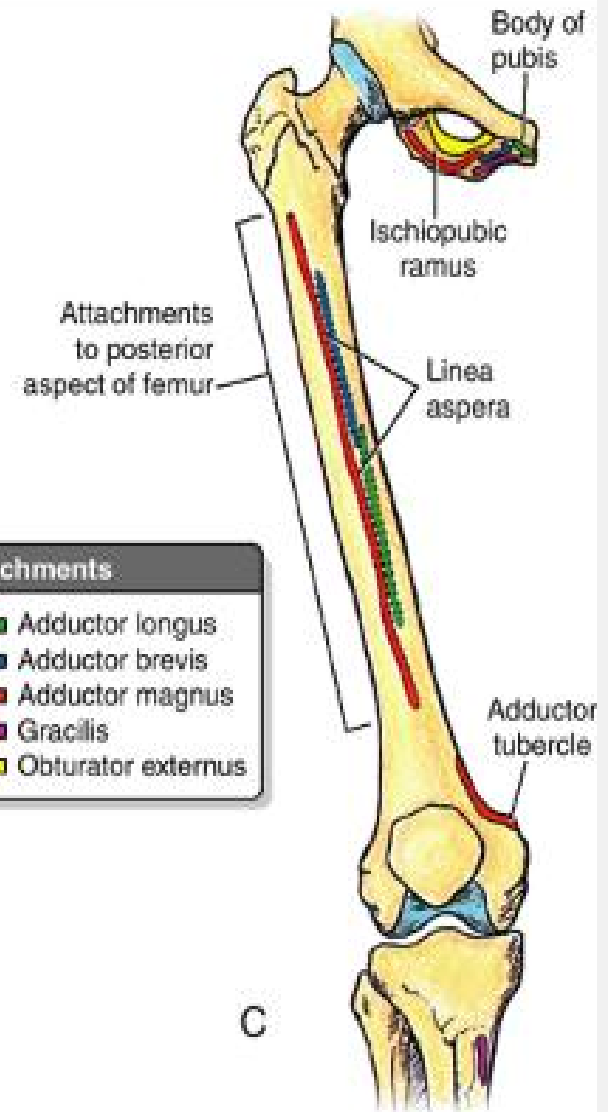


A

Anterior views



B



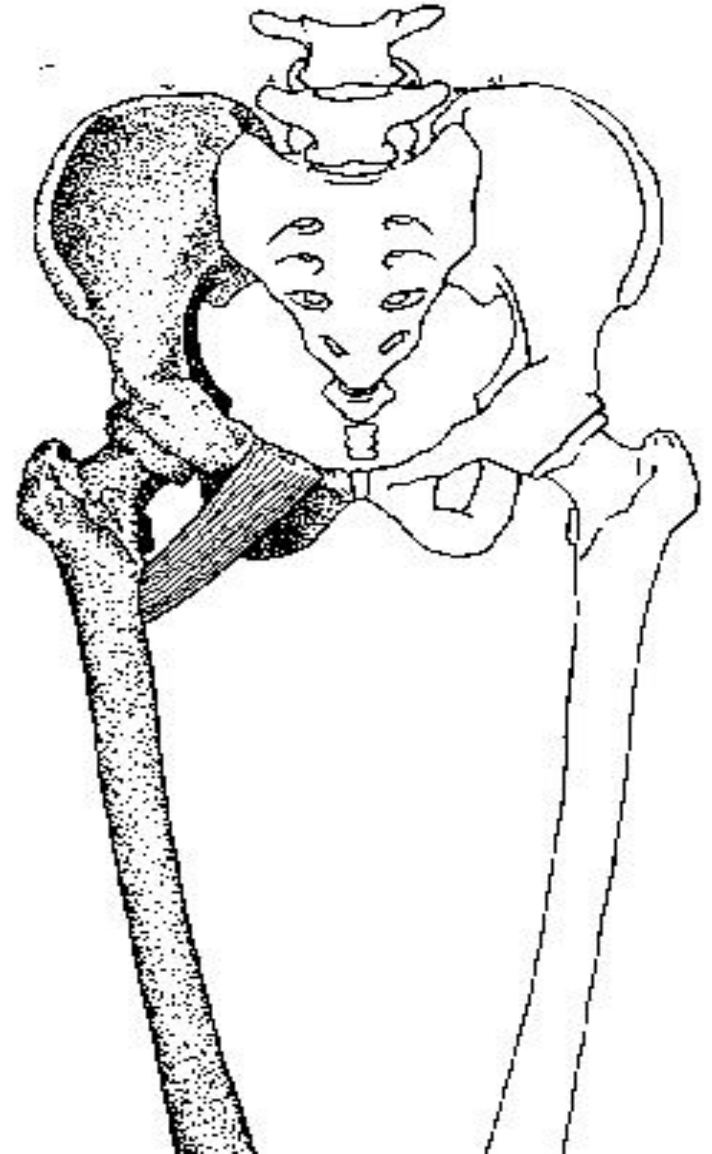
C

Attachments

- Adductor longus
- Adductor brevis
- Adductor magnus
- Gracilis
- Obturator externus

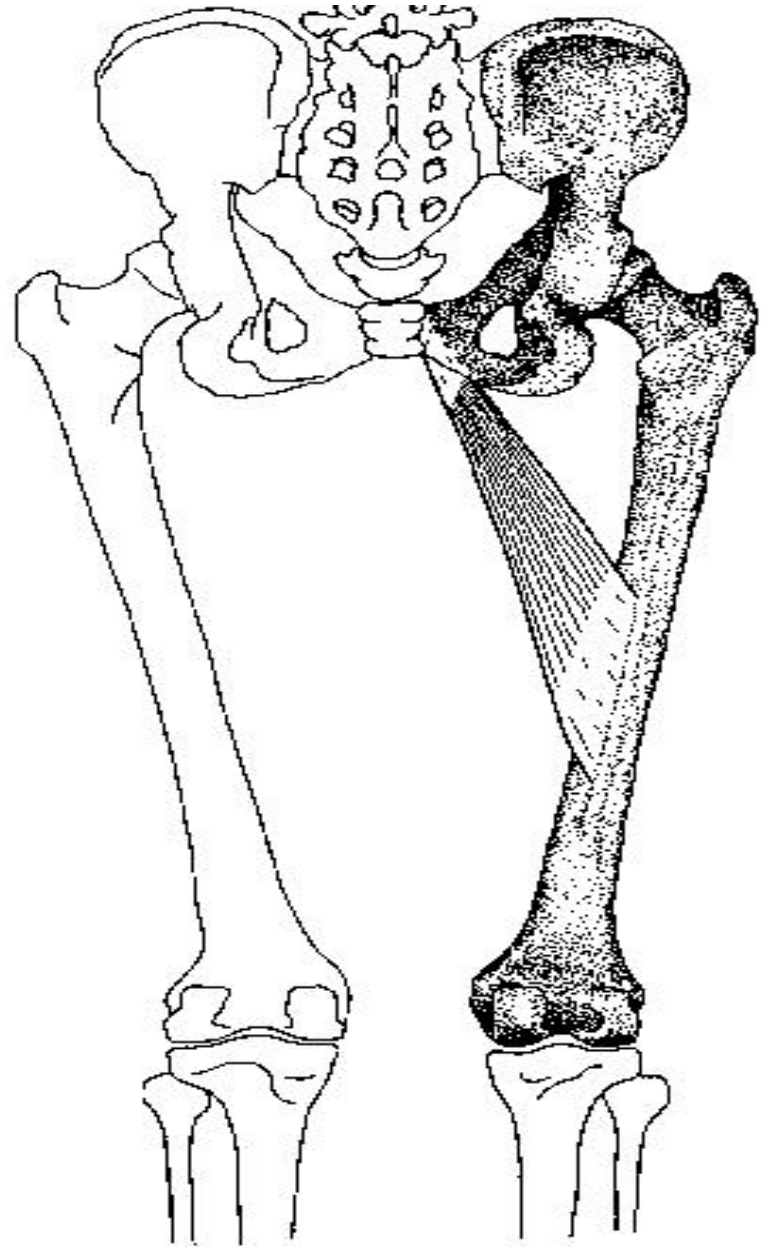
Pectineus

- Origin - pectineal line of pubis
- Insertion - lesser trochanter of femur
- Action - adducts, flexes, and laterally rotates thigh
- Innervation - femoral nerve



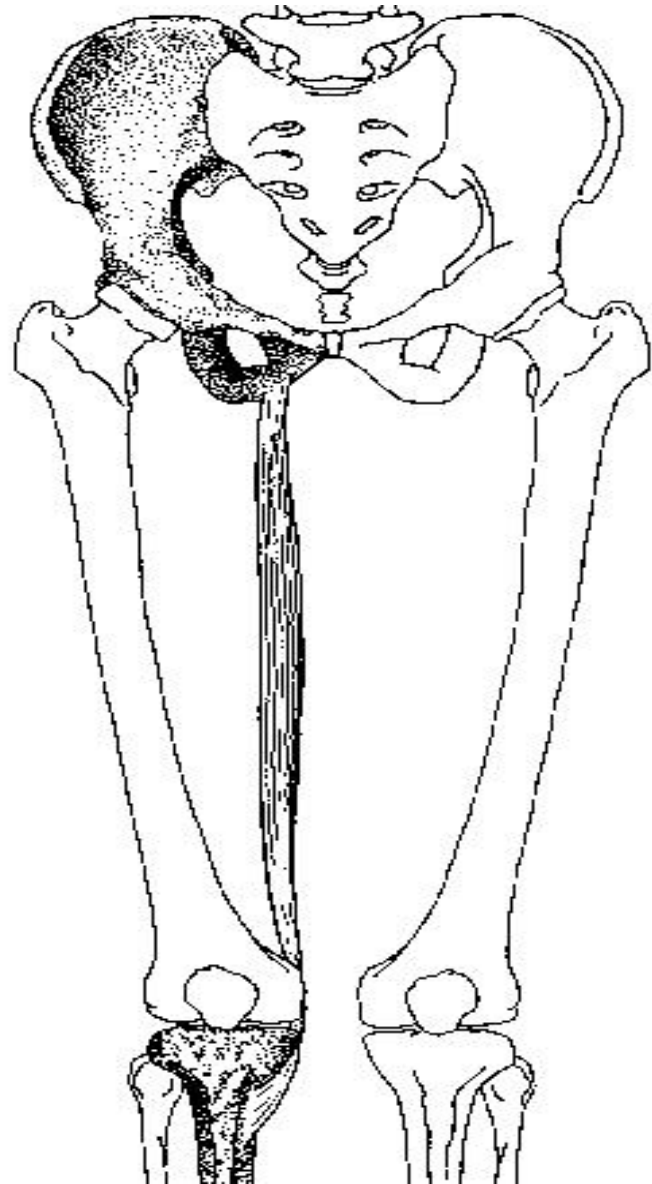
Adductor longus

- Origin - pubis near pubic symphysis
- Insertion - linea aspera
medial lip of middle 1/3
- Action - adducts thigh
- Innervation - Obturator
nerve (anterior branch)



Gracilis

- Origin - inferior ramus and body of pubis
- Insertion - medial surface of tibia just inferior to medial condyle
- Action - adducts the thigh, flexes and medially rotates the leg
- Innervation - anterior branch of obturator nerve

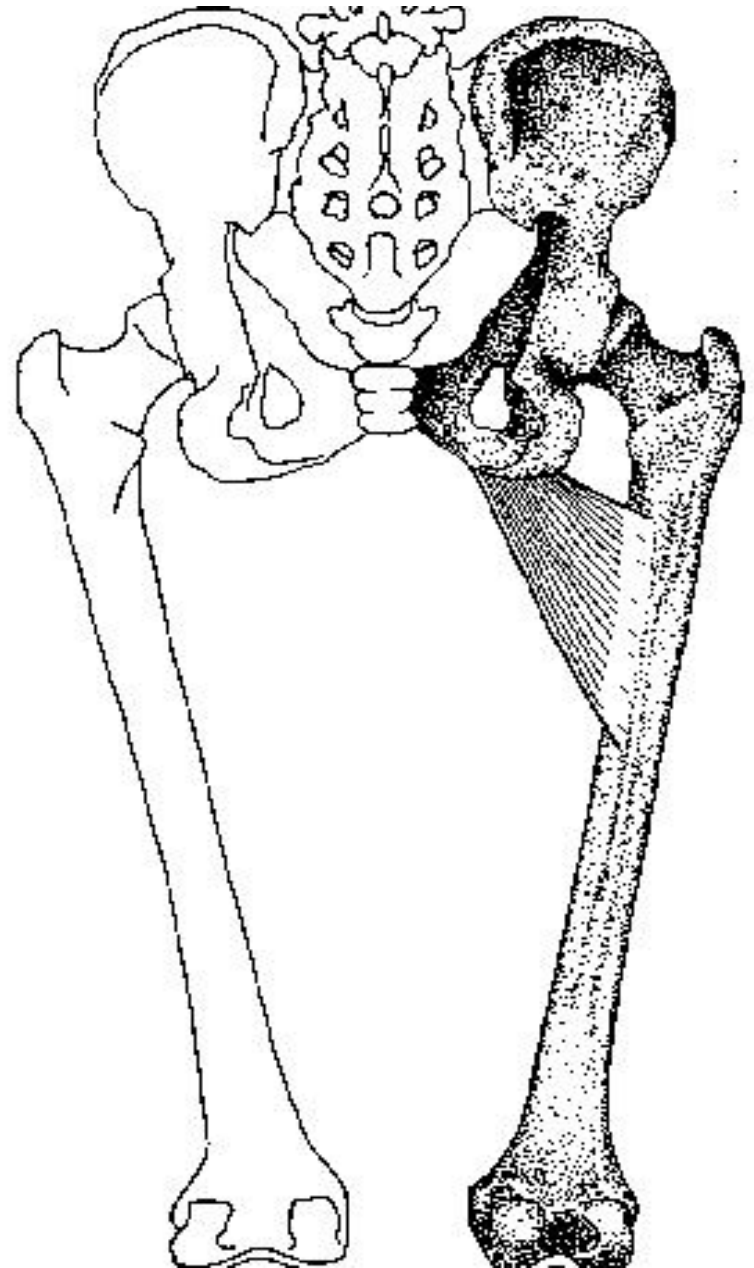


Transplantation of gracilis

- ✓ Because gracilis is relatively weak adductor, it can be removed without noticeable loss of its actions
- ✓ Hence used to replace a damaged muscle in other areas

Adductor brevis

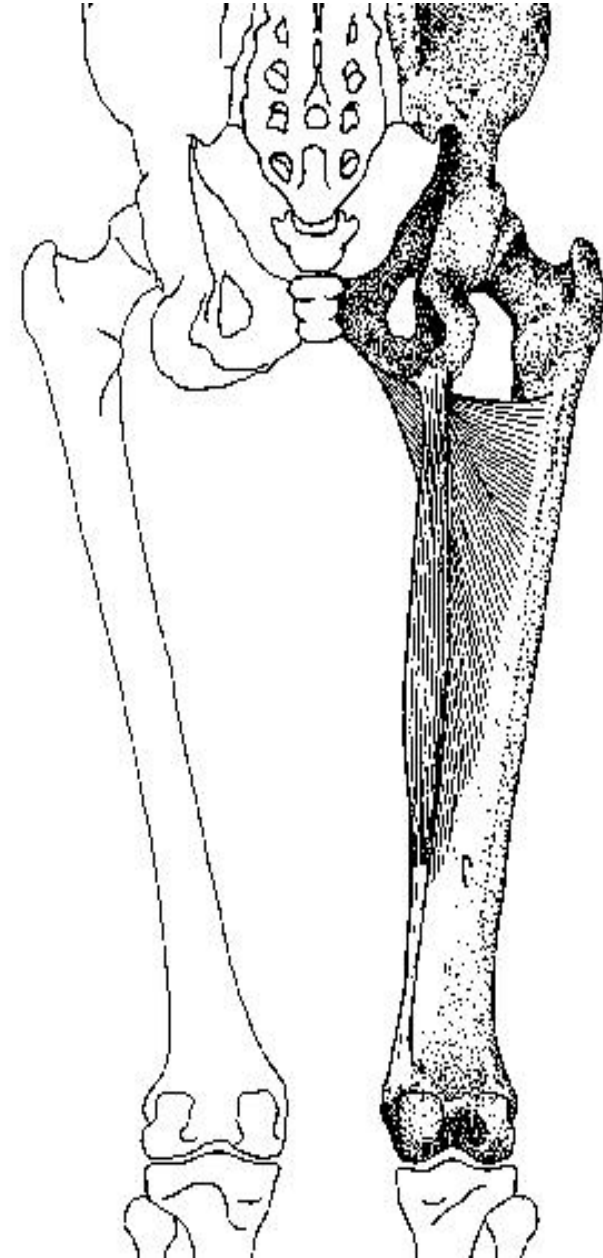
- Origin - body and inferior ramus of pubis
- Insertion - linea aspera (medial lip)
- Action – adducts, laterally rotates, and flex (weak) the thigh
- Innervation - anterior branch of obturator nerve

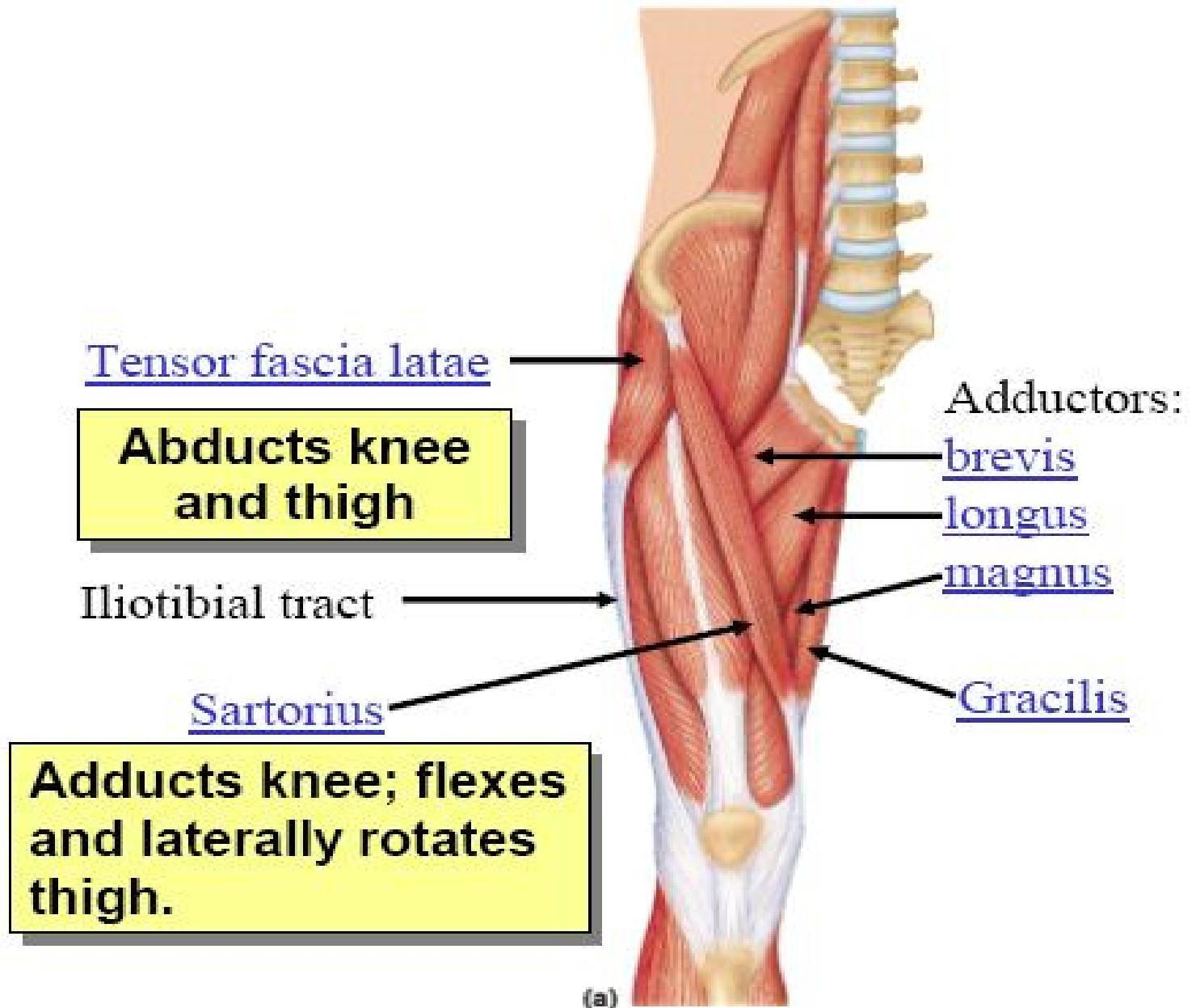


Adductor magnus

Origin

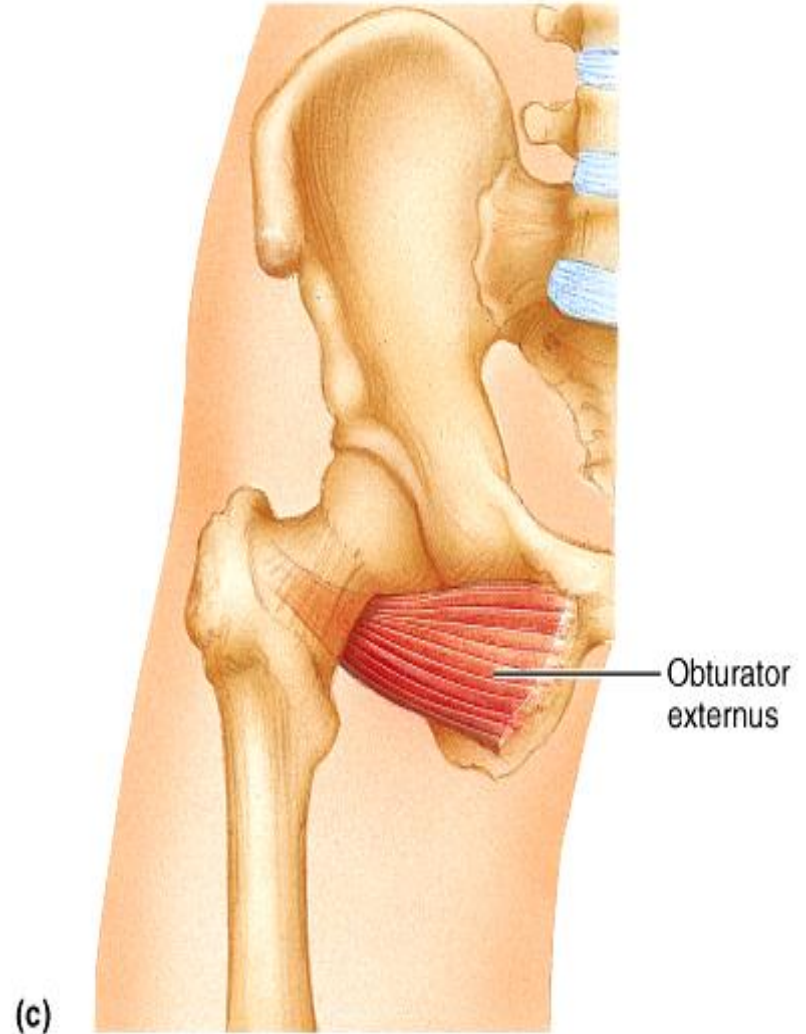
- Adductor part: ischial and pubic rami
- Hamstring part: ischial tuberosity
- Insertion
 - Adductor part: linea aspera (medial lip) and supracondylar line
 - Hamstring part: adductor tubercle
- Action - pubic part adducts and ischial part flexor of the hip
- Innervation
 - Adductor part: obturator nerve
 - Hamstring part: sciatic nerve





Obturator externus

- Origin:- margins of obturator foramen and obturator membrane
- Insertion:- trochanteric fossa of femur
- Innervation:- obturator nerve
- Action:- lateral rotators of the thigh; steadies head of femur



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Groin pull

- Strain of proximal attachment of anteromedial thigh muscles
- Usually occur in sports that require quick start

Femoral triangle

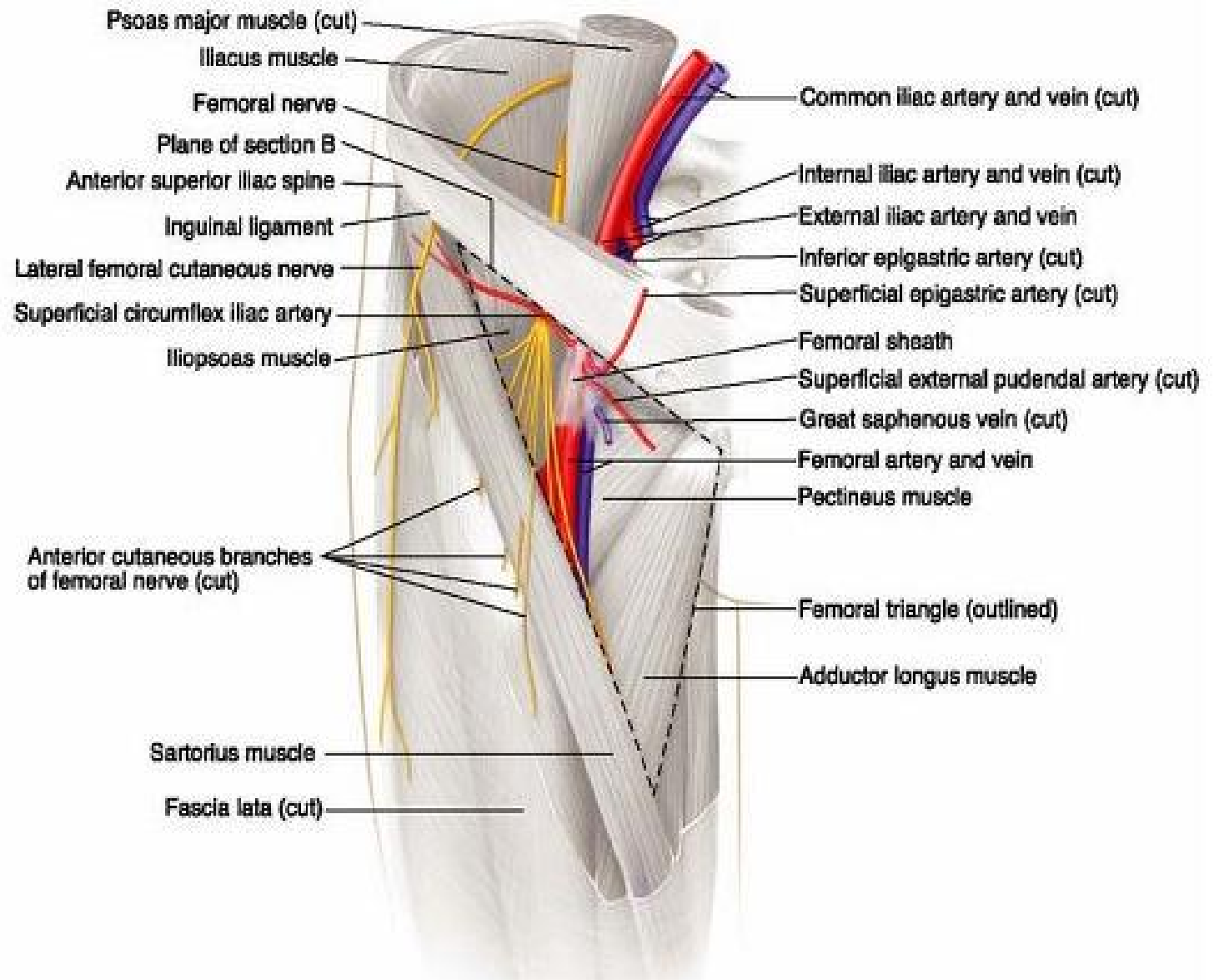
Boundaries

- **Medial** - medial boarder of adductor longus
- **Lateral** - medial boarder of sartorius
- **Superior** - inguinal ligament
- **Floor** - adductor longus, pectineus, psoas and iliacus
- **Roof** - formed by
 - superficial fascia, containing superficial inguinal lymph nodes and great saphenous vein with its tributaries
 - deep fascia (fascia lata), which is pierced by the saphenous vein at the saphenous opening

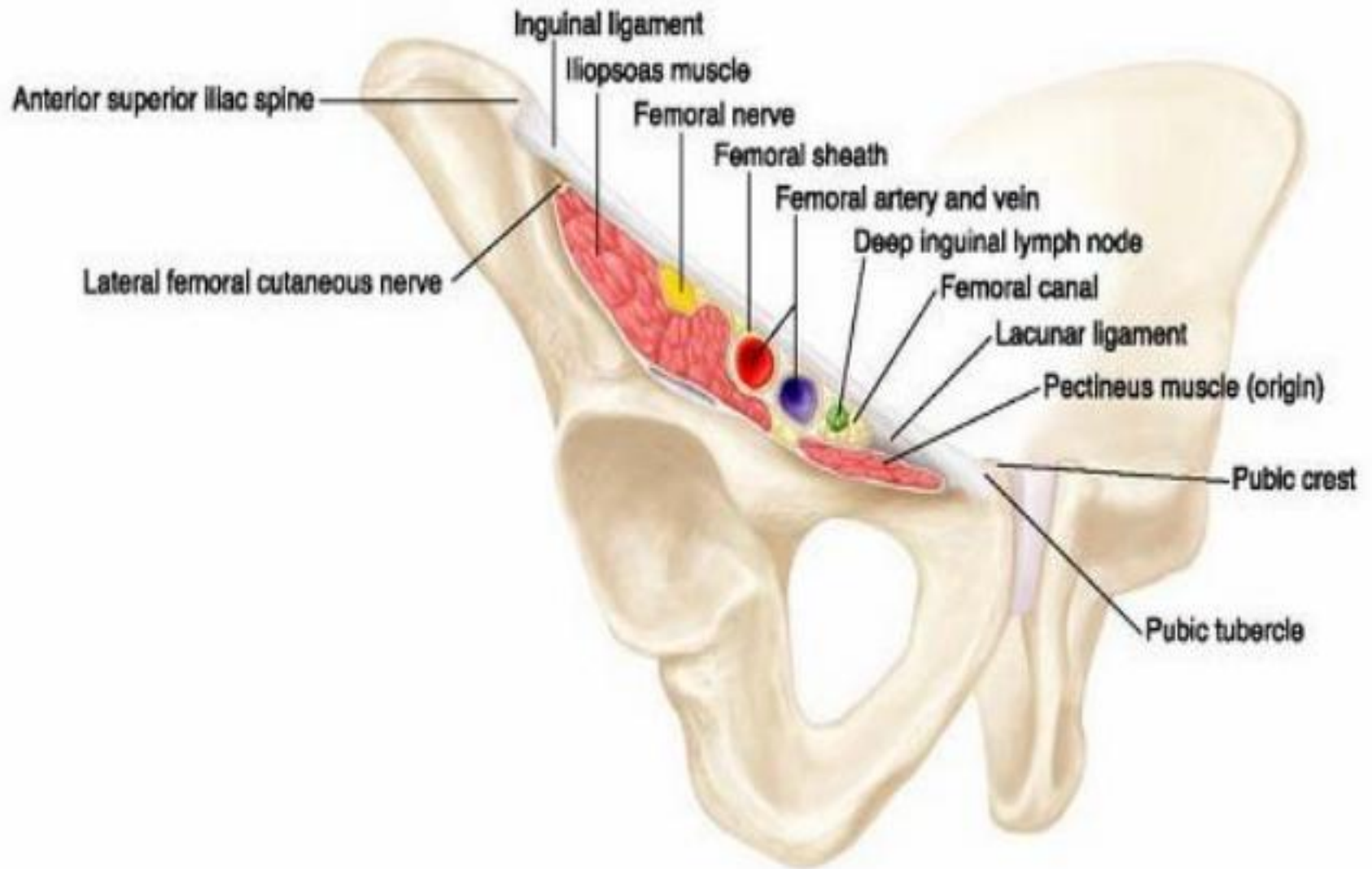
Contents

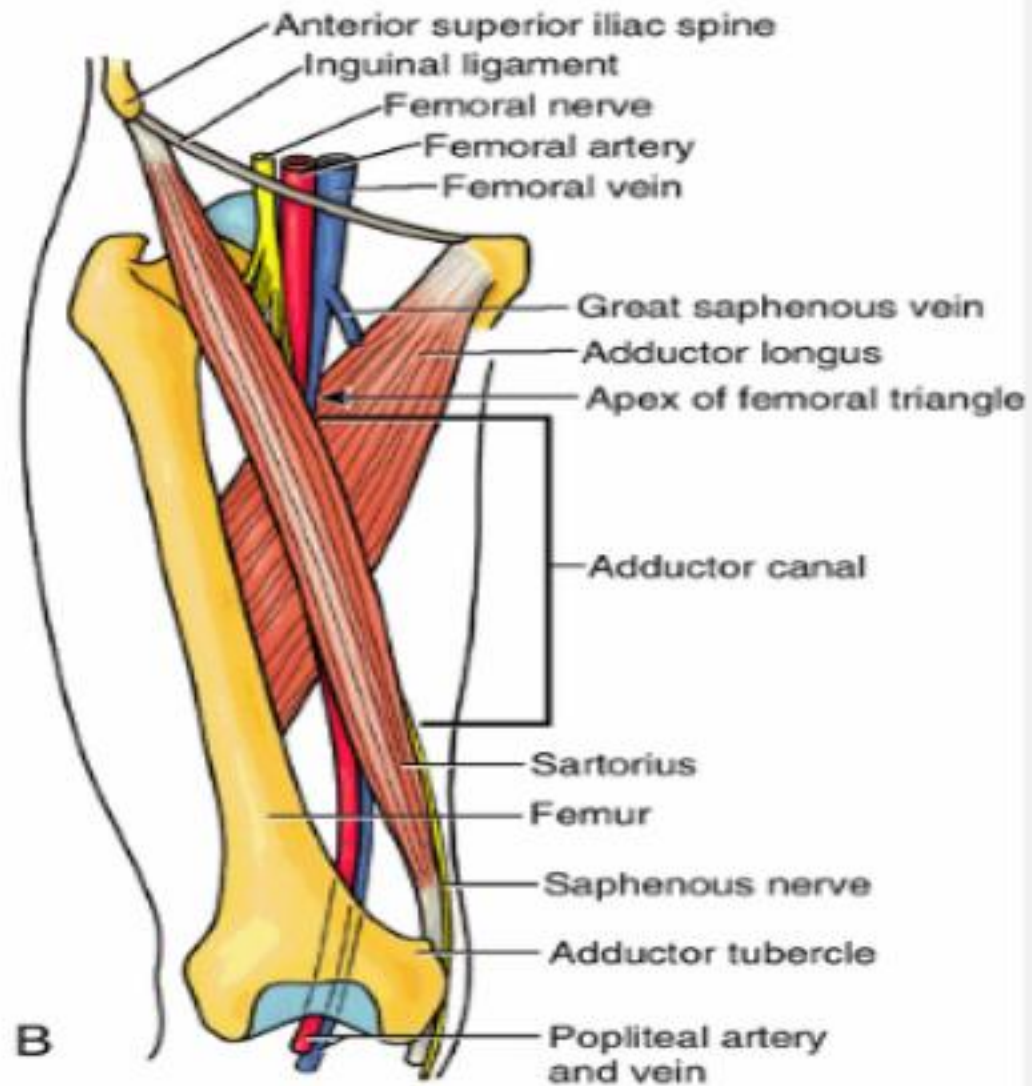
- ✓ femoral sheath
- ✓ femoral artery and its branch
- ✓ femoral vein and its tributaries
- ✓ femoral nerve and its branches
- ✓ deep inguinal lymph nodes
- ✓ profunda femoris artery and vein and their circumflex branches

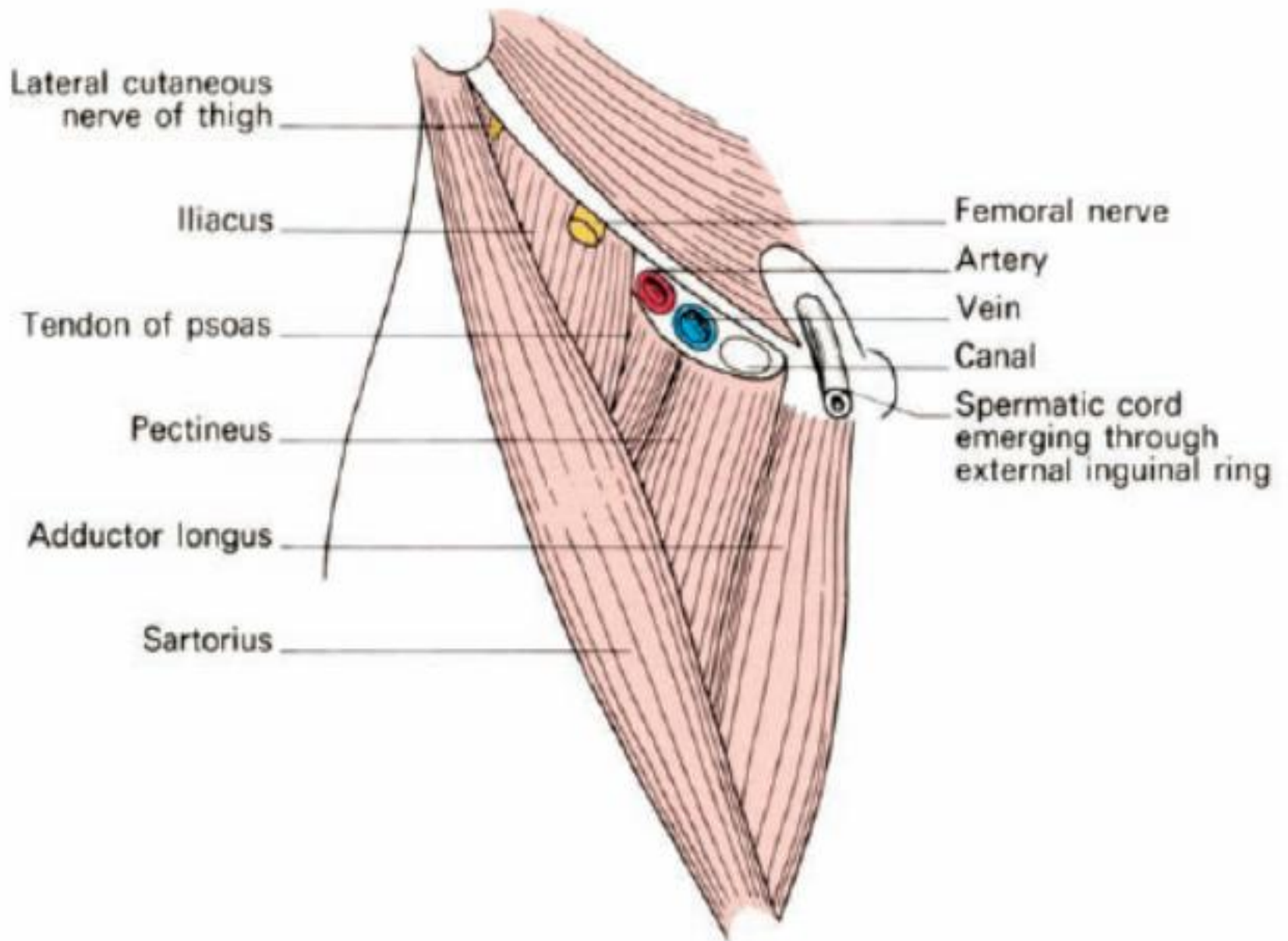
A. Anterior view

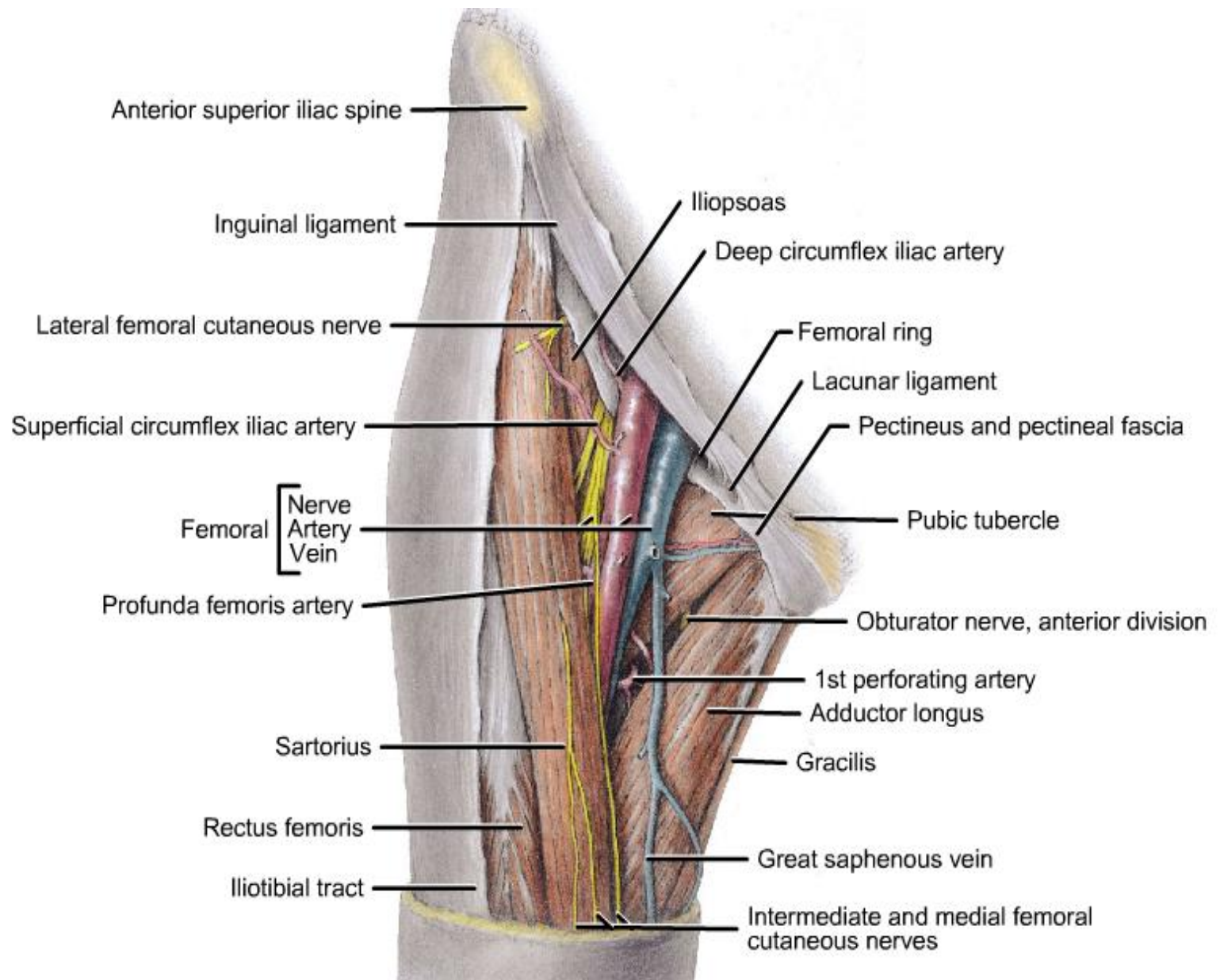


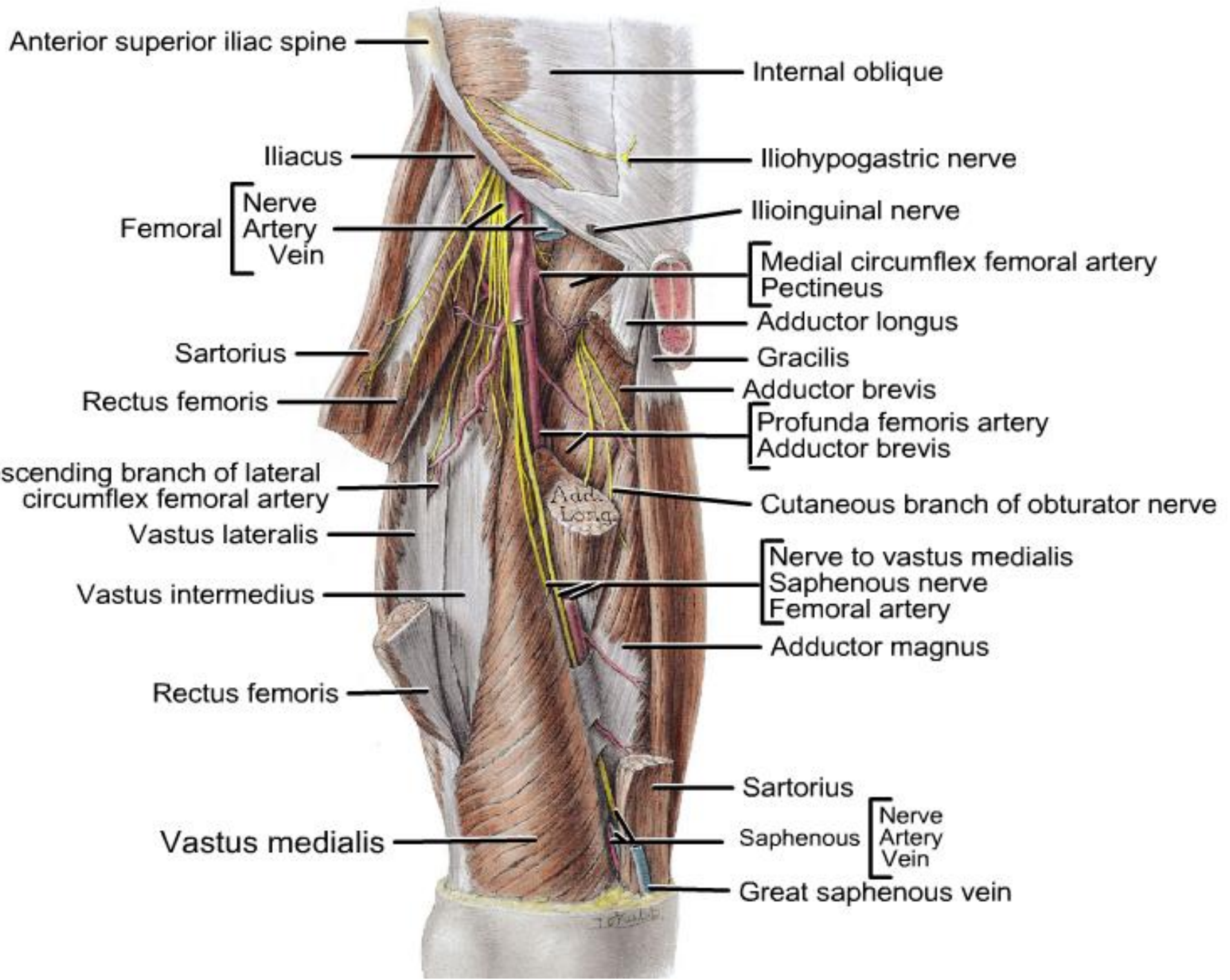
B. Section through the femoral sheath











Femoral sheath

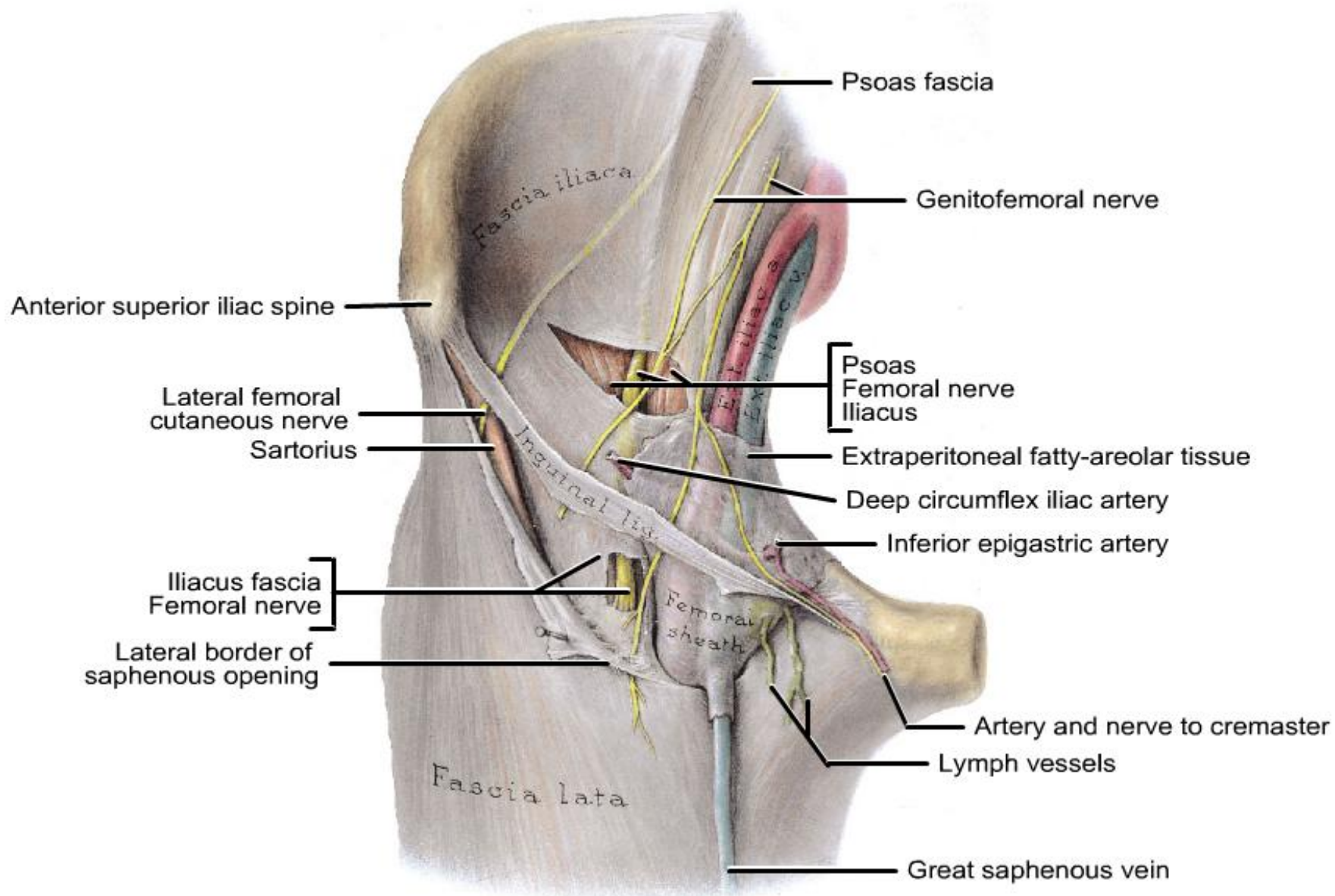
Prolongation of extraperitoneal areolar tissue

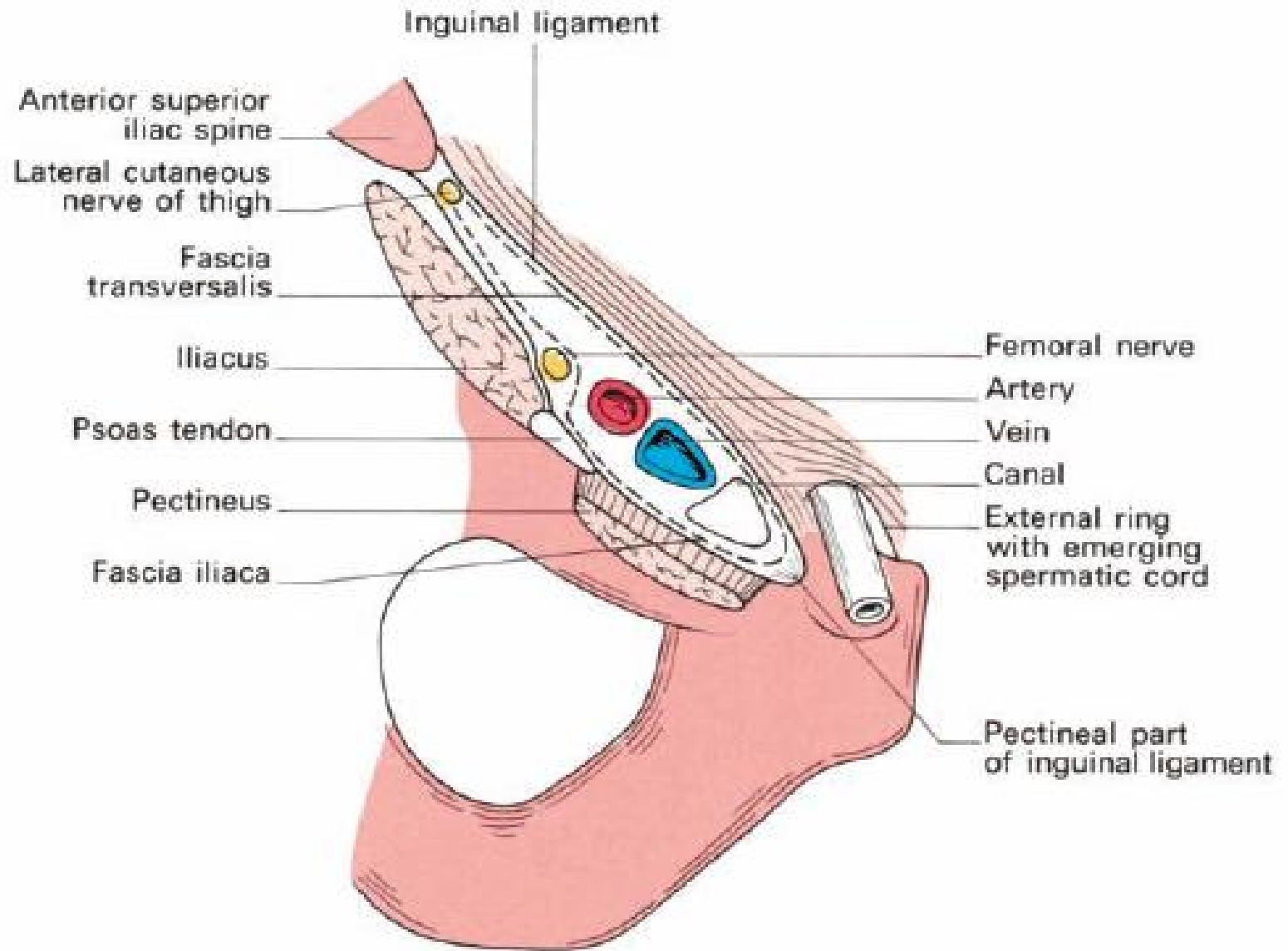
Derived from the extraperitoneal intra-abdominal fascia

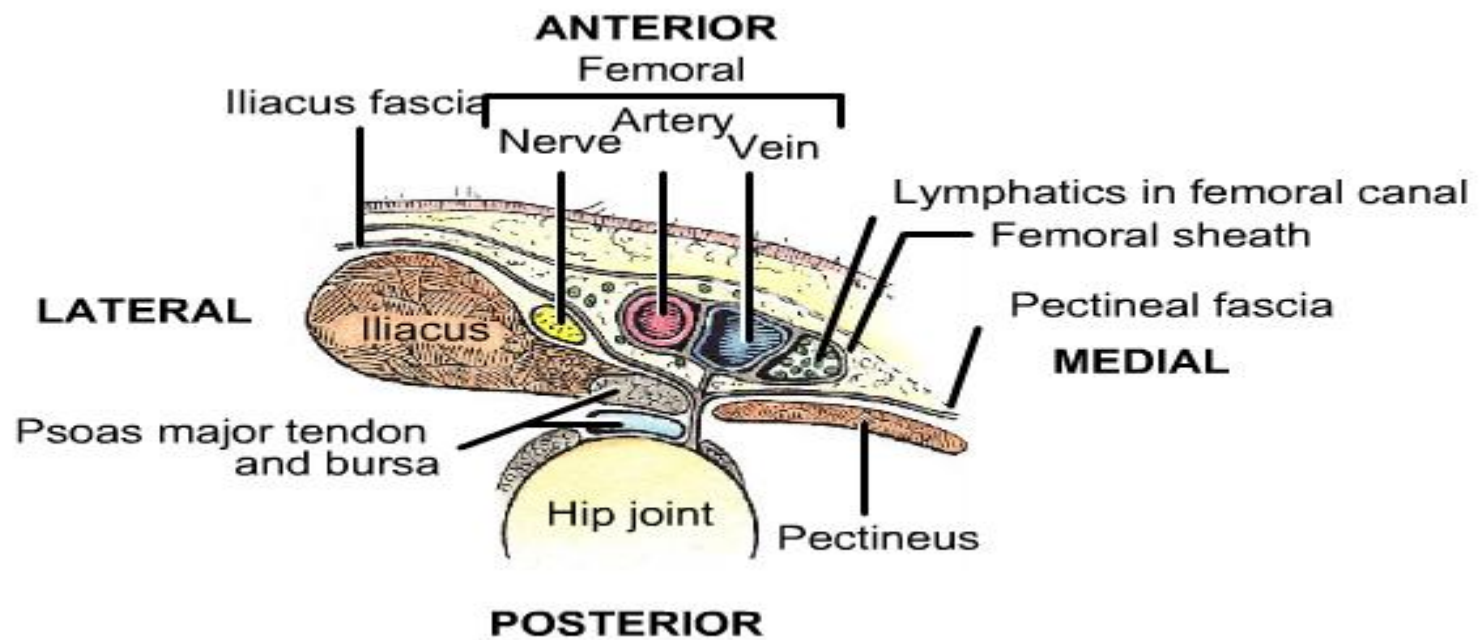
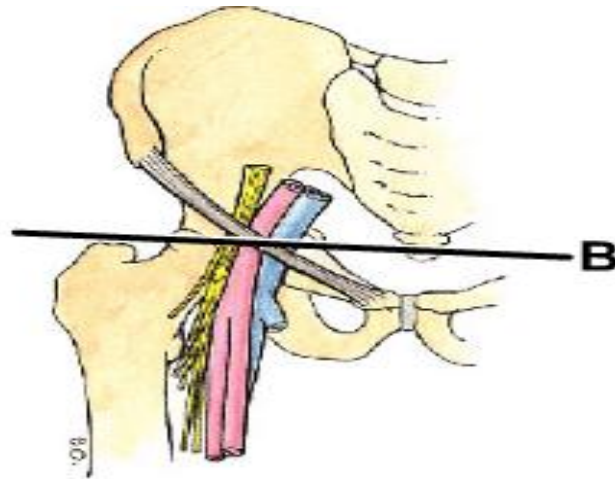
- its anterior wall arising from the transversalis fascia
- its posterior wall from the fascia covering the Iliacus

Has 3 compartments and two septa

- lateral for the femoral artery and genital branch of genitofemoral nerve
- middle for the femoral vein
- medial for the femoral canal
 - contain lymph vessels and deep lymph nodes
 - the site of femoral hernia





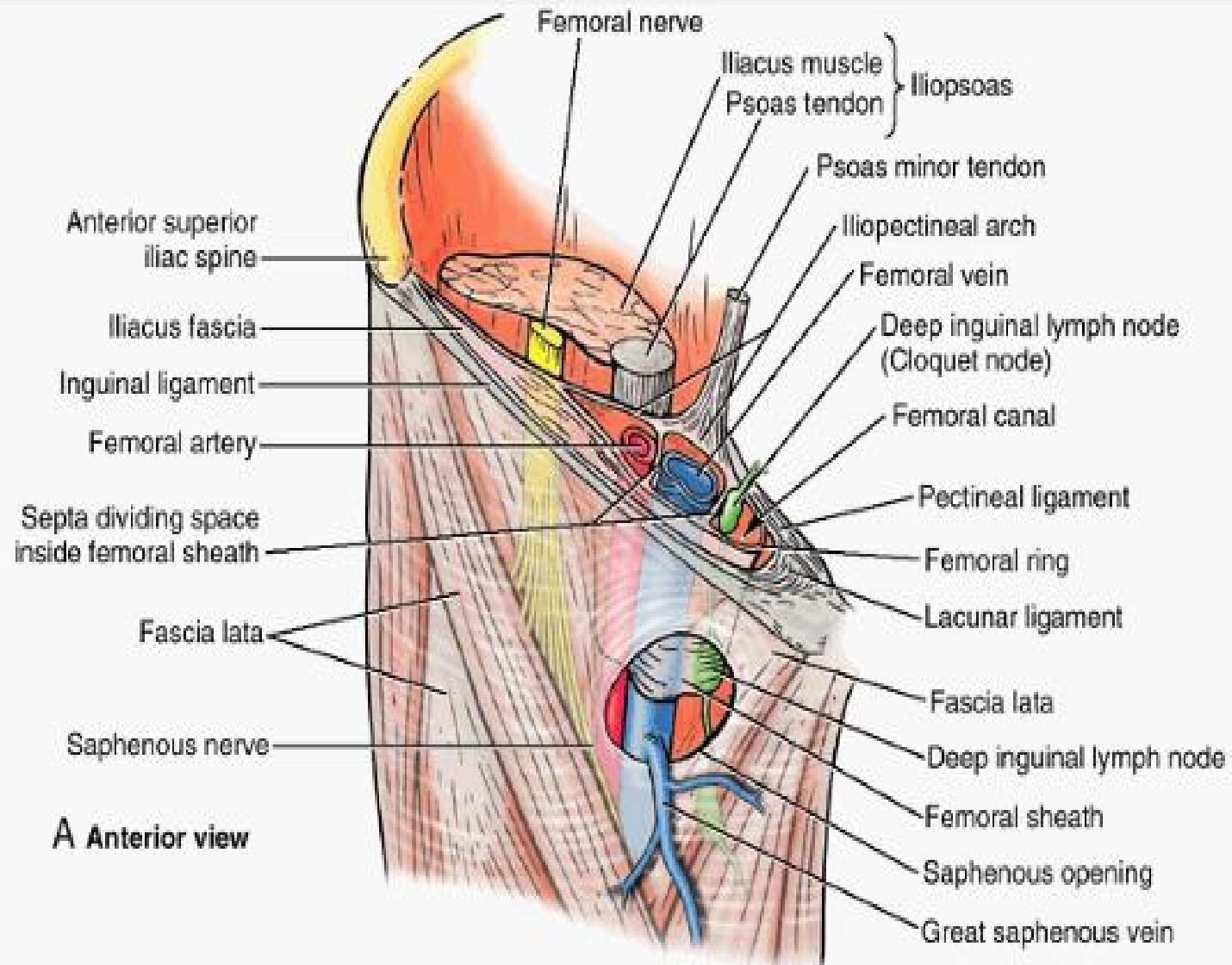


Femoral canal

- ✓ medial part of the femoral sheath
- ✓ small, vertically placed gap
- ✓ about 0.5 inch in length, admits the tip of little finger
- ✓ larger in female due to the greater width of the female pelvis
 - ✓ femoral hernias are more common in female

Boundaries

- Anteriorly - inguinal ligament
- Medially - the sharp free edge of the pectineal part of the inguinal ligament, termed the lacunar ligament
- Laterally - femoral vein
- Posteriorly - pectineal ligament
 - the thickened periosteum along the pectineal border of the superior pubic ramus; which continues medially with the pectineal part of the inguinal ligament



Contents

- ✓ fat
- ✓ lymph nodes
- ✓ lymph vessels

Functions

- dead space for expansion of the distended femoral vein
- lymphatic pathway from the lower limb to the external iliac nodes

Femoral hernia

- The femoral ring is a weak area in the lower anterior abdominal wall
- It is a site of femoral hernia (protrusion of abdominal viscera) through the femoral ring to femoral canal. More common in women than in men.
- Initially, the hernia is relatively small because it is contained within the femoral canal, but it can enlarge by passing through the saphenous opening into the subcutaneous tissue of the thigh

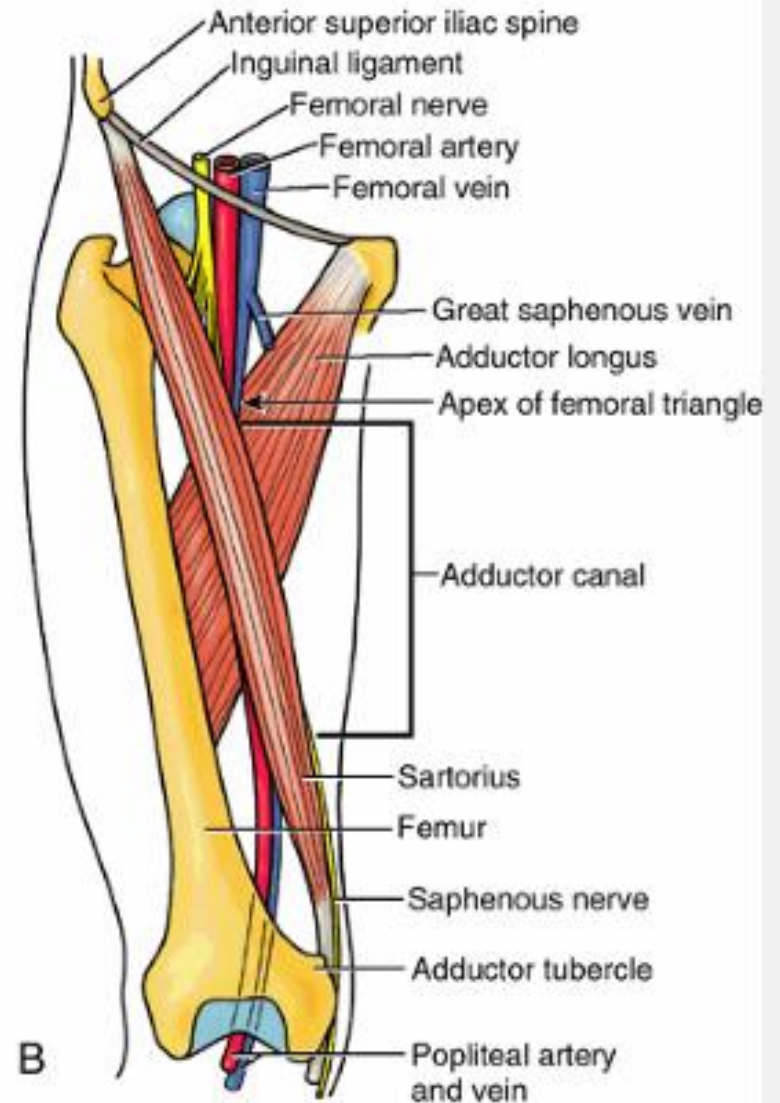
- As the hernia sac enlarges, it emerges through the saphenous opening then turns upwards along the pathway presented by the superficial epigastric and superficial circumflex iliac vessels so that it may come to project above the inguinal ligament
- The neck of the femoral canal is narrow and bears a particular sharp medial border; for this reason, strangulation occur more commonly at this site.
- Strangulation of a femoral hernia interfere with the blood supply to the herniated intestine, and vascular impairment may result in death of the tissues.

Adductor canal (of Hunter) or sub-sartorial canal

- Leads on from the apex of the femoral triangle
- Ends 10cm above the adductor tubercle
- Triangular
- Its walls (boundaries) are:
 - Posteriorly - adductor longus and magnus
 - Anterolaterally - vastus medialis
 - Anteromedially - sartorius, which lies on a fascial sheet forming the roof of the canal

Contents of the adductor canal

- ✓ femoral artery
- ✓ femoral vein (which lies behind the artery)
- ✓ saphenous nerve
- ✓ nerve to vastus medialis from the femoral nerve



Blood vessels

Femoral artery

Origin

- Main continuation of external iliac distal to inguinal ligament

Course

- It traverses the femoral triangle and the adductor canal
- Terminates a hand's breadth above the adductor tubercle by passing through the hiatus in adductor magnus to become the popliteal artery
- Throughout its course, it is accompanied by its vein, which lies first on the medial side of the artery and then passes posteriorly to it at the apex of the femoral triangle

Distribution: anterior and anteromedial thigh

Branches

- In the groin, the femoral artery gives off: Superficial branches
 - ✓ Superficial external pudendal artery
 - crosses anterior to the spermatic cord to supply scrotum and labium major
 - ✓ Superficial epigastric artery
 - passes towards the umbilicus
 - ✓ Superficial circumflex iliac artery
 - passes laterally just inferior to the inguinal ligament
- Their corresponding veins drain into the great saphenous vein

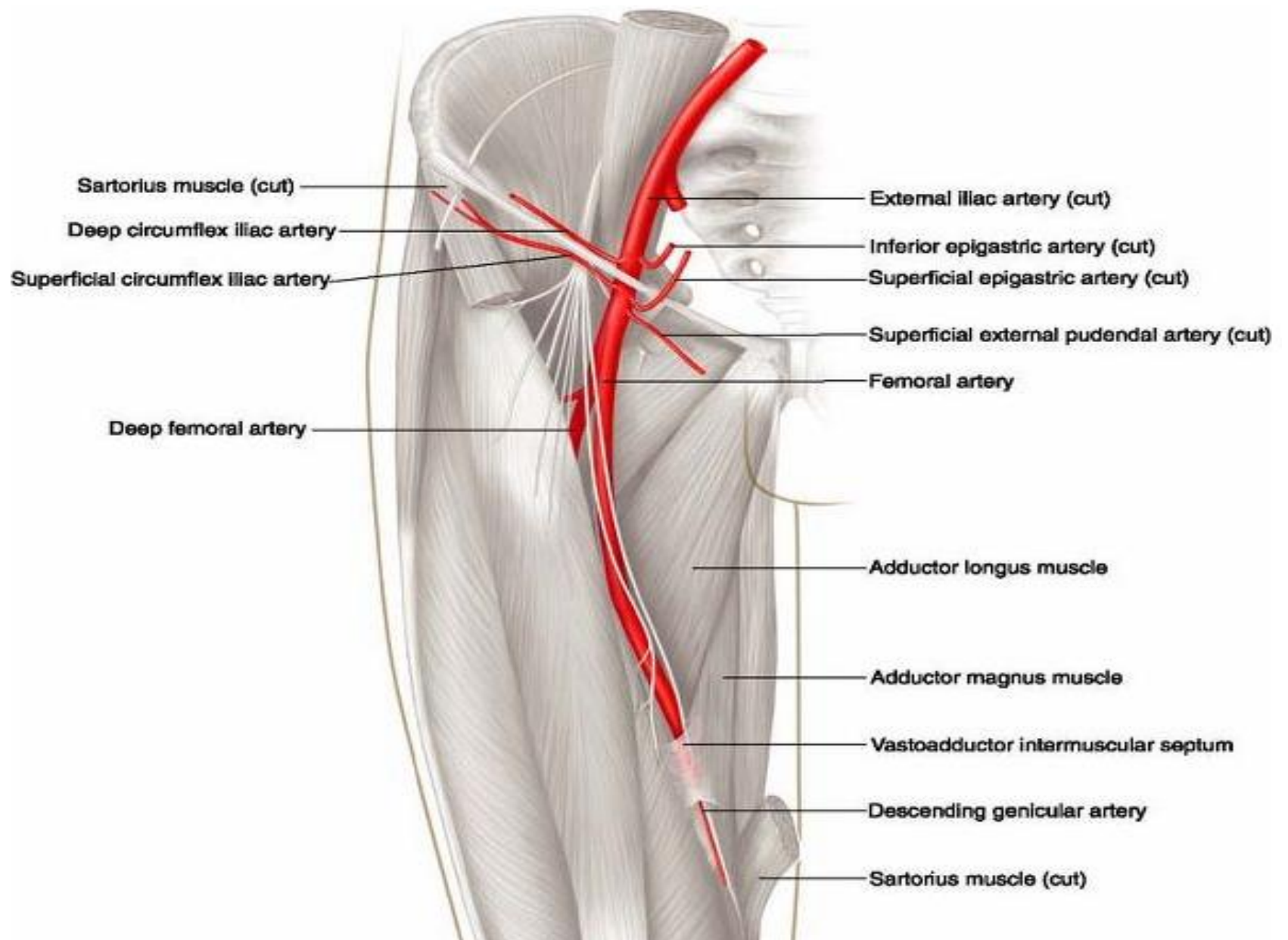
Deep femoral (profunda femoris) artery

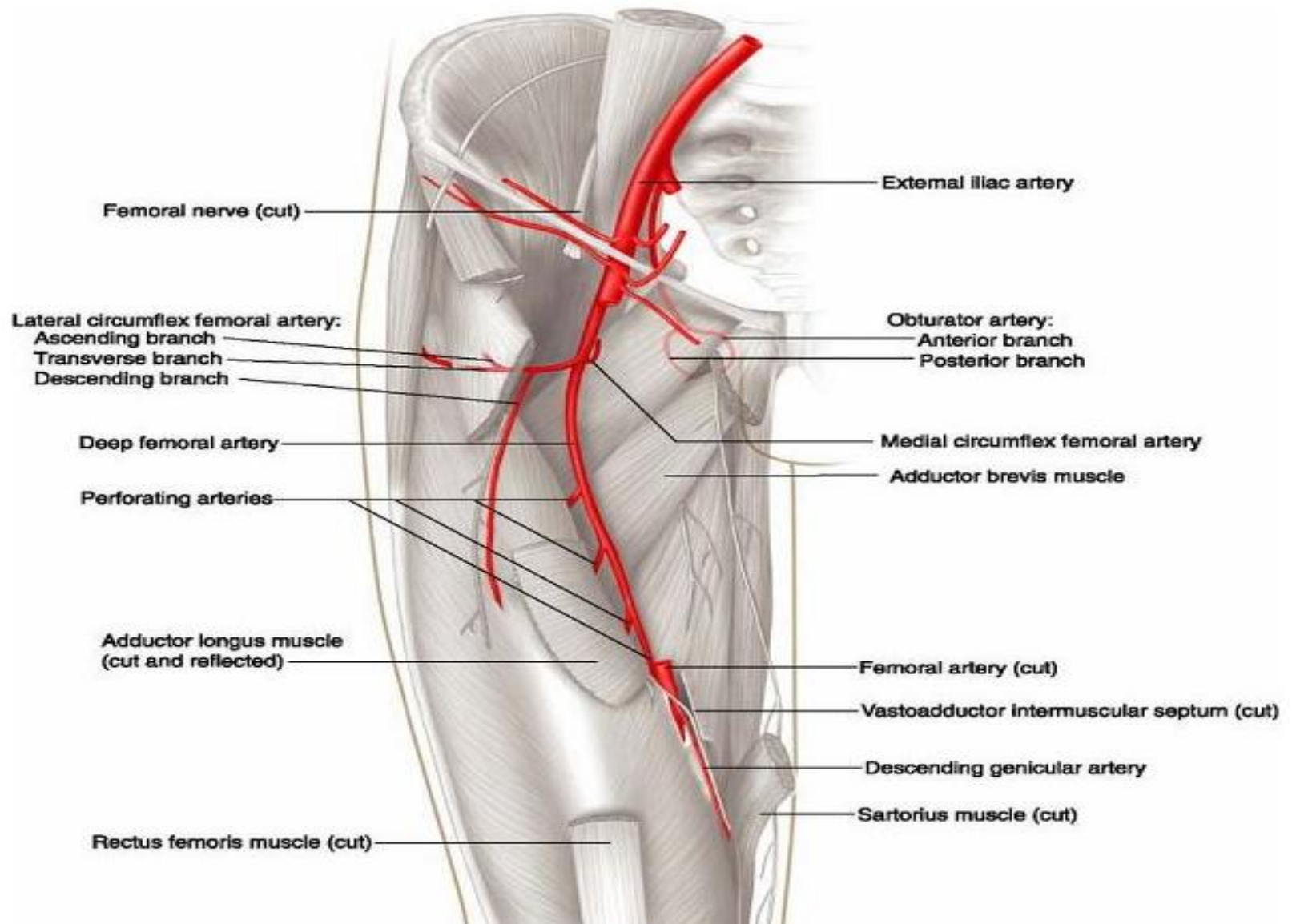
- ✓ Origin: arises posterolaterally from the femoral artery 2 inch distal to the inguinal ligament
- ✓ Course: passes deep to adductor longus
- ✓ Distribution: perforating branches supply muscles in medial, posterior and lateral part of anterior compartment.

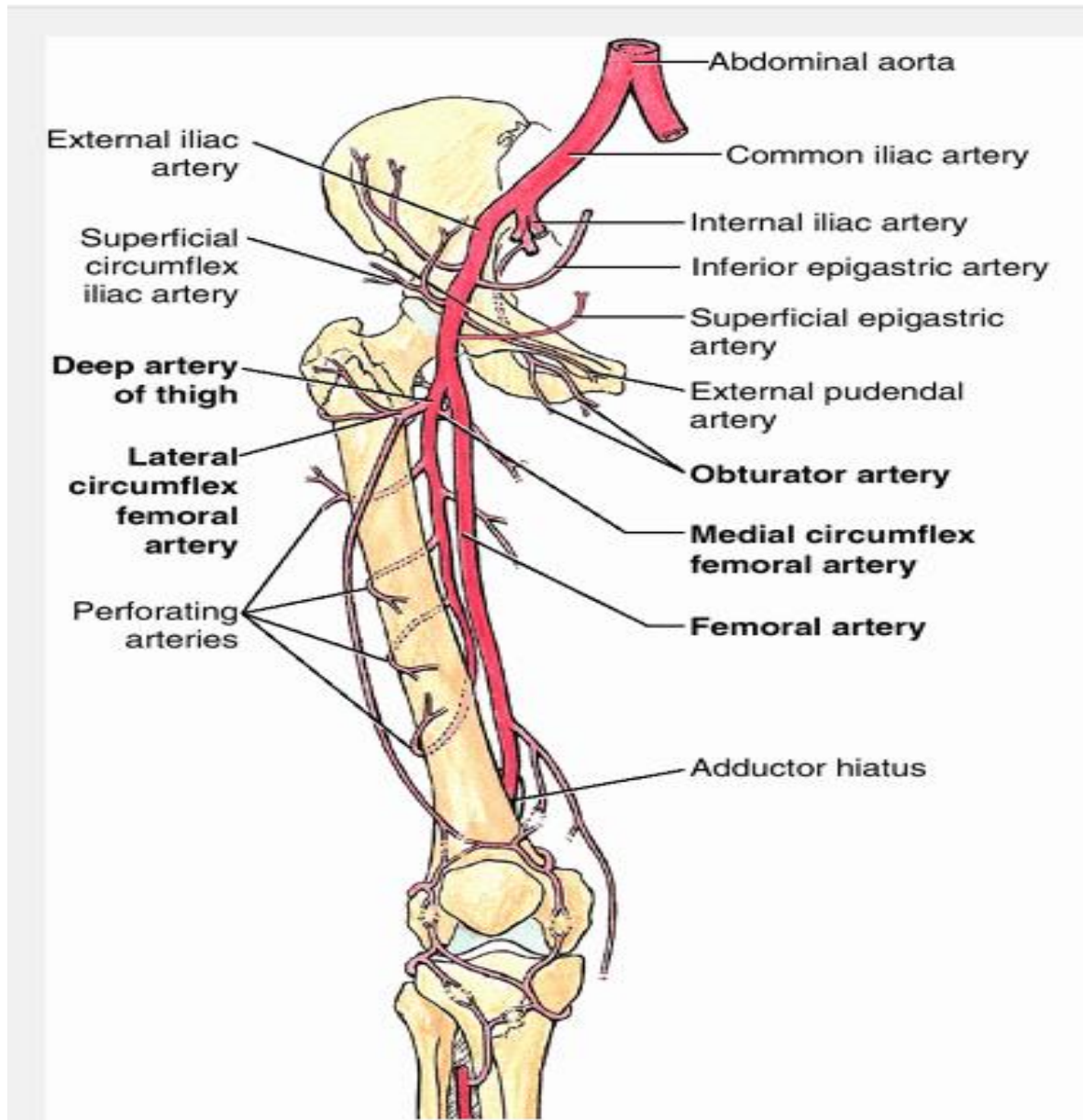
Branches

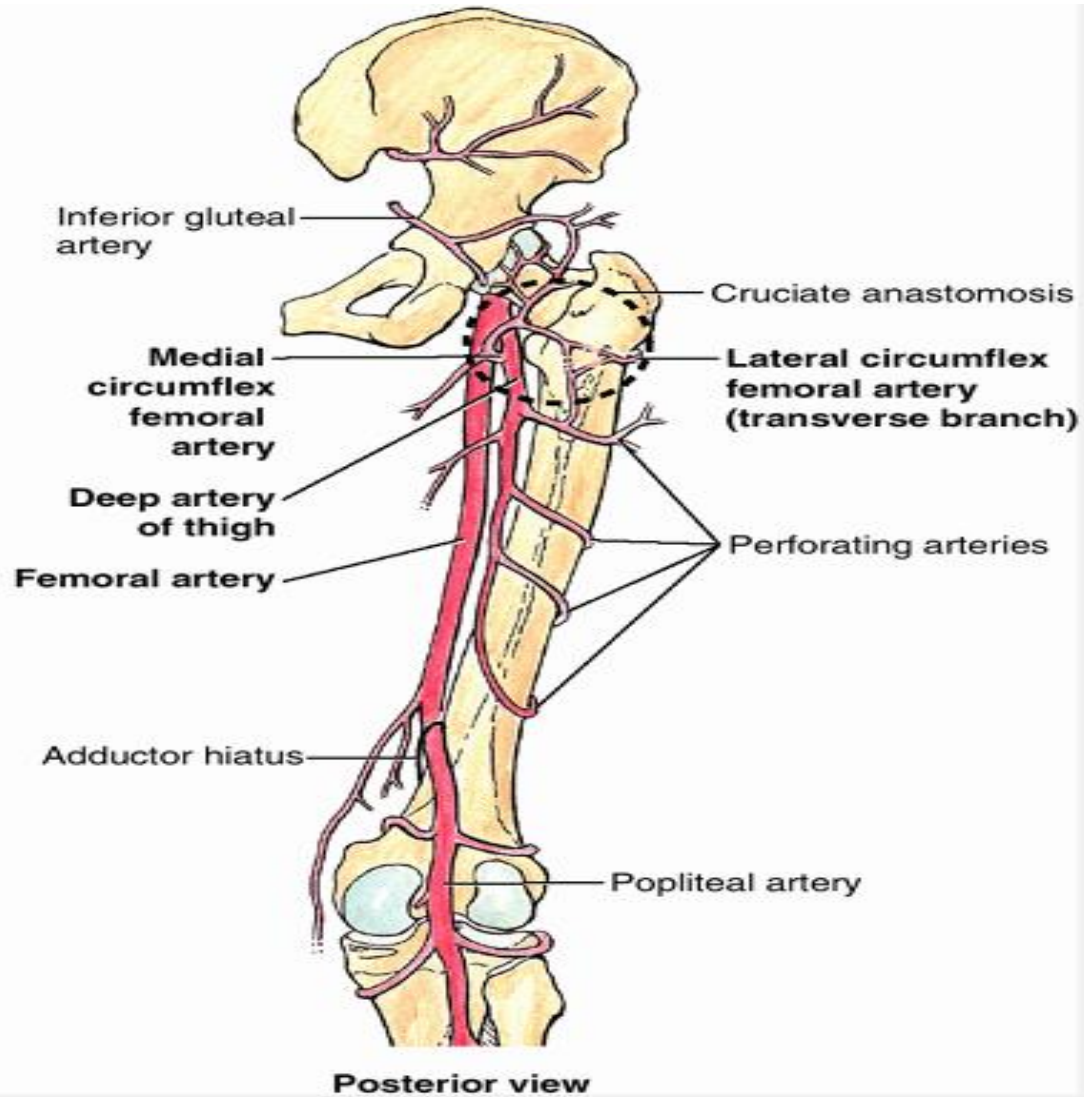
- Medial circumflex femoral
 - Passes medially and posteriorly between the pectineus and iliopsoas
 - Enters gluteal region and divides into 2 branches; transverse and ascending branches
- Lateral circumflex femoral
 - Passes laterally deep to sartorius and rectus femoris muscle
 - Divide into ascending, transverse and descending branches

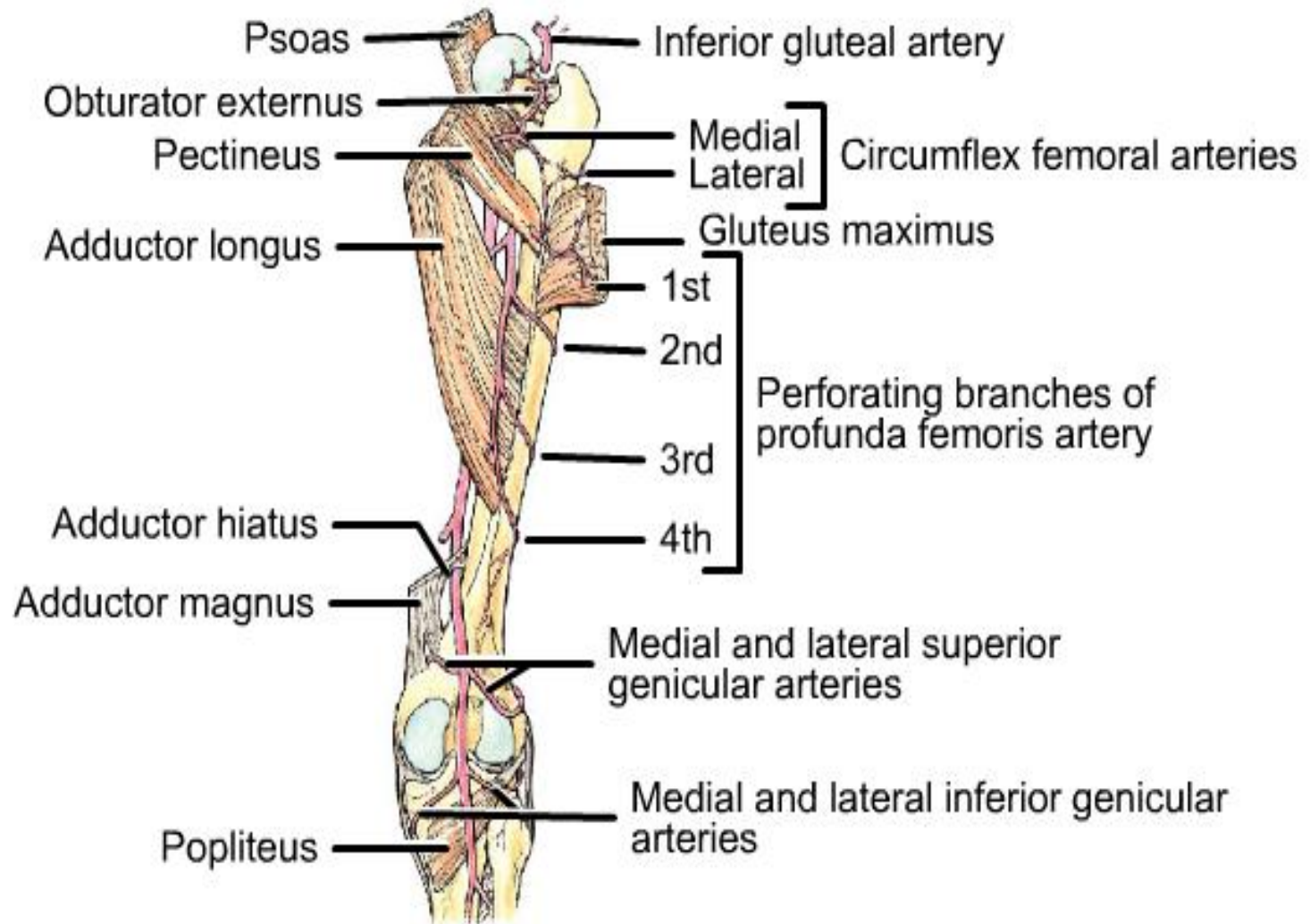
- Three perforating branches and terminates at the fourth perforating
 - Perforating branches hug the shaft of the femur
 - They supply muscles and a nutrient artery to the femur
- The anastomosis at the hip joint is formed by the medial and lateral circumflex arteries with the inferior gluteal artery and the first perforating artery











Femoral pulse and cannulation of the femoral artery

- ✓ The pulse of the femoral artery is usually palpable just inferior to the midpoint of the inguinal ligament
- ✓ Normally, the pulse is strong; however, if the common or external iliac arteries are partially occluded, the pulse may be diminished
- ✓ The femoral artery may be compressed here to control arterial bleeding after lower limb trauma
- ✓ The femoral artery may be cannulated just inferior to the midpoint of the inguinal ligament (e.g., for cardioangiography, radiography of the heart and great vessels after the introduction of contrast material)
- ✓ For left cardiac angiography, a long slender catheter is inserted percutaneously into the femoral artery and passed superiorly in the aorta to the openings of the coronary arteries

Obturator artery

- ✓ Arises from internal iliac artery
- ✓ Passes through obturator foramen
- ✓ Enters medial compartment of the thigh
- ✓ Divides into anterior and posterior branches
- ✓ Supplies obturator externus and adductors of thigh
- ✓ Posterior branch gives off acetabular branch to head of femur

Deep veins

Femoral vein

A continuation of popliteal vein at adductor hiatus

Course:

- ✓ ascends through adductor canal
- ✓ lies medial and then posterior to femoral artery
- ✓ enters femoral sheath lateral to femoral canal
- ✓ ends posterior to inguinal ligament, becomes external iliac vein

Tributaries

- ✓ Descending genicular
- ✓ Profunda femoris
- ✓ Lateral circumflex femoral
- ✓ Medial circumflex femoral
- ✓ Deep external pudendal
- ✓ Great saphenous

Cannulation of the femoral vein

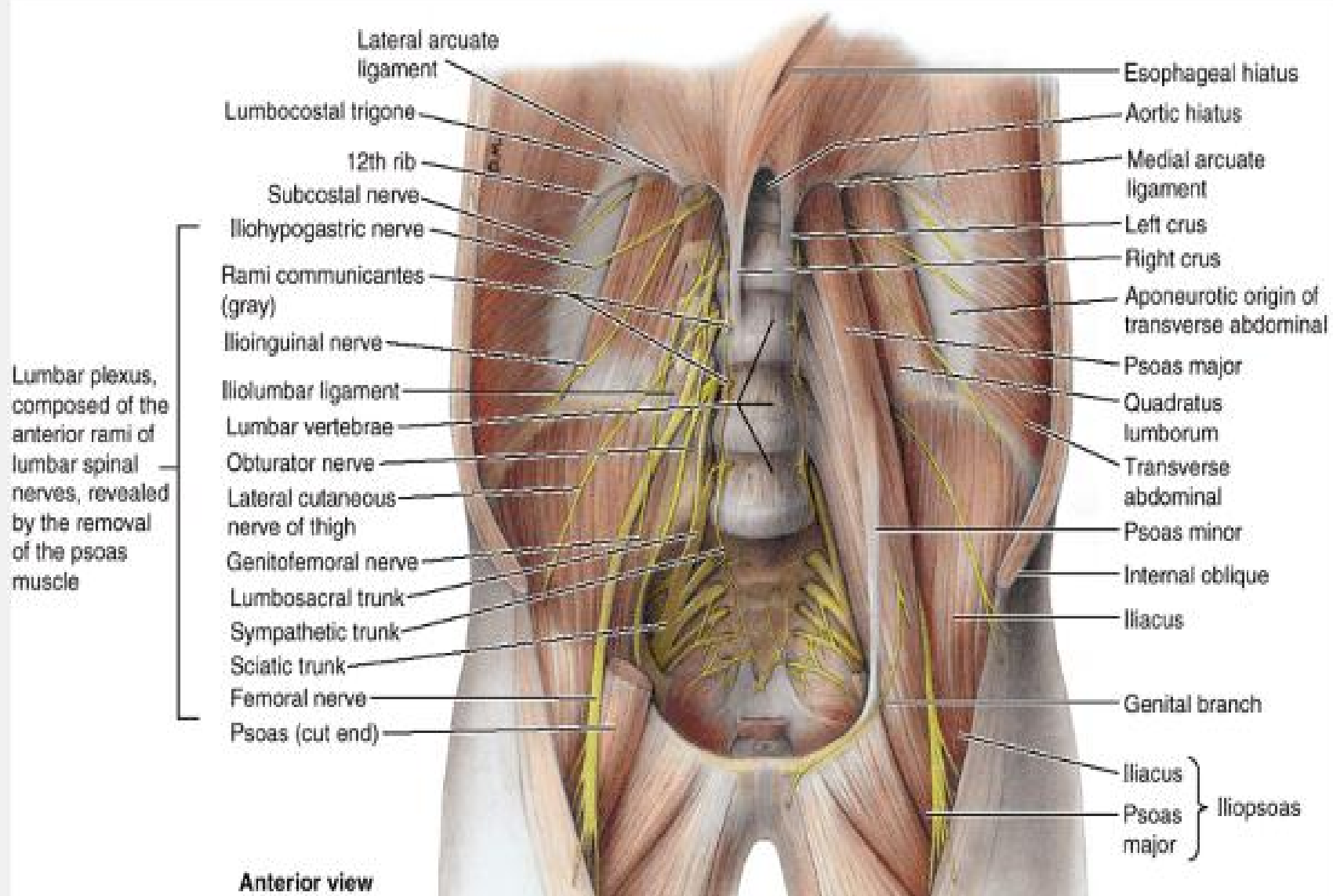
- ✓ The femoral vein usually is not palpable, but its position can be located by feeling the pulsations of the femoral artery, which lies just lateral to it
- ✓ In thin people, the femoral vein may be close to the surface and may be mistaken for the great saphenous vein
- ✓ It is thus important to know that the femoral vein has no tributaries at this level, except for the great saphenous vein that joins it approximately 3 cm inferior to the inguinal ligament
- ✓ To secure blood samples and take pressure recordings from the chambers of the right side of the heart and/or from the pulmonary artery and to perform right cardiac angiography, a long slender catheter is inserted into the femoral vein as it passes through the femoral triangle

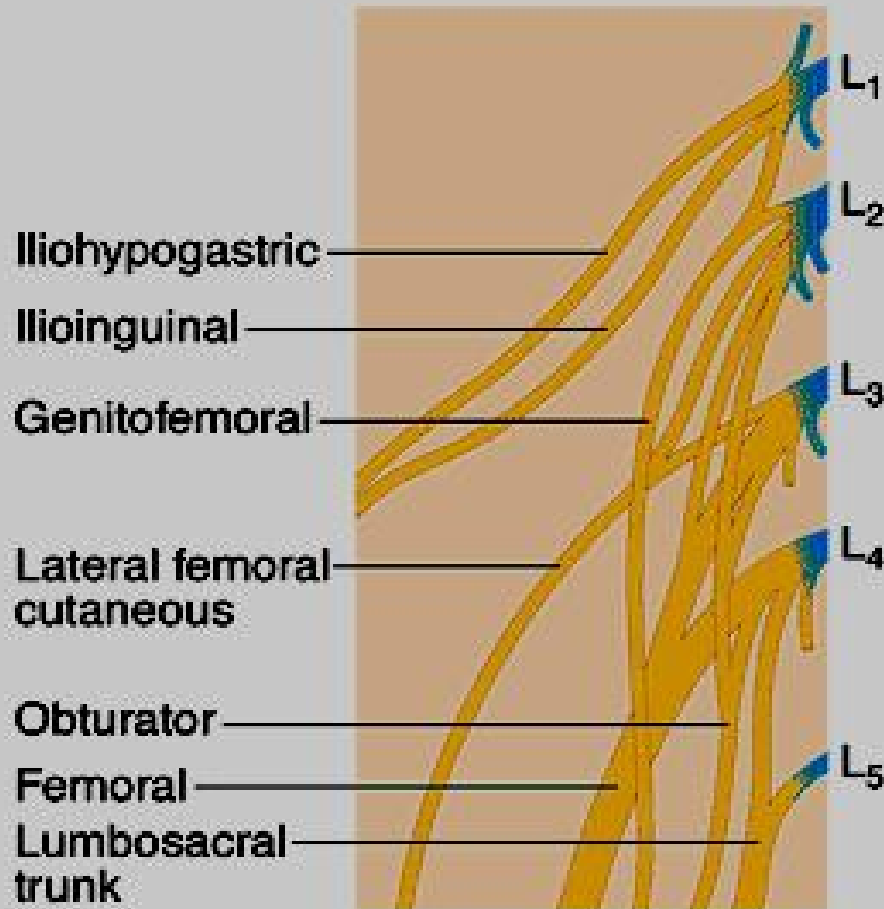
Lumbar plexus

- ✓ Formed anterior to the lumbar transverse processes, within the proximal attachment of the psoas major
- ✓ The plexus originates from the anterior rami of L1 - L4
- ✓ The trunks of the plexus traverse psoas major and emerge from its lateral border
- ✓ There are two exceptions:
 - obturator nerve appears at the medial border of psoas tendon
 - genitofemoral nerve emerges on the anterior aspect of the muscle

Branches of the lumbar plexus

- Obturator nerve
- Femoral nerve
- Lumbosacral trunk
- Ilioinguinal nerve
- Iliohypogastric nerve
- Genitofemoral nerve
- Lateral cutaneous nerve of the thigh



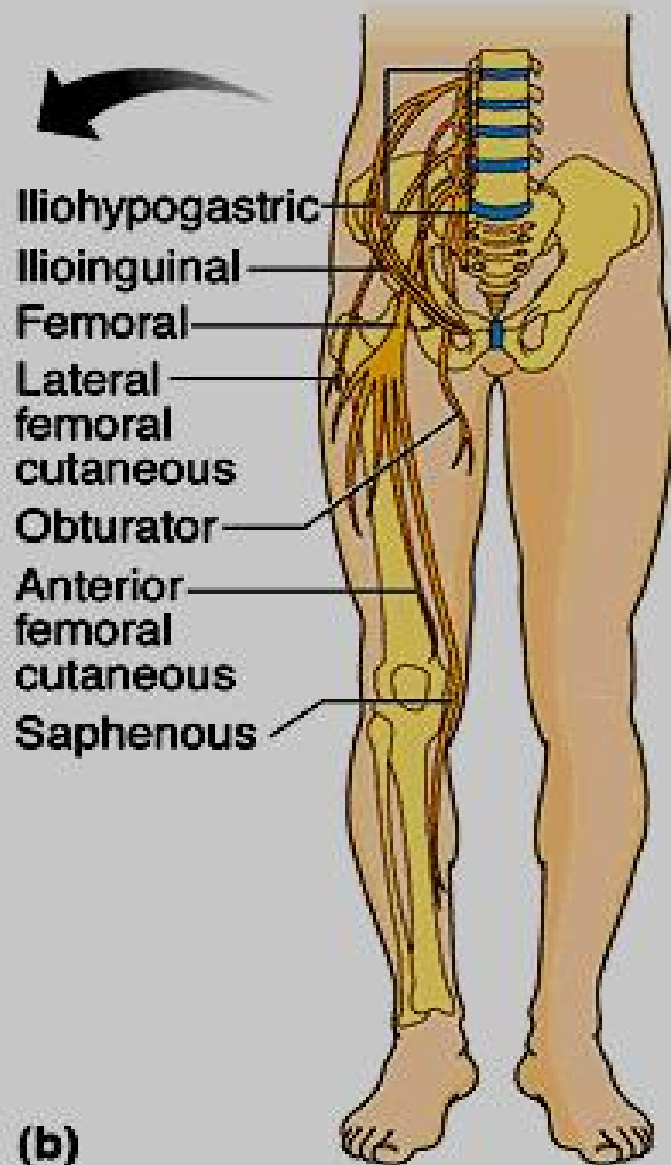


(a)

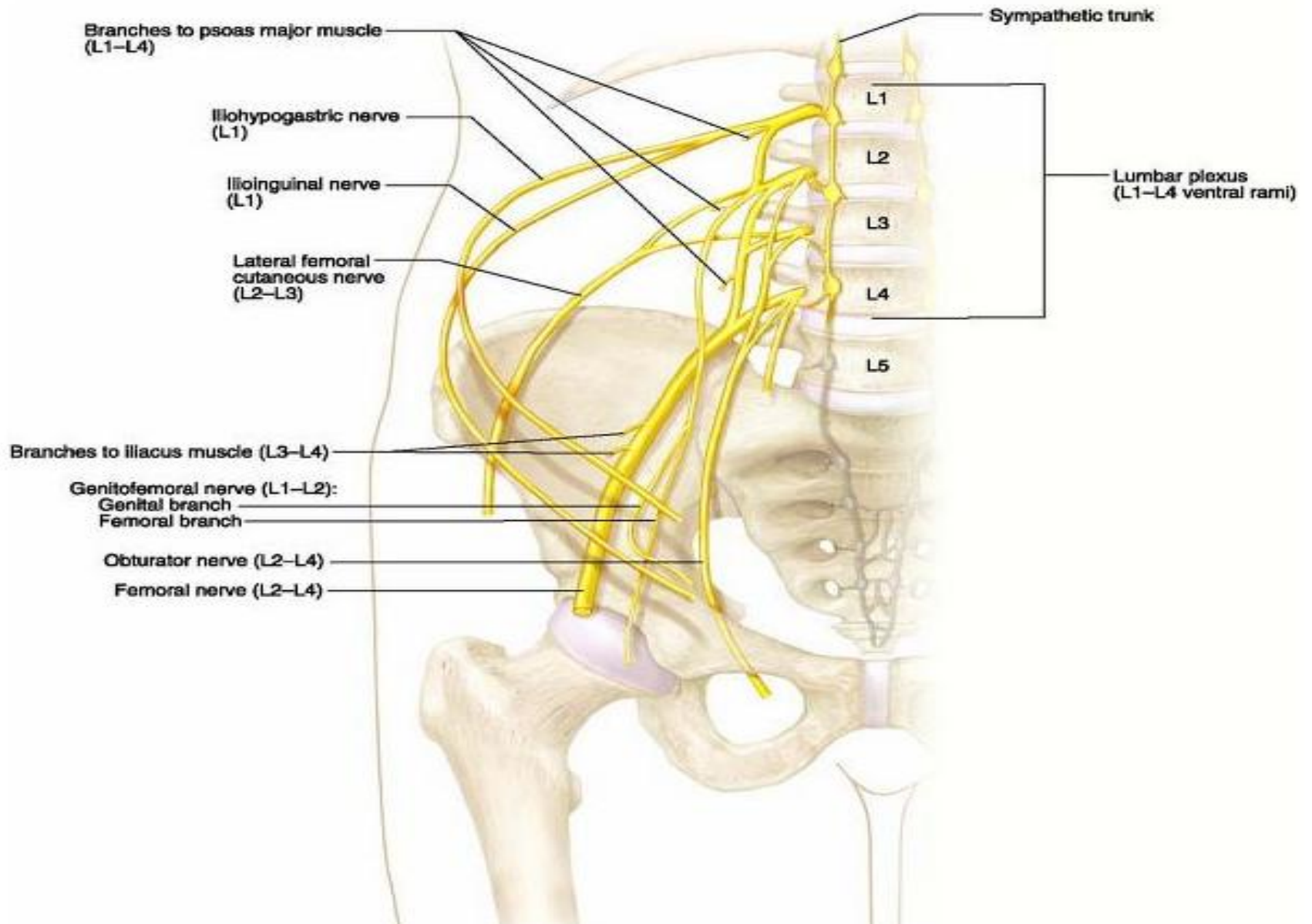
Key:

■ = Roots

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(b)



Obturator nerve (L2- L4)

- ✓ Emerges from the medial border of the psoas major
- ✓ Passes into the lesser pelvis
- ✓ Passing inferior to the superior pubic ramus (through the obturator foramen) to the medial thigh
- ✓ Divides into anterior and posterior divisions past the obturator foramen and supplies adductor muscles
- ✓ The nerve to gracilis becomes the only cutaneous branch
- ✓ Has articular branches for both the hip and knee joint

Two divisions – anterior and posterior

Anterior division

- Adductor longus
- Adductor brevis
- Gracilis
- Articular branch to hip joint
- Cutaneous branch to skin overlying adductor canal & knee joint

Posterior division

- Obturator externus
- Adductor magnus (part)

Femoral nerve (L2 - L4)

- Emerges from the lateral border of the psoas major
- Lies in front of the iliacus; innervates it
- Passes deep to the inguinal ligament/iliopubic tract to the anterior thigh, lies lateral to the artery, deep to the fascia lata
- Breaks up into motor and sensory branches
- Supply the flexors of the hip and extensors of the knee
- 2 of the branches follow the artery on its lateral side into the adductor canal:
 - nerve to vastus medialis (motor)
 - saphenous nerve (sensory)

Branches

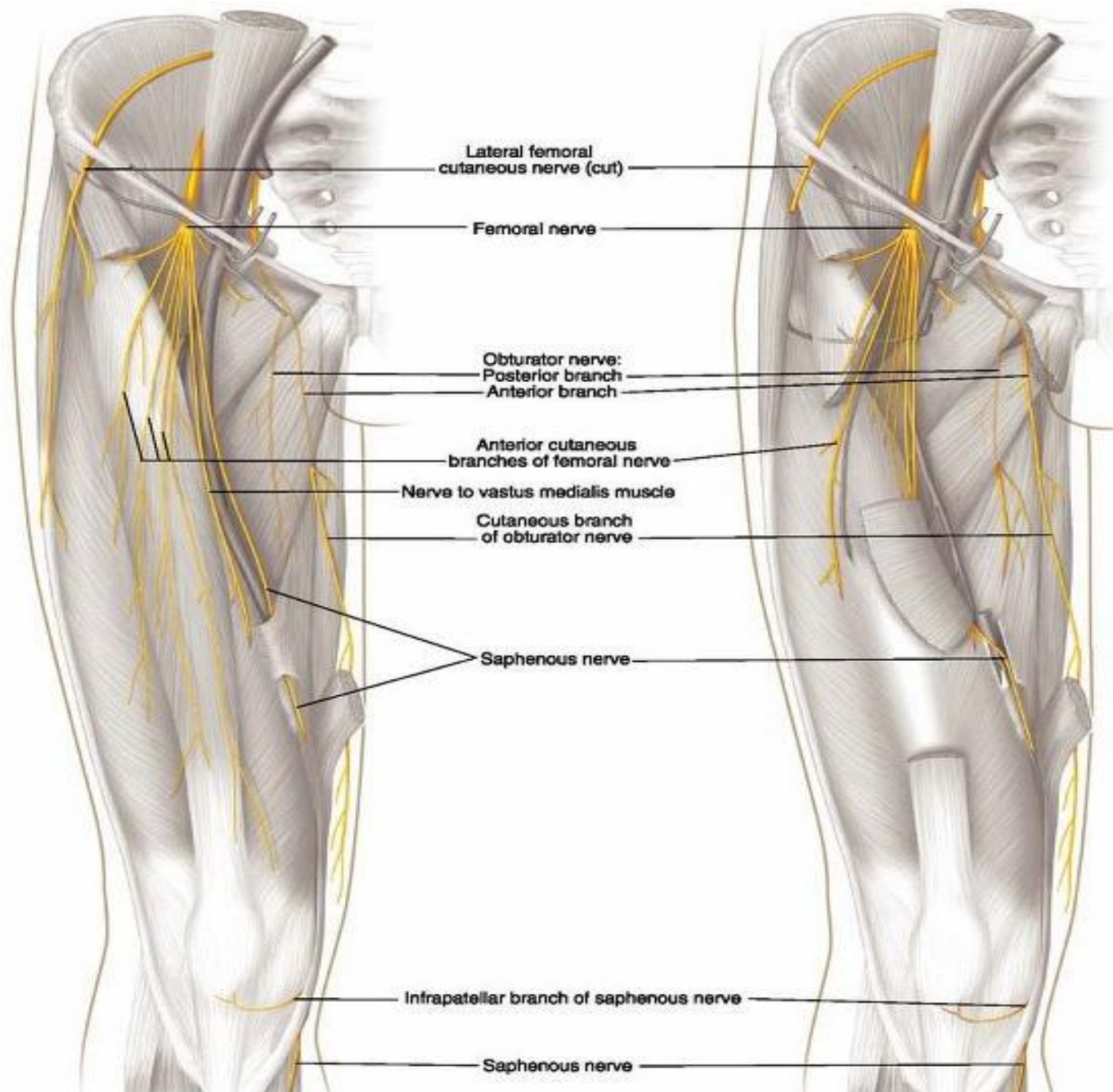
Muscular

- to the anterior compartment of the thigh (quadriceps, sartorius and pectineus)

Cutaneous

- medial and intermediate cutaneous nerves of the thigh
 - the lateral aspect is supplied by the lateral cutaneous nerve of the thigh (L2–3)
- saphenous nerve
 - traverses the adductor canal
 - supply the skin of the medial side of the leg, ankle and foot to the great toe

Articular - to the hip and knee joints



Lumbosacral trunk (L4 & L5)

- Passes over the ala (wing) of the sacrum
- Descends into the pelvis
- Participate in the formation of the sacral plexus with the anterior rami of S1 - S4 nerves

Ilioinguinal and iliohypogastric nerves (L1)

- Arise from the anterior ramus of L1
- Entering the abdomen posterior to the medial arcuate ligament and passing inferolaterally, anterior to the quadratus lumborum
- They run superior and parallel to the iliac crest, piercing the transverse abdominal near the ASIS
- They then pass through the internal and external obliques to supply the abdominal muscles and skin of the inguinal and pubic regions
- The division of the L1 anterior ramus may occur as far distally as the ASIS, so that often only one nerve (L1) crosses the posterior abdominal wall instead of two

Genitofemoral nerve (L1 & L2)

- Pierces the psoas major and runs inferiorly on its anterior surface, deep to the psoas fascia
- Divides lateral to the common and external iliac arteries into femoral and genital branches

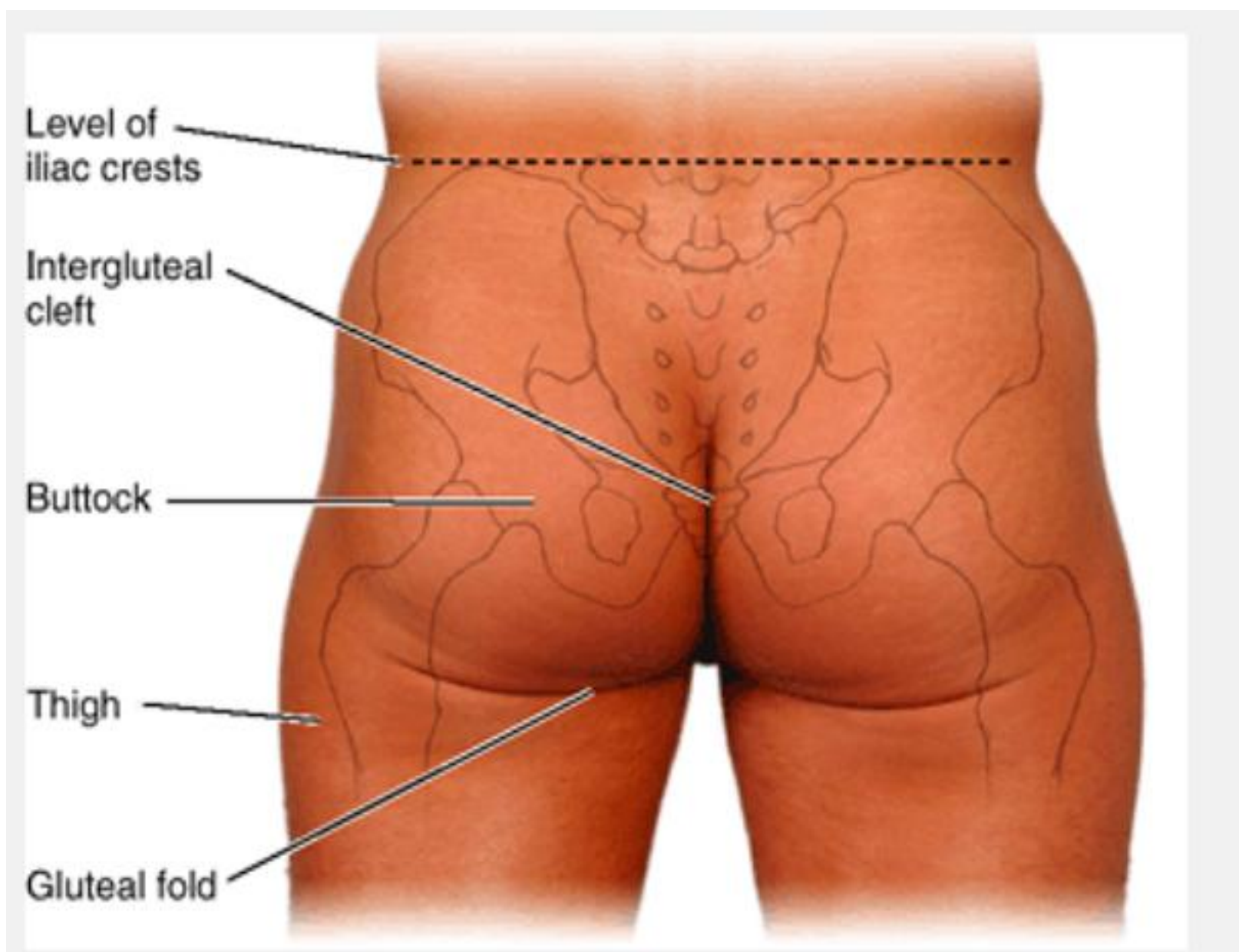
The lateral cutaneous nerve of the thigh, or lateral femoral cutaneous nerve (L2, L3)

- Runs inferolaterally on the iliacus and enters the thigh deep to the inguinal ligament/iliopubic tract, just medial to the ASIS
- Supplies skin on the anterolateral surface of the thigh

Hip and Buttocks: The Gluteal Region

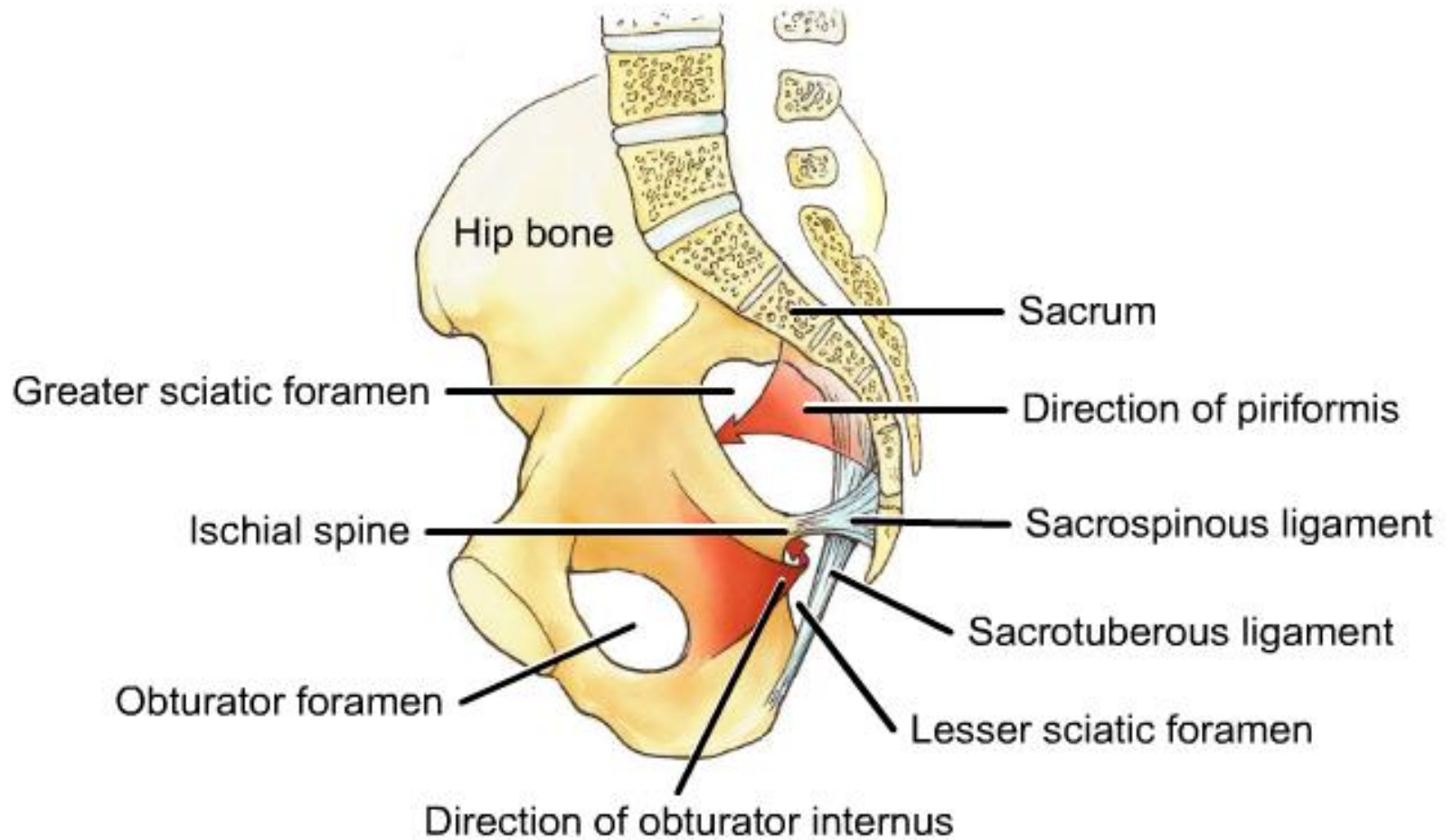
- Although the demarcation of trunk and limb is abrupt anteriorly at the inguinal ligament, posteriorly the gluteal region is a large transitional zone between trunk and limb
- Physically part of the trunk, functionally the gluteal region is definitely part of the limb
- The gluteal region is the prominent area posterior to the pelvis and inferior to the level of the iliac crests (the buttocks) and extending laterally and anteriorly to the greater trochanter according to some definitions and to the ASIS according to others

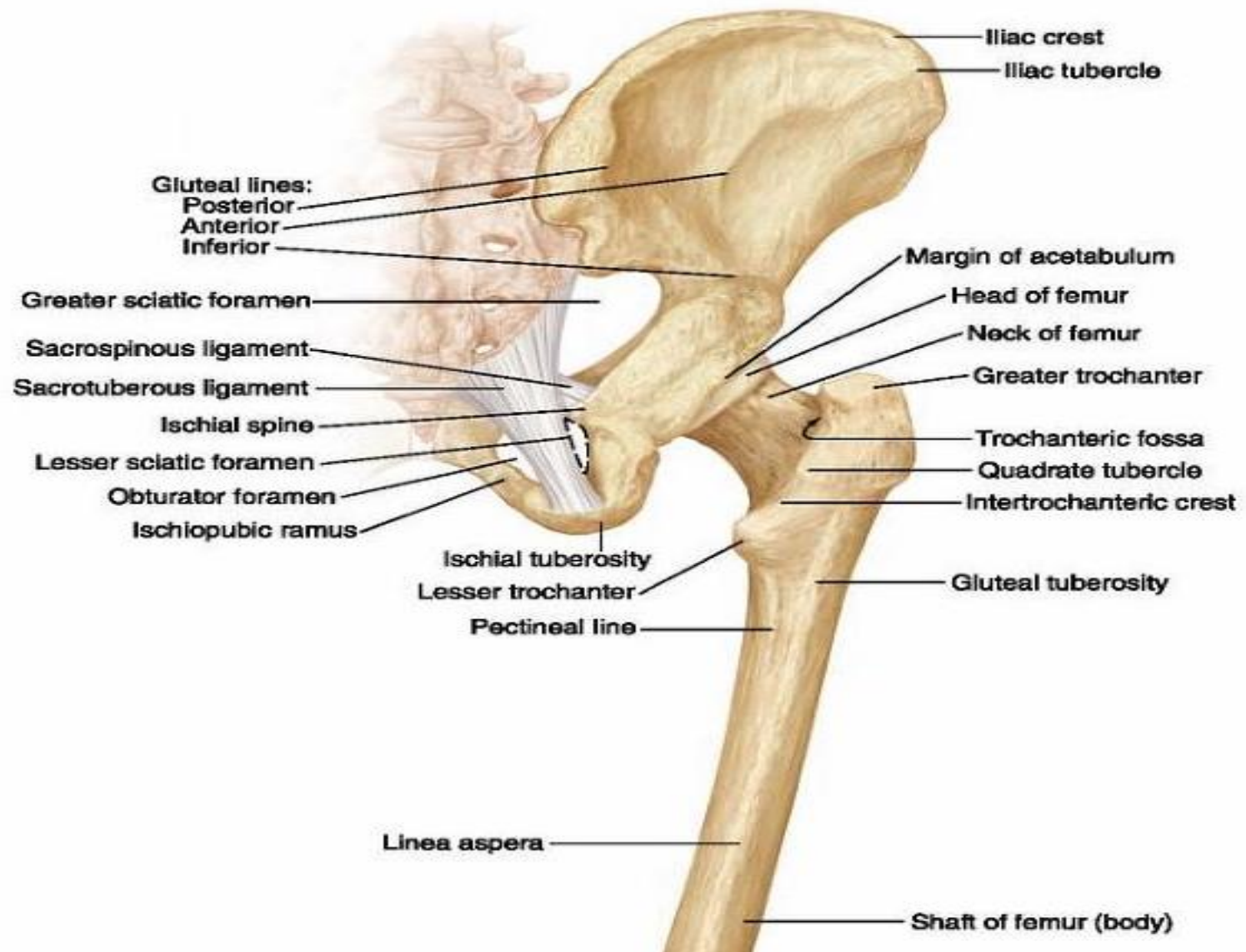
- The intergluteal cleft separates the buttocks from each other
- The gluteal muscles (gluteus maximus, medius, and minimus and tensor of the fascia lata) form the bulk of the region
- The gluteal fold demarcates the inferior boundary of the buttock and the superior boundary of the thigh

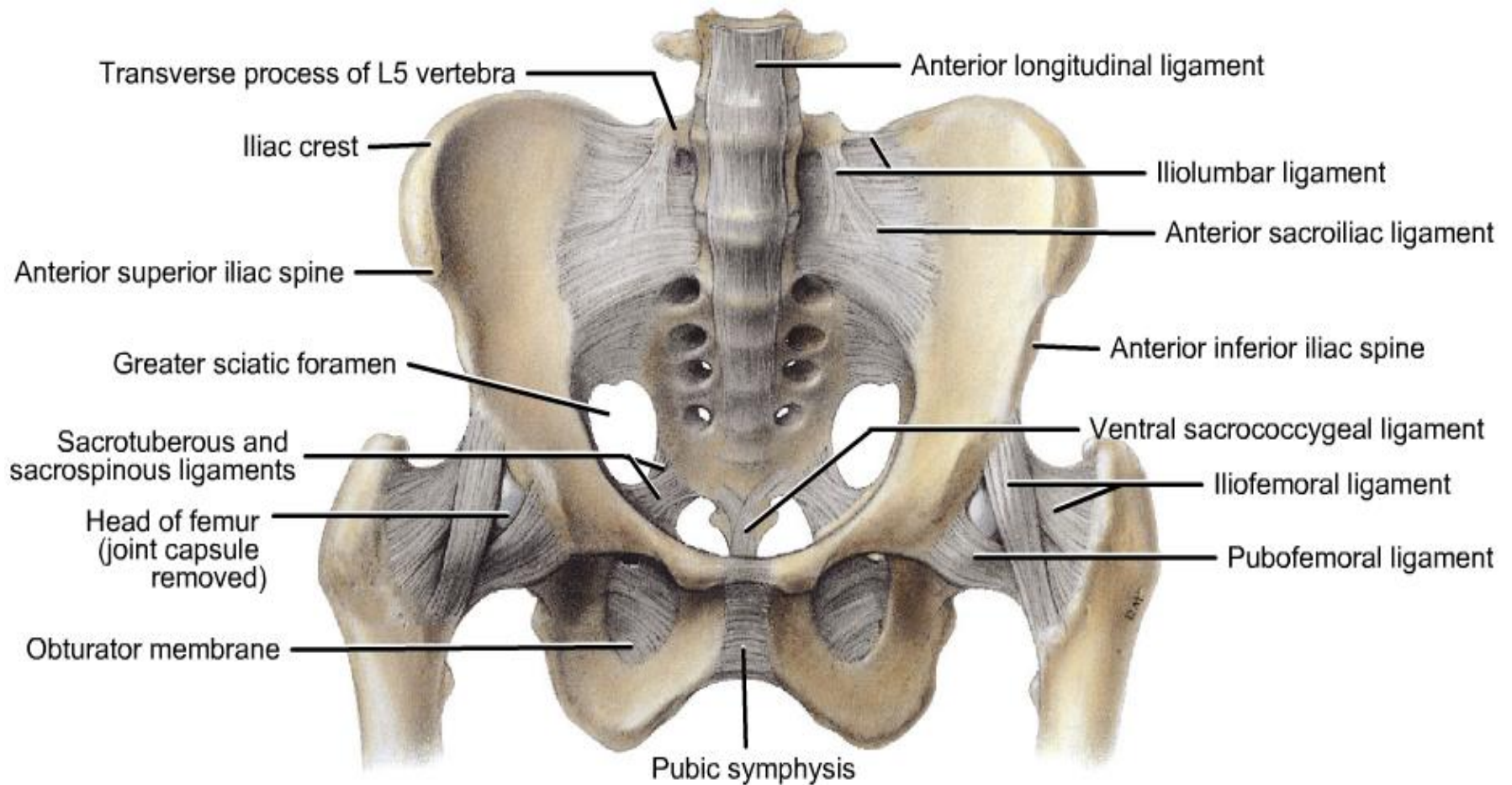


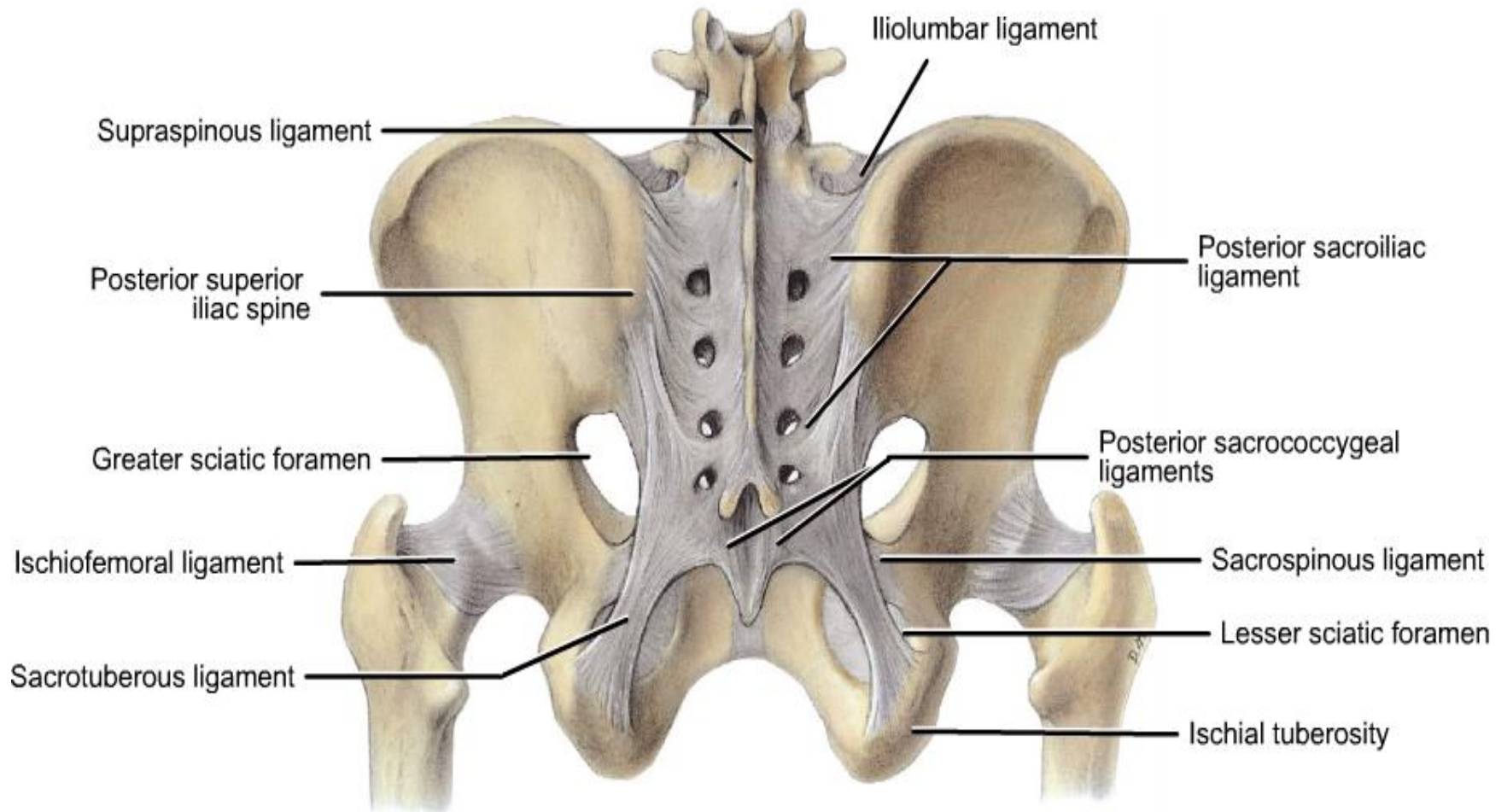
Gluteal Ligaments

- The parts of the bony pelvis hip bones, sacrum, and coccyx are bound together by dense ligaments
- The **sacrospinous** and **sacrotuberous** ligaments convert the sciatic notches in the hip bones into the greater and lesser sciatic foramina









Gluteal Muscles

Organized into 2 layers: superficial and deep

Superficial layer: 3 gluteal muscles and tensor of fascia lata

Deep layer: piriformis, obturator internus, superior and inferior gemilli, and quadratus femoris

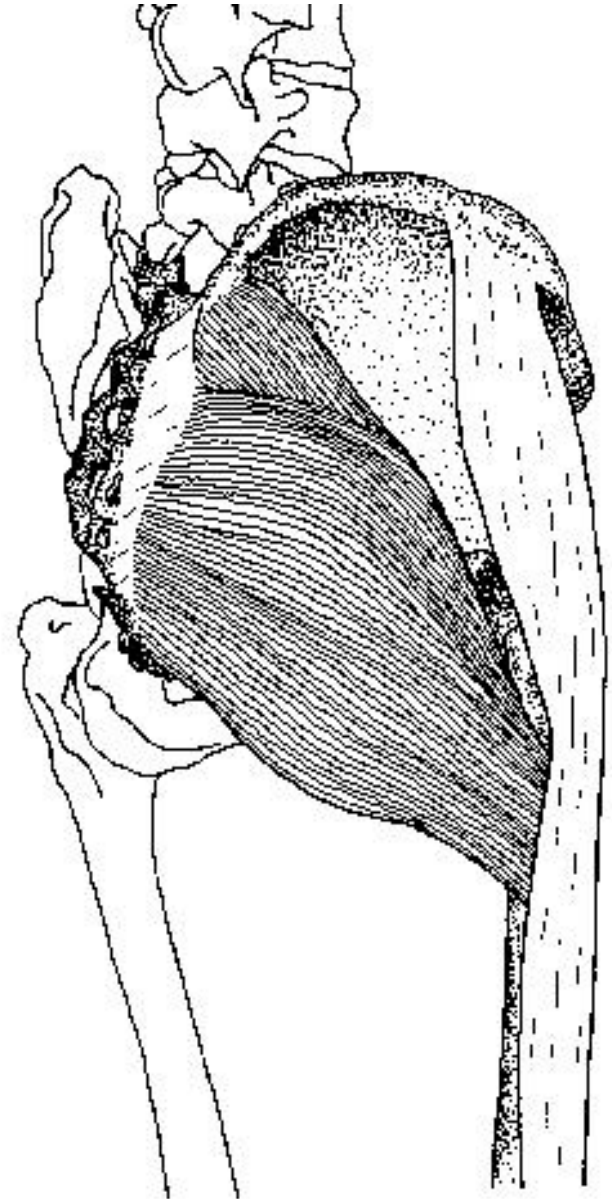
Gluteus maximus

Origin - dorsal surface of ilium, sacrum and coccyx

Insertion - iliotibial tract and gluteal tuberosity of femur

Action - thigh extension, lateral rotation

Innervation: inferior gluteal nerves



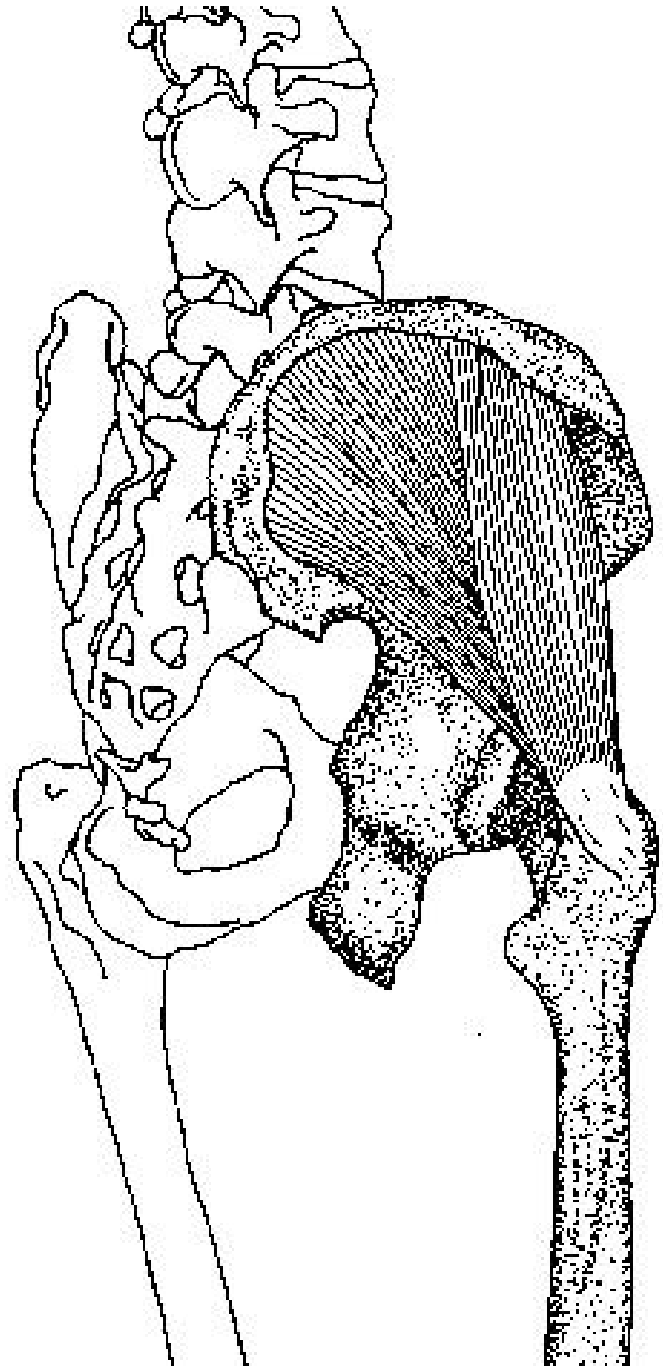
Gluteus medius

Origin - between anterior and posterior gluteal lines on lateral surface of ilium

Insertion - greater trochanter of femur

Action - abducts and medially rotates thigh

Innervation - superior gluteal nerve



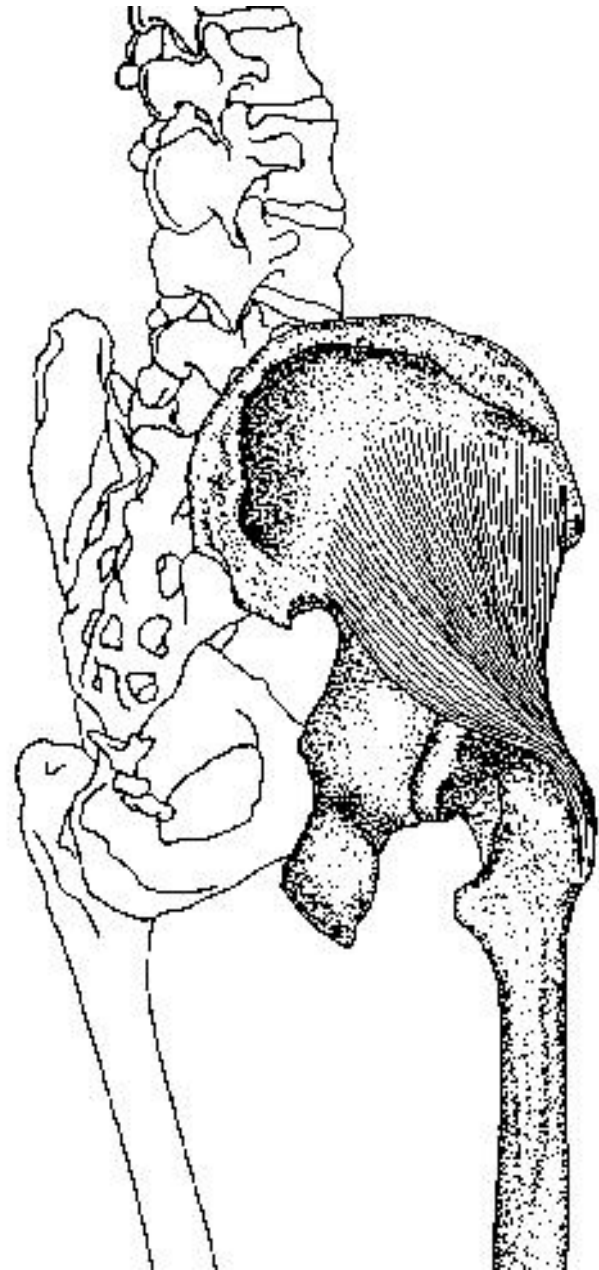
Gluteus minimus

Origin - between anterior and inferior gluteal lines on external surface of ilium

Insertion - greater trochanter of femur

Action - abducts and medially rotates thigh and with gluteus medius prevent tilting of the pelvis

Innervation - superior gluteal nerve



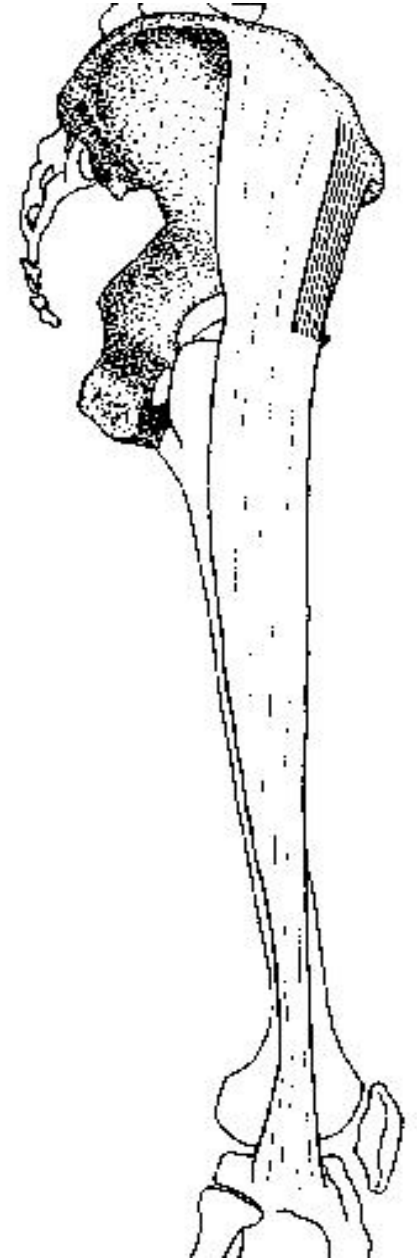
Tensor fascia latae

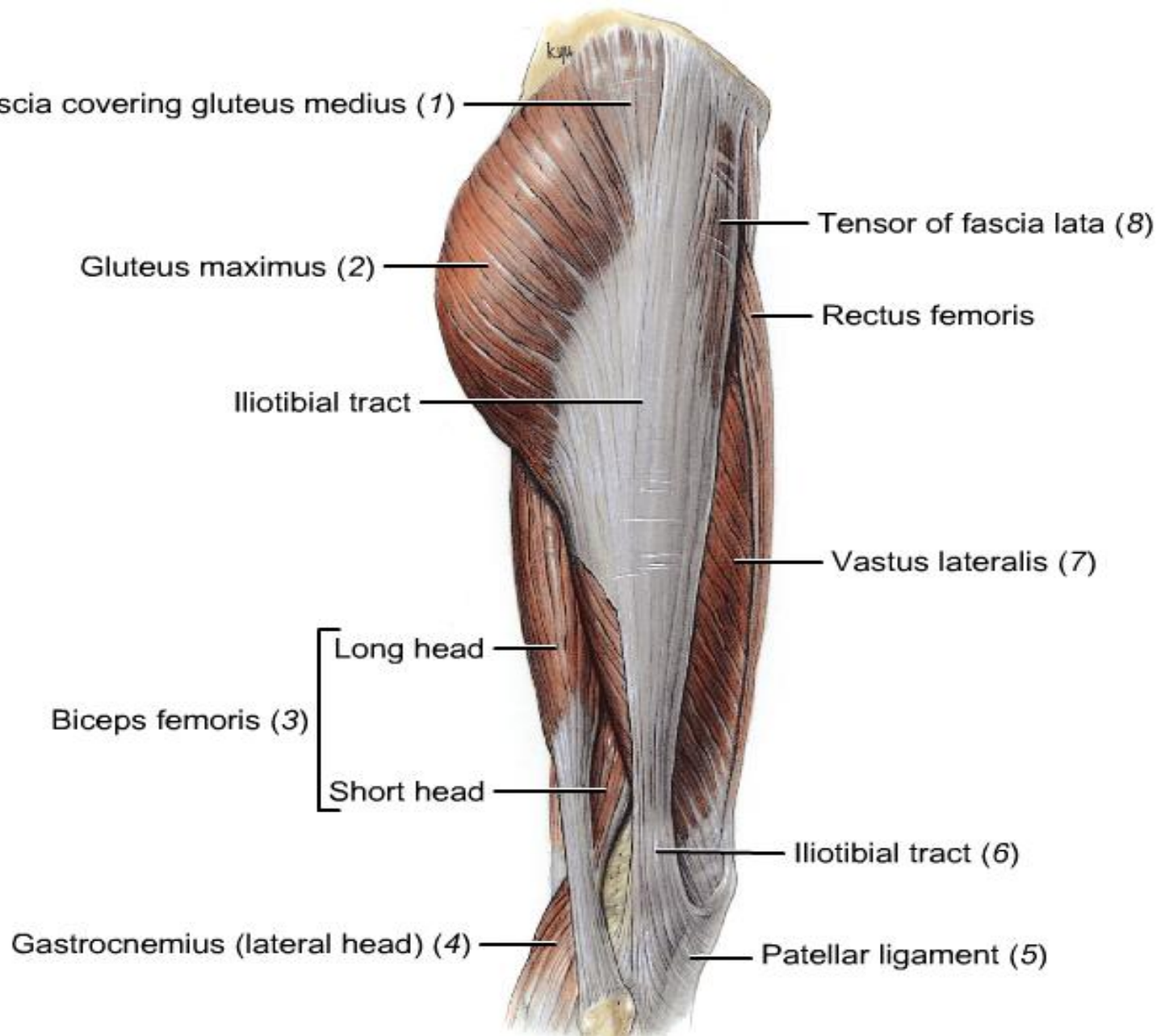
Origin - anterior superior iliac spine
and the outer lip of the iliac crest

Insertion - iliotibial tract

Action - flexes and abducts thigh

Innervation - superior gluteal nerve

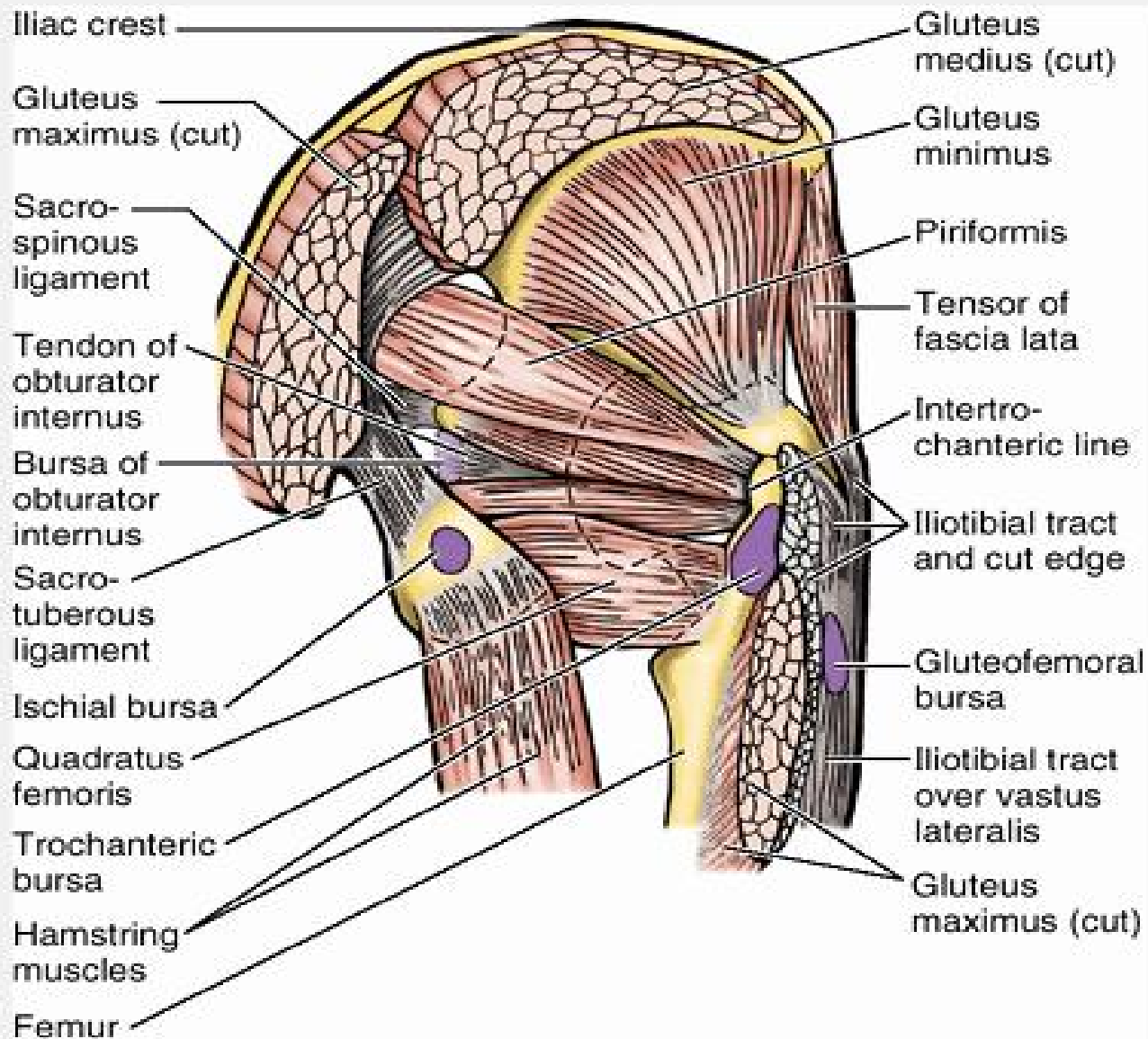




Gluteal bursae

Gluteal bursae, membranous sacs containing a capillary layer of synovial fluid, separate the gluteus maximus from adjacent structures

The bursae are located in areas subject to friction for example, between a muscle and a bony prominence to reduce friction and permit free movement



Posterior view

Usually, three bursae are associated with the gluteus maximus:

Trochanteric bursa

- separates the superior fibers of the gluteus maximus from the greater trochanter of the femur

Ischial bursa

- separates the inferior part of the gluteus maximus from the ischial tuberosity

Gluteofemoral bursa

- separates the iliotibial tract from the superior part of the proximal attachment of the vastus lateralis

Lateral rotators of the hip joint

Piriformis

Key muscle – divide the greater sciatic foramen

- structures passing above the muscle

superior gluteal nerve and vessels

- structures passing below the muscle

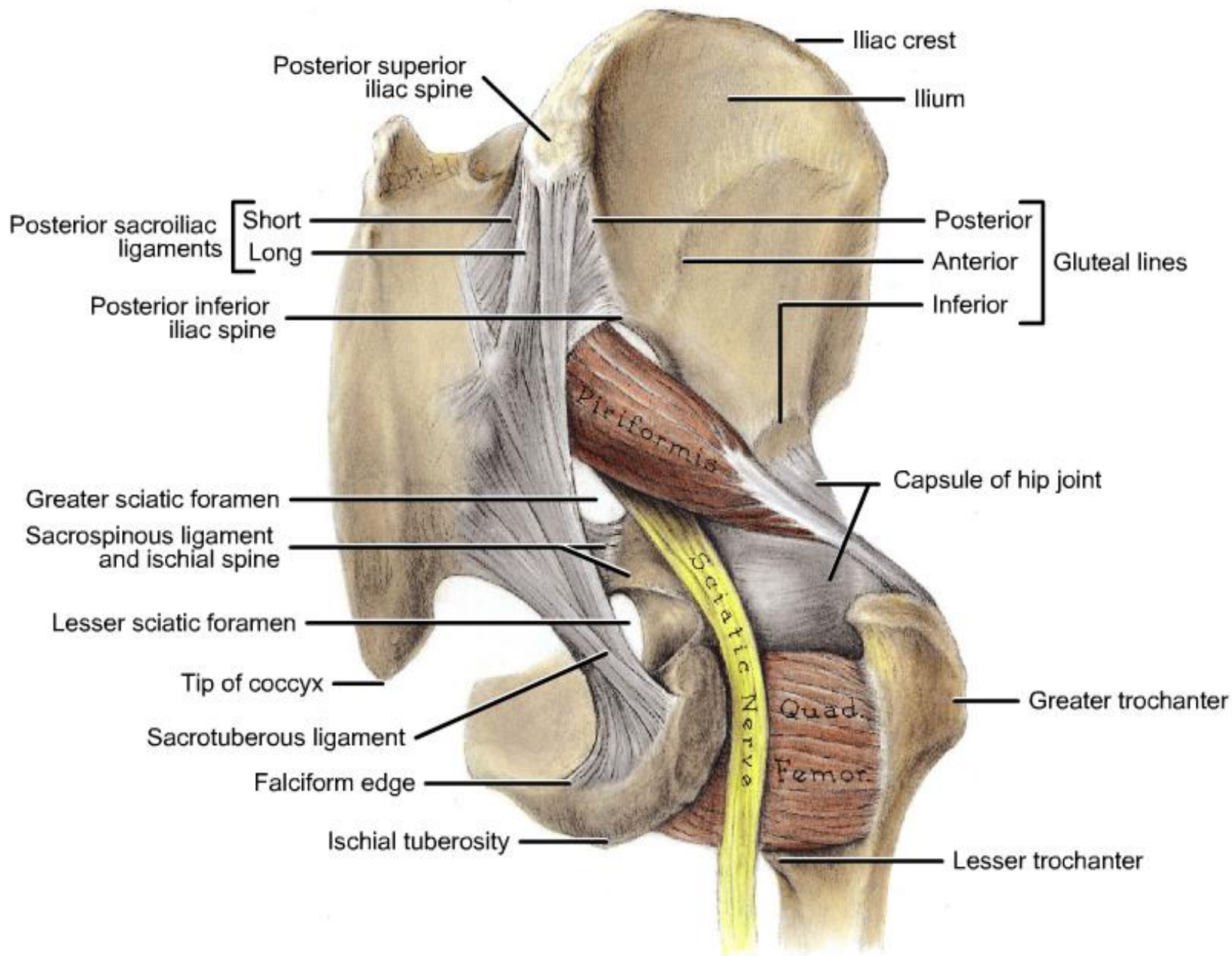
nerves - inferior gluteal, sciatic, posterior femoral cutaneous, pudendal, nerve to quadratus femoris and obturator internus

vessels - inferior gluteal and internal pudendal

Origin: pelvic surface of sacrum

Insertion: greater trochanter

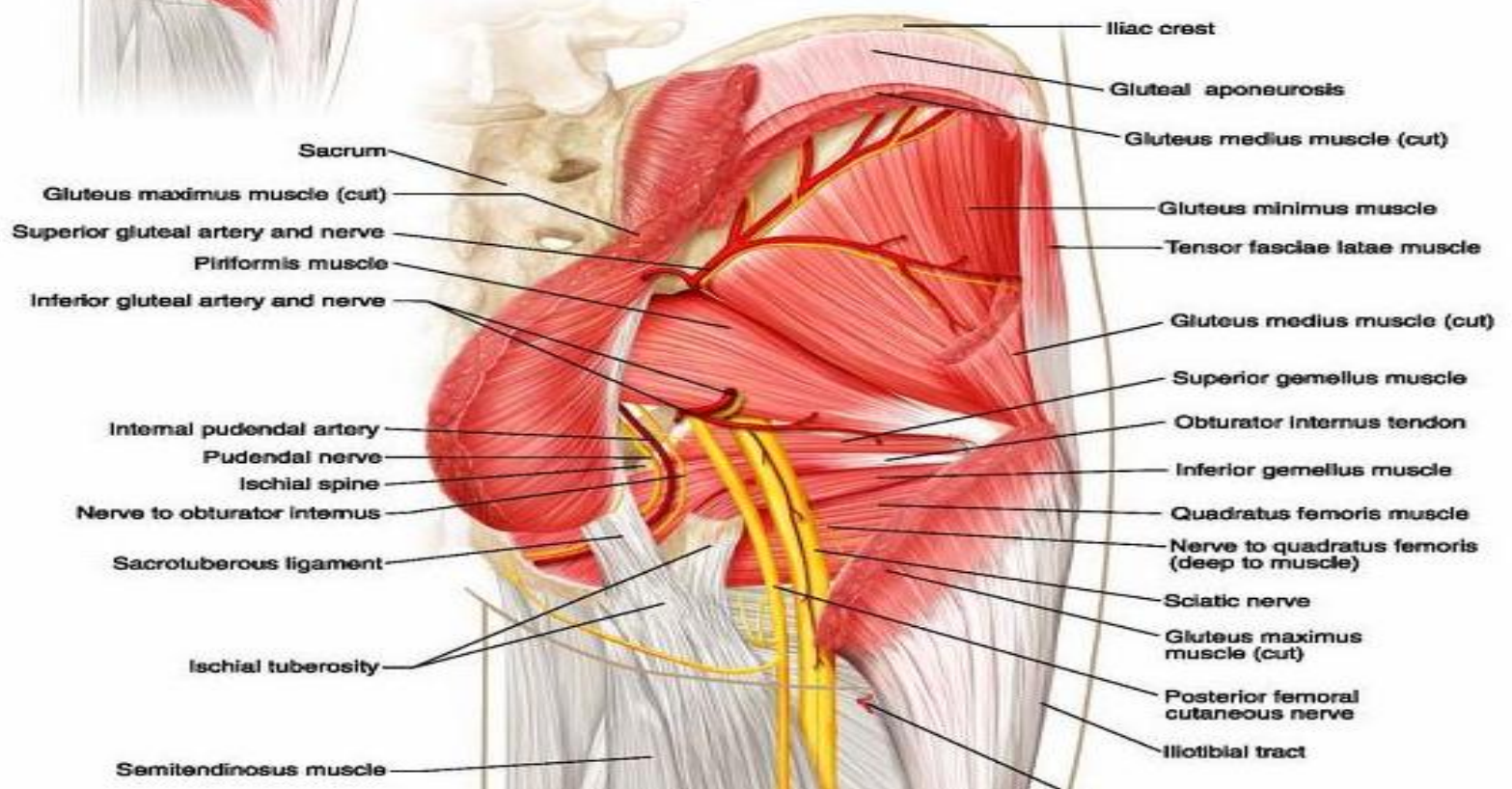
Innervation - branches of ventral rami of S1 and S2



A. Superficial dissection



B. Deep dissection



Quadratus femoris

Origin: ischial tuberosity

Insertion: intertrochanteric crest of femur

Innervation: nerve to quadratus femoris (L4, 5, S1)

Obturator internus

Origin: pelvic surface of the obturator membrane

Insertion: greater trochanter

Innervation: nerve to obturator internus (L5, S1, 2)

Superior gemellus

Origin: ischial spine

Insertion: greater trochanter

Innervation: nerve to obturator internus

Inferior gemellus

Origin: ischial tuberosity

Insertion: greater trochanter

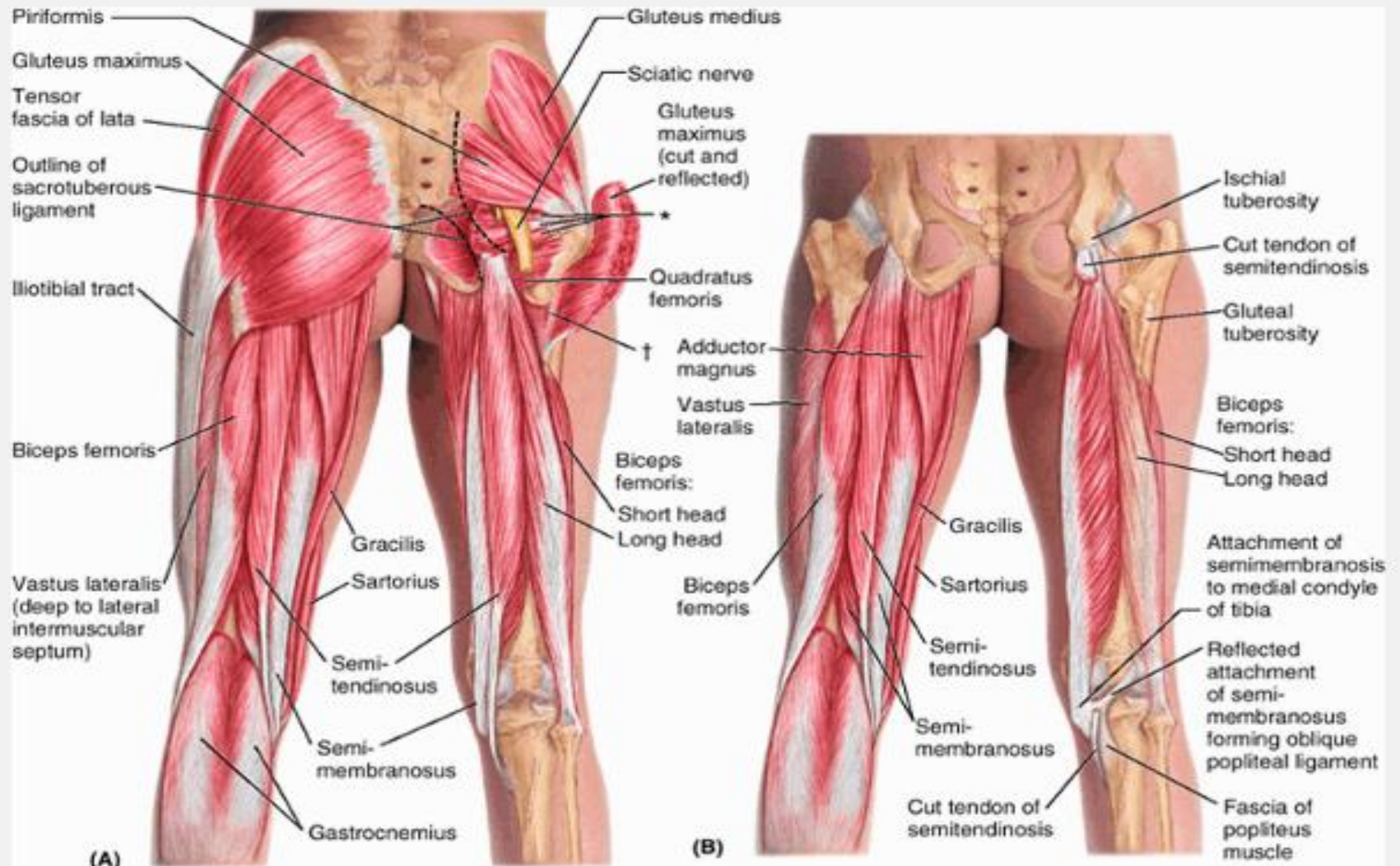
Innervation: nerve to quadratus femoris

Obturator externus

Origin: obturator membrane and attachments

Insertion: back of the femoral neck

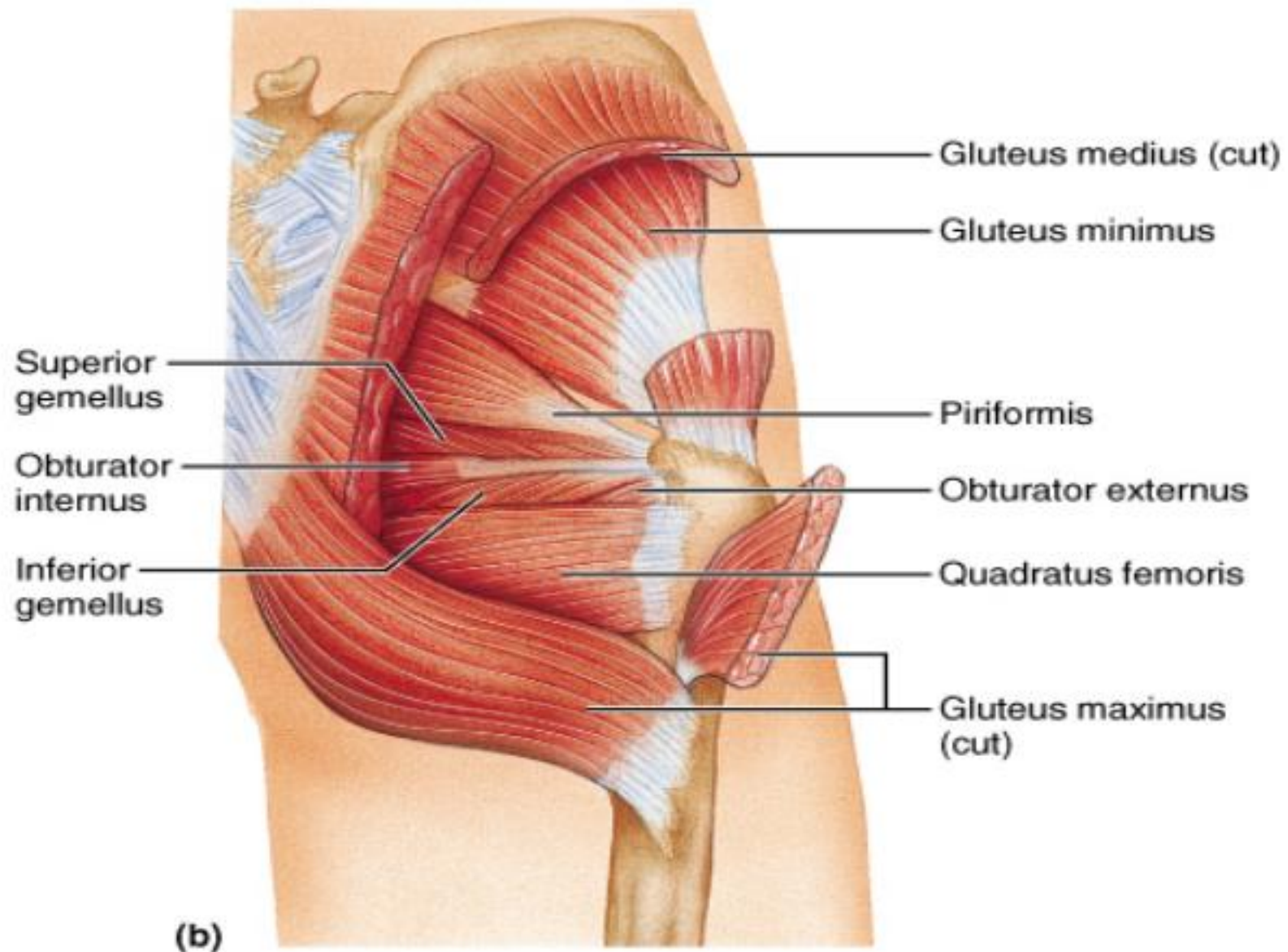
Innervation: posterior division of obturator nerve



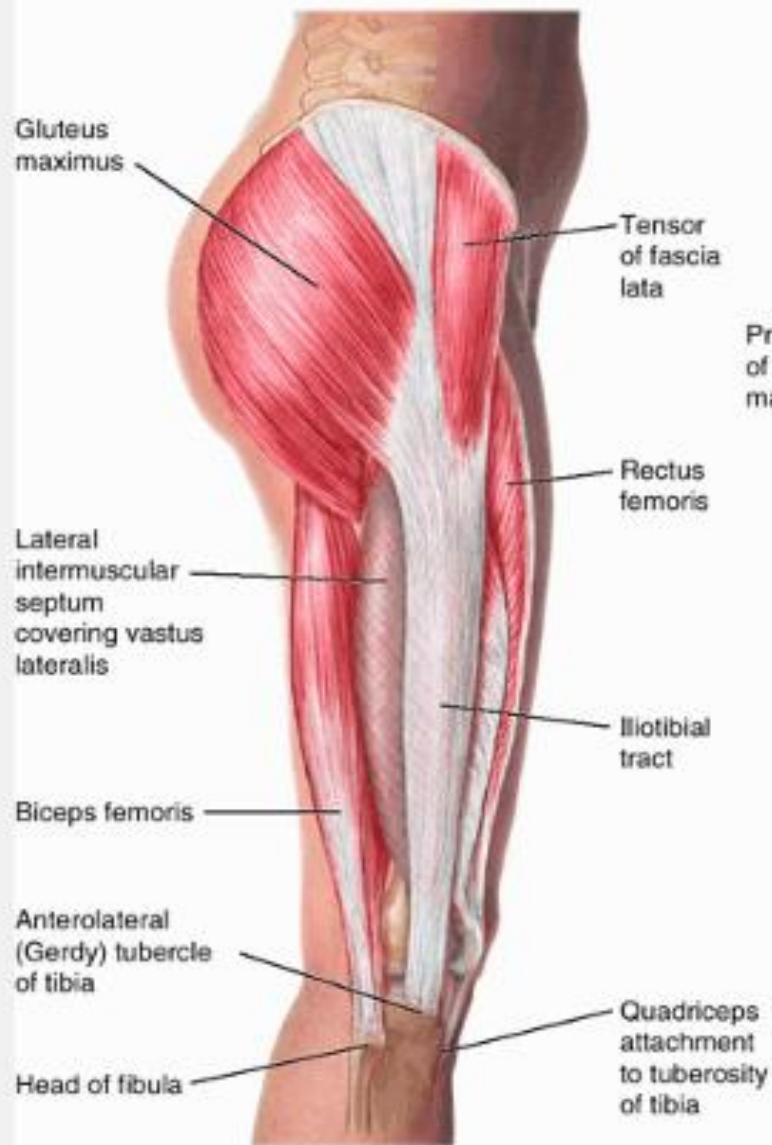
Posterior views

† Attachment of inferior half of deep part of gluteus maximus to gluteal tuberosity

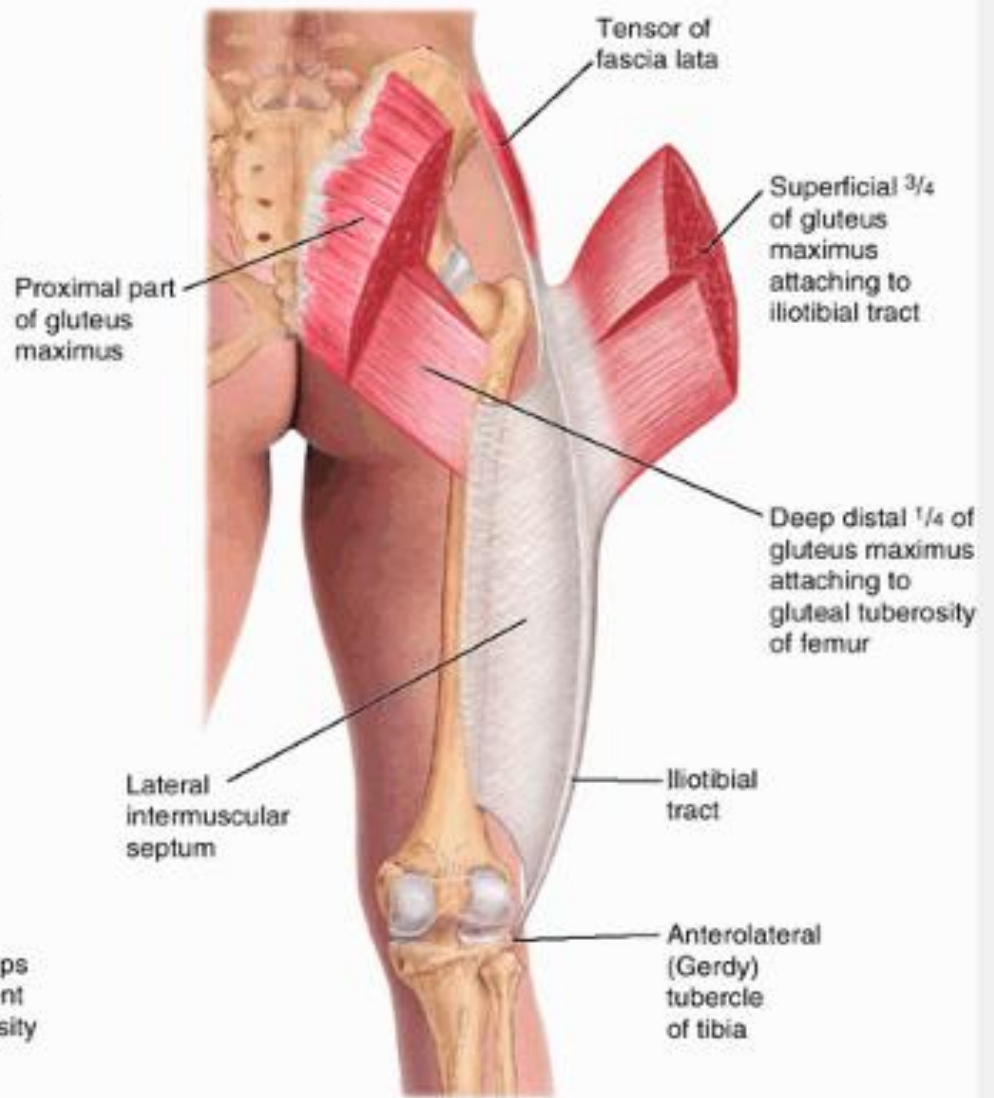
* Triceps coxae:
Tendon of obturator internus
Superior gemellus
Inferior gemellus



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(C) Lateral view



(D) Posterior view

Structures deep to the gluteus maximus

- ✓ Communication between the pelvis and the gluteal region is done via the greater sciatic foramen
- ✓ The piriformis muscle is the key to this region
- ✓ Superior gluteal vessels and nerve emerge above the piriformis

2 groups of nerve and vessels emerge inferior to piriformis

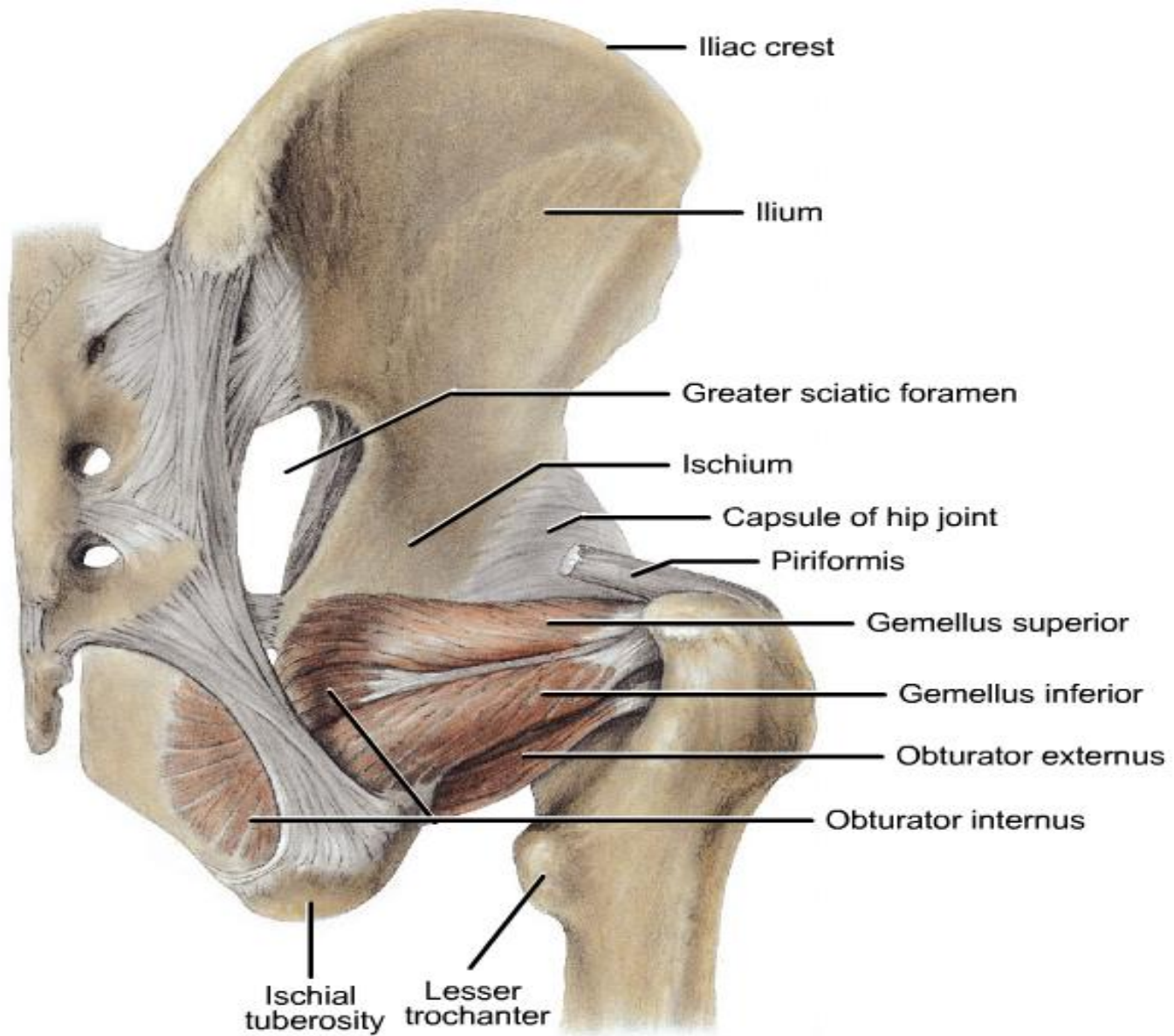
A group of 5 structures

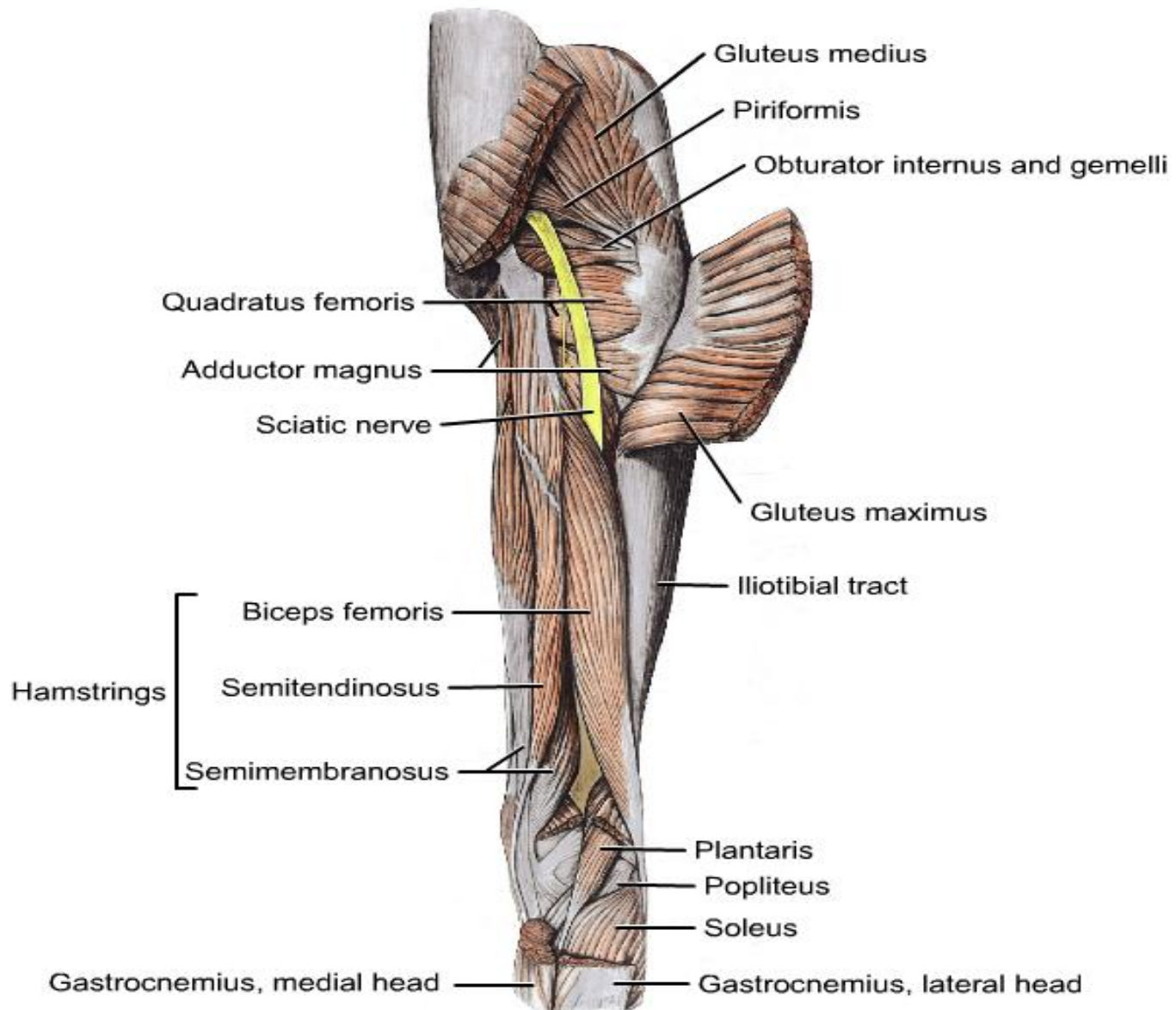
- Sciatic nerve
 - lies between ischial tuberosity and greater trochanter
 - Its "safe" side for gluteal injection is the suprolateral side where it has no branches
 - It may divide in this region into common peroneal and tibial nerves
- Inferior gluteal nerve
- Inferior gluteal artery
- Inferior gluteal vein
- Posterior femoral cutaneous nerve

- A group of 3 structures
 - Nerve to obturator internus
 - Internal pudendal vessels
 - Pudendal nerve
- The group of 3 emerges from the greater sciatic foramen and runs through the lesser sciatic foramen into the ischioanal fossa
- The obturator internus is the only structure entering the gluteal region via the lesser sciatic foramen

The greater trochanter serves as attachment point for 5 muscles

- Gluteus medius
- Gluteus minimus
- Piriformis
- Obturator internus
- Gemelli





Posterior superior
iliac spine

Superior gluteal
artery and nerve

Sacrospinous
ligament

Pudendal nerve

Internal pudendal artery

Nerve to obturator
internus

Tip of coccyx

Inferior gluteal
nerve and artery

Posterior cutaneous
nerve of thigh

Biceps femoris, long head

Semimembranosus

Semitendinosus

Adductor magnus

Gracilis

Semimembranosus

Semitendinosus

Biceps femoris, long head

Gluteus minimus

Piriformis

Gluteus medius

Sciatic nerve

Obturator internus and gemelli

Greater trochanter

Obturator externus

Medial circumflex femoral artery

Artery to sciatic nerve

Quadratus femoris

Gluteus maximus

Posterior cutaneous
nerve of thigh

1st perforating artery

Intermuscular septum

Sciatic nerve

2nd perforating artery

Biceps femoris, short head

(A) Posterior view

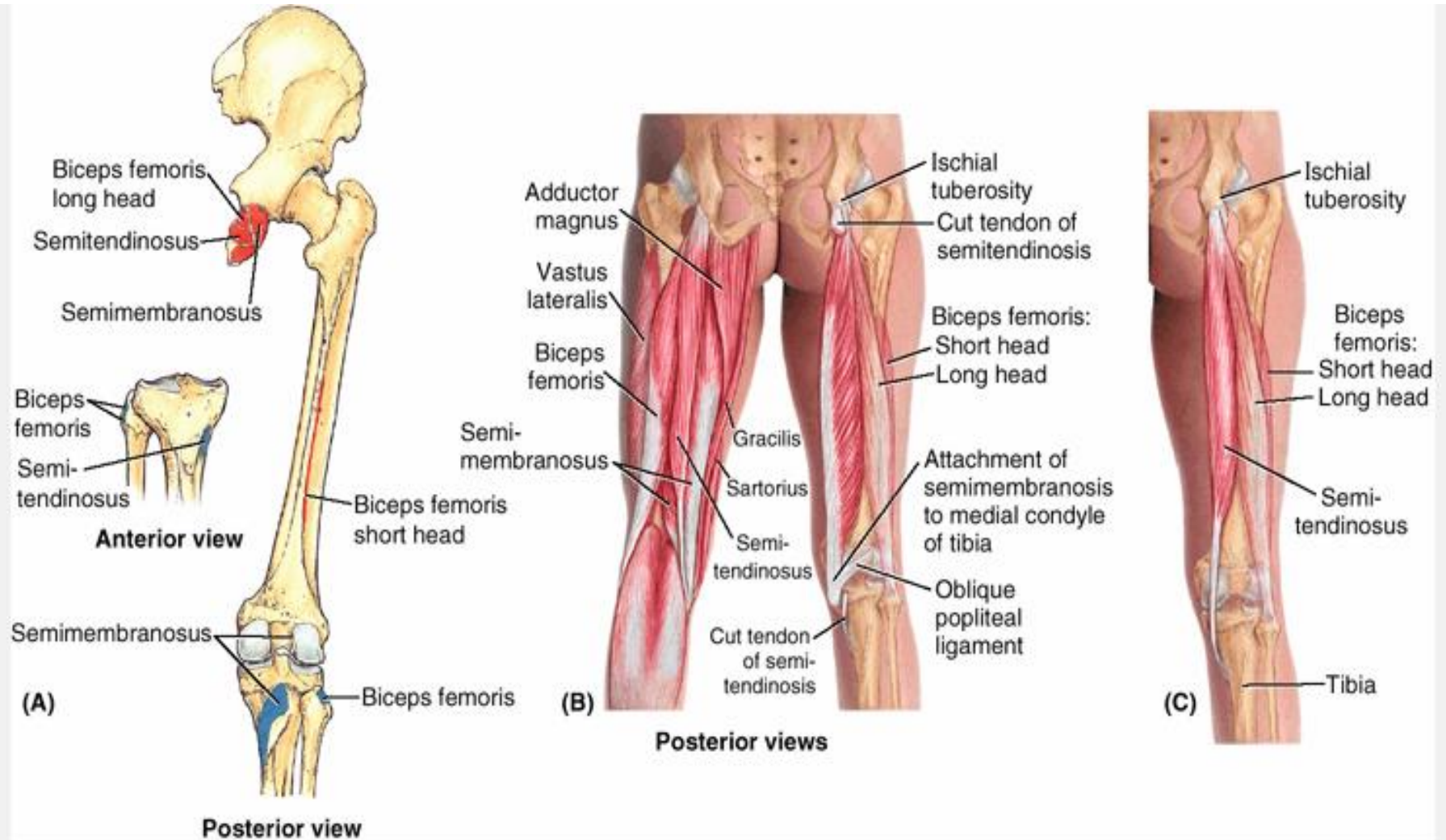
Posterior thigh muscles

Three of the four muscles in posterior thigh are hamstrings

- Semitendinosus
- Semimembranosus
- Biceps femoris (long head)

Short head of biceps femoris

- not hamstring as it crosses only knee joint
- innervated by fibular division of sciatic nerve



Hamstring muscles

Arise from **ischial tuberosity**

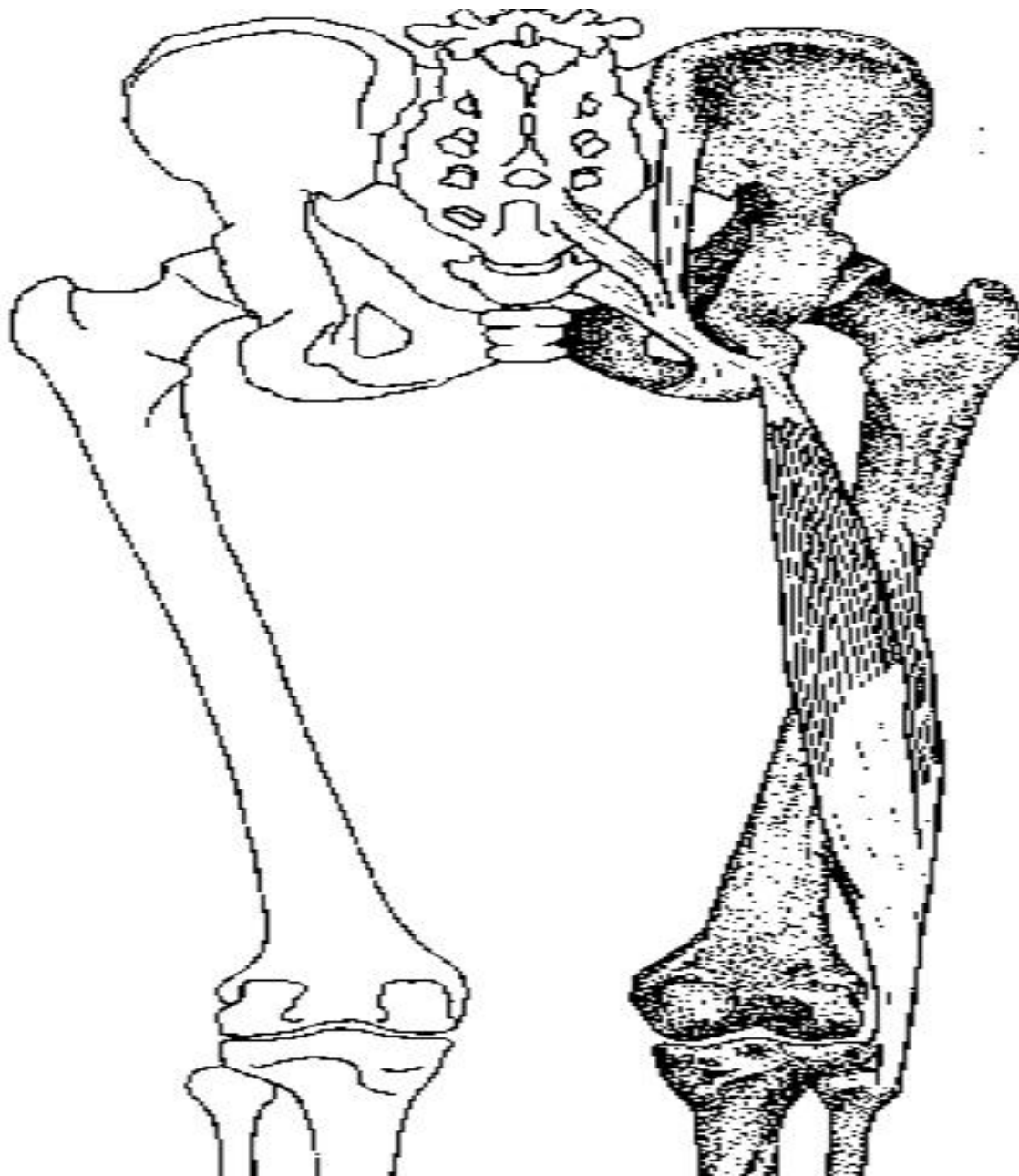
Insert **into tibia or fibula**

- Innervated by the tibial division of sciatic nerve
 - The nerve to the short head of the biceps femoris comes from the lateral (peroneal) division whereas branches to other hamstrings arise from the medial (tibial) division
- Extend hip and flex knee joint
- The **semimembranosus and semitendinosus** also rotate the leg medially, being attached to the medial head of the tibia.

E
Orig

Inse
Act

Inne



of tibia

the thigh

rotates leg

erve

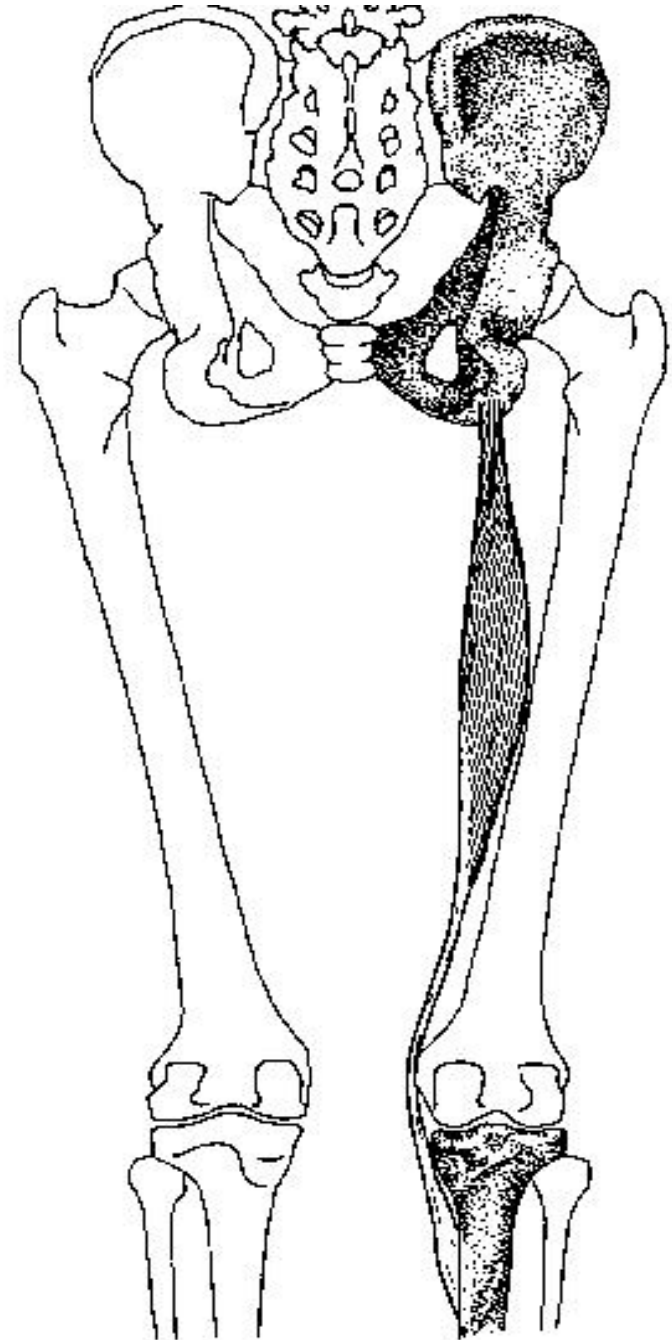
Semitendinosus

Origin - ischial tuberosity

Insertion - medial aspect of upper tibial shaft

Action - extends thigh at hip; flexes knee, medially rotates leg

Innervation - tibial division of sciatic nerve



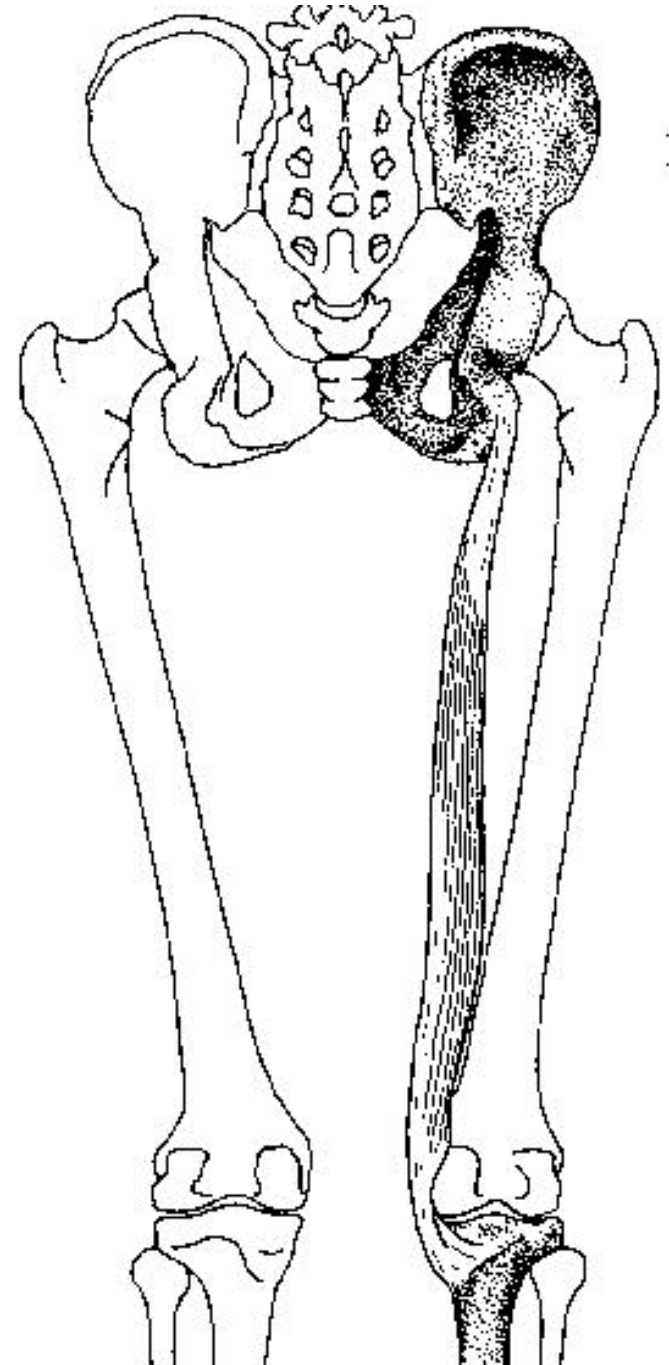
Semimembranosus

Origin - ischial tuberosity

Insertion - medial condyle of tibia

Action - extends thigh and flexes knee, medially rotates leg

Innervation - tibial division of sciatic nerve



Hamstring injuries

- Hamstrings **strains** (pulled and/or torn hamstrings) are common in people who run and/or kick hard (e.g., in running, jumping and **quick-start** sports such as baseball and soccer)
- The violent muscular exertion required to excel in these sports may tear part of the proximal attachments of the hamstrings to the ischial tuberosity
- Hamstring injuries may result from **inadequate warming up** before practice or competition

Clunial (superficial gluteal) nerves

- The **skin of the gluteal region** is richly innervated by superior, middle, and inferior clunial nerves (L. clunes, **buttocks**)
- These superficial nerves supply the skin over the iliac crest, between the posterior superior iliac spines and over the iliac tubercles
- Consequently, these nerves are **vulnerable to injury** when bone is taken from the **ilium for grafting**

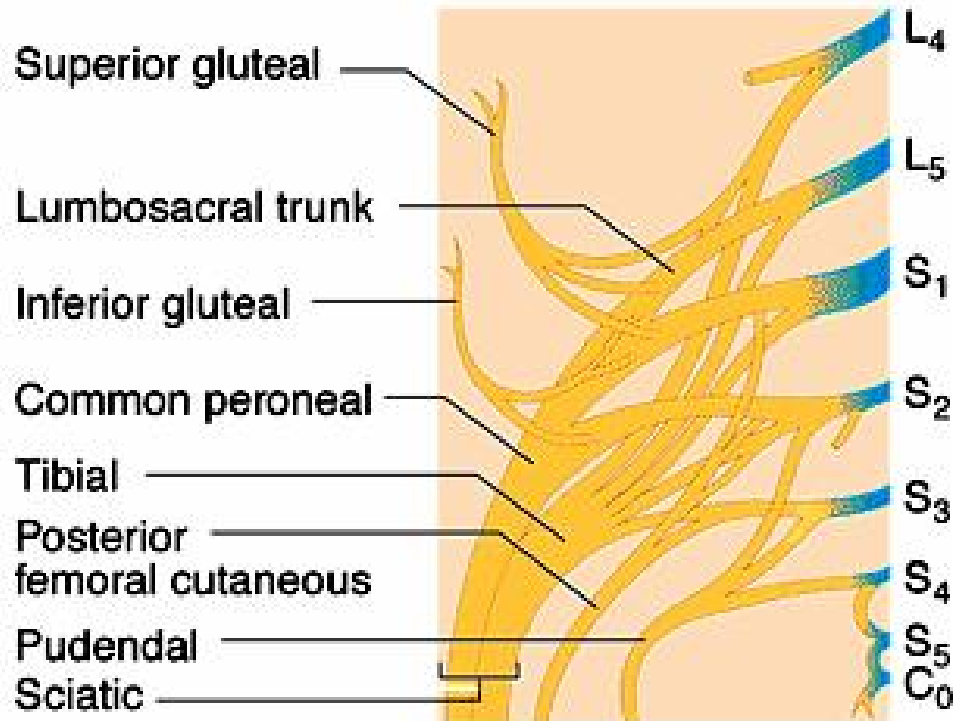
Deep gluteal nerves

- ✓ The deep gluteal nerves are the **superior and inferior gluteal nerves**, sciatic nerve, nerve to quadratus femoris, posterior cutaneous nerve of the thigh, nerve to obturator internus, and pudendal nerve
- ✓ All of these nerves are branches of the sacral plexus and leave the pelvis through the greater sciatic foramen
- ✓ Except for the **superior gluteal nerve**, they all emerge **inferior to the piriformis**

Sacral plexus

- Originates from the anterior rami of L4, L5, S1–S4
- L4 is shared by both plexuses, a branch from it joining L5 to form the **lumbosacral trunk**
- The sacral nerves emerge from the anterior sacral foramina and unite in front of piriformis where they are joined by the lumbosacral trunk

Sacral Plexus

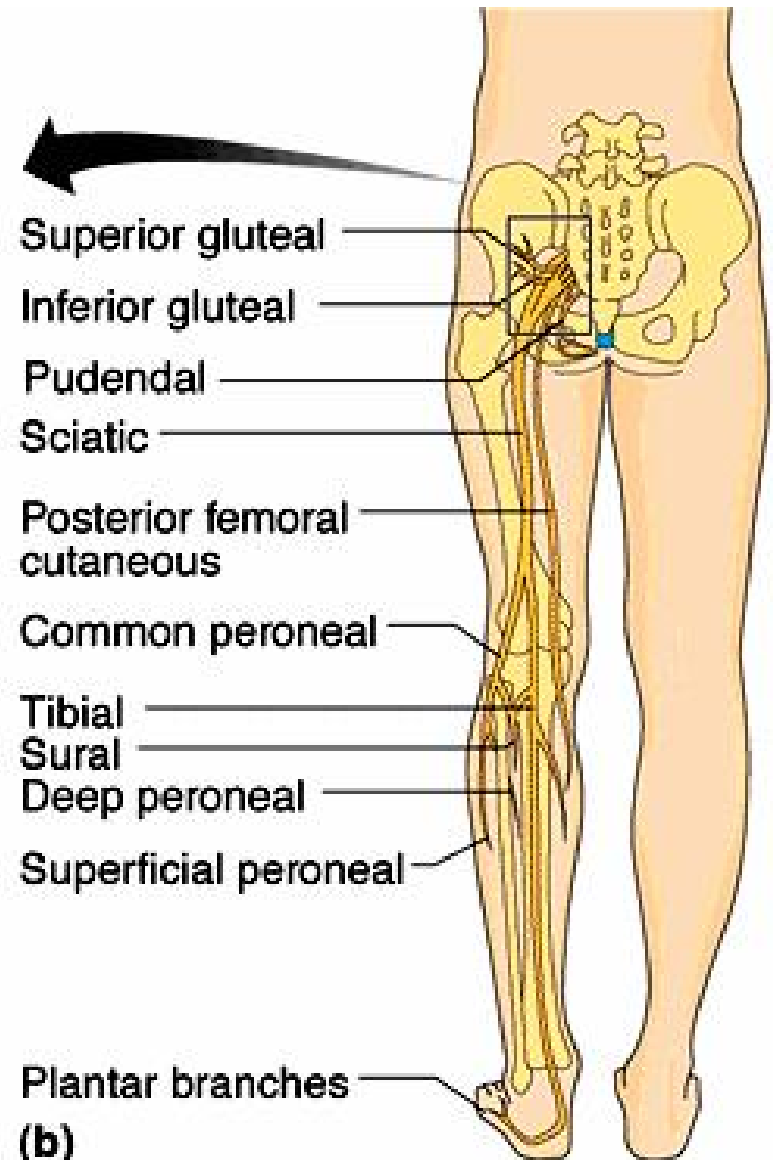


(a)

Key:

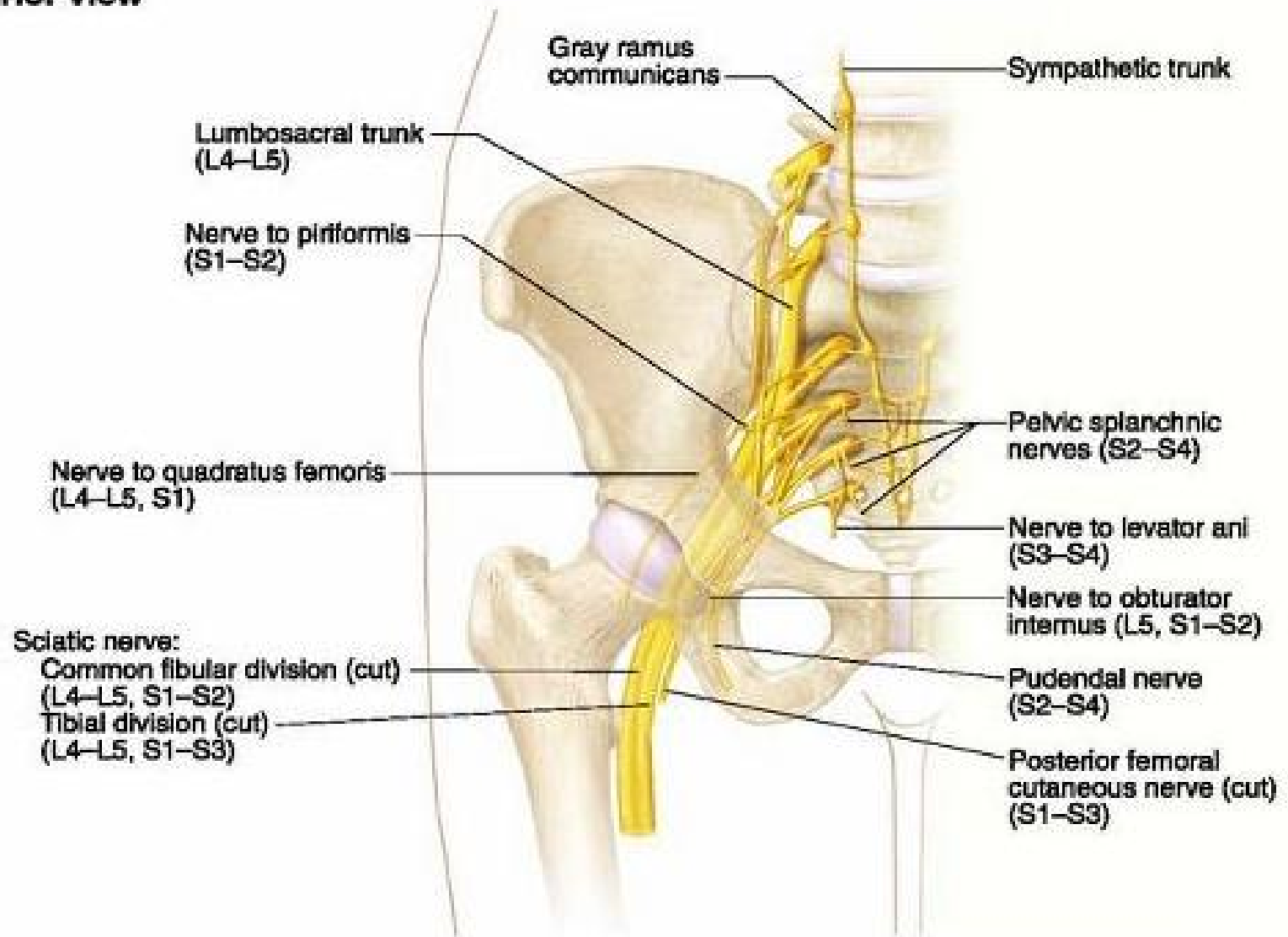
■ = Roots

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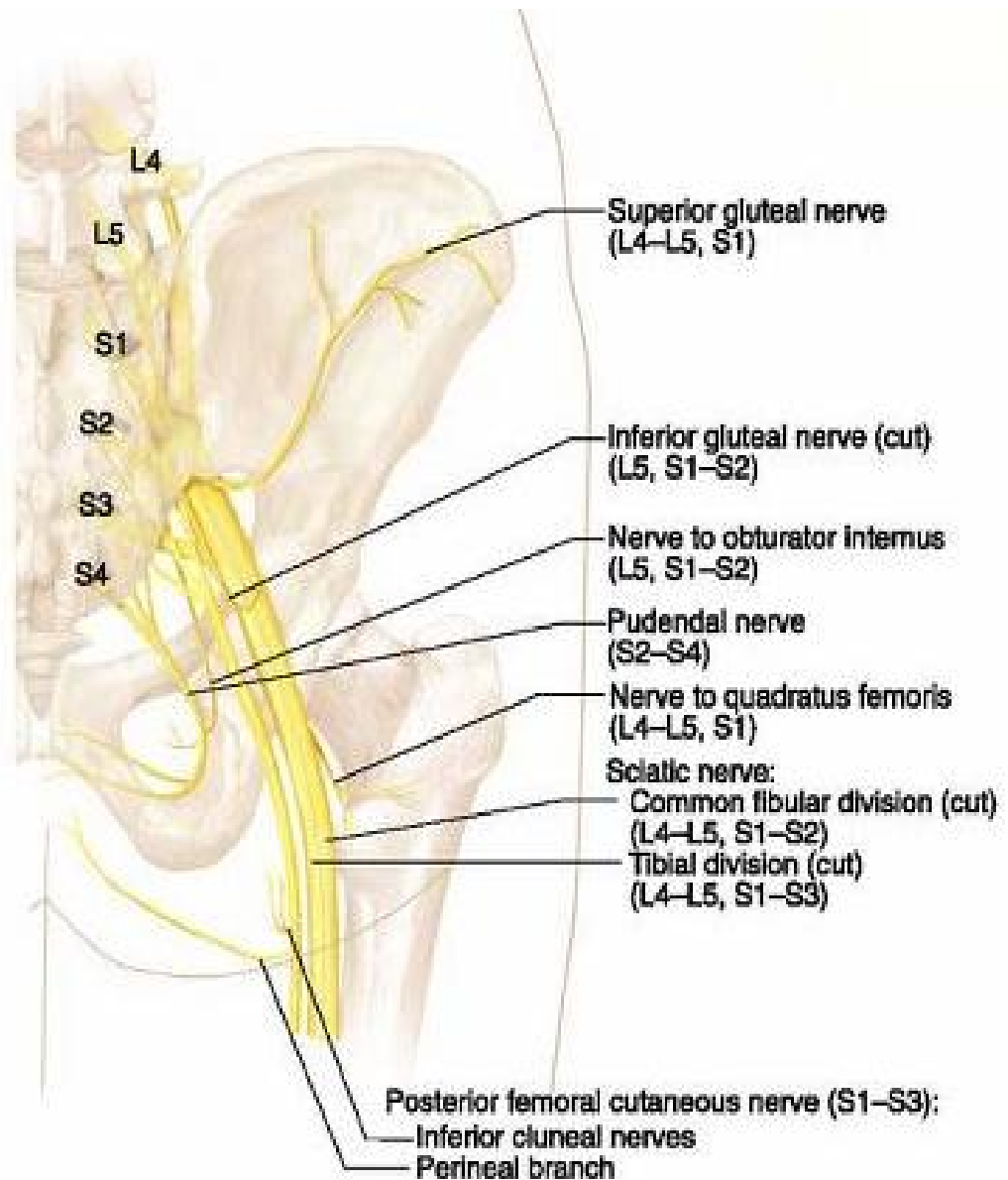


(b)

A. Anterior view

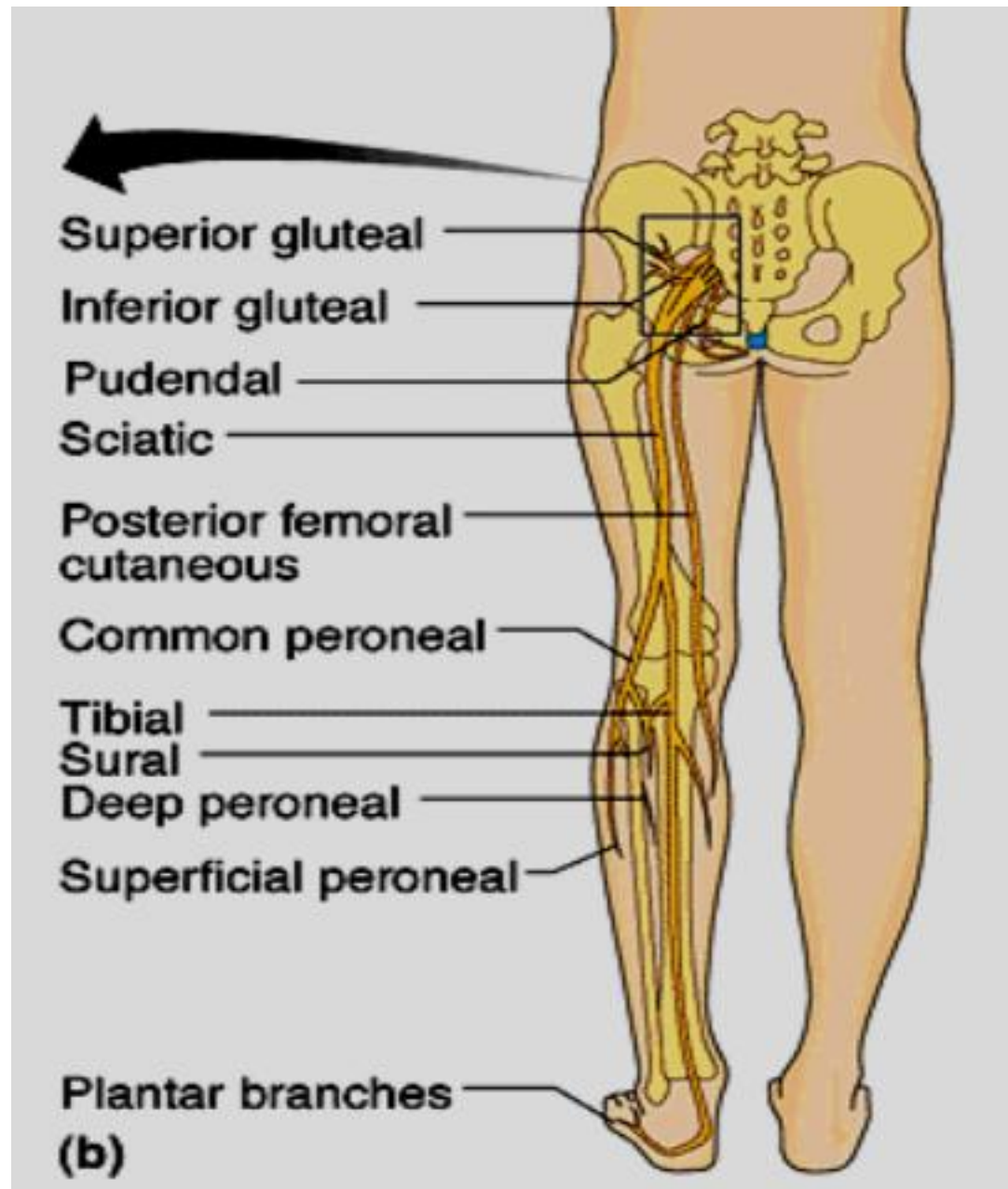


B. Posterior view



Branches

- Superior gluteal nerve - gluteus medius, minimus, tensor fascia lata
- Inferior gluteal nerve – gluteus maximus
- Nerve to quadratus femoris
- Nerve to piriformis
- Nerve to obturator internus
- Posterior cutaneous nerve of the thigh
- Pudendal nerve
- Sciatic nerve

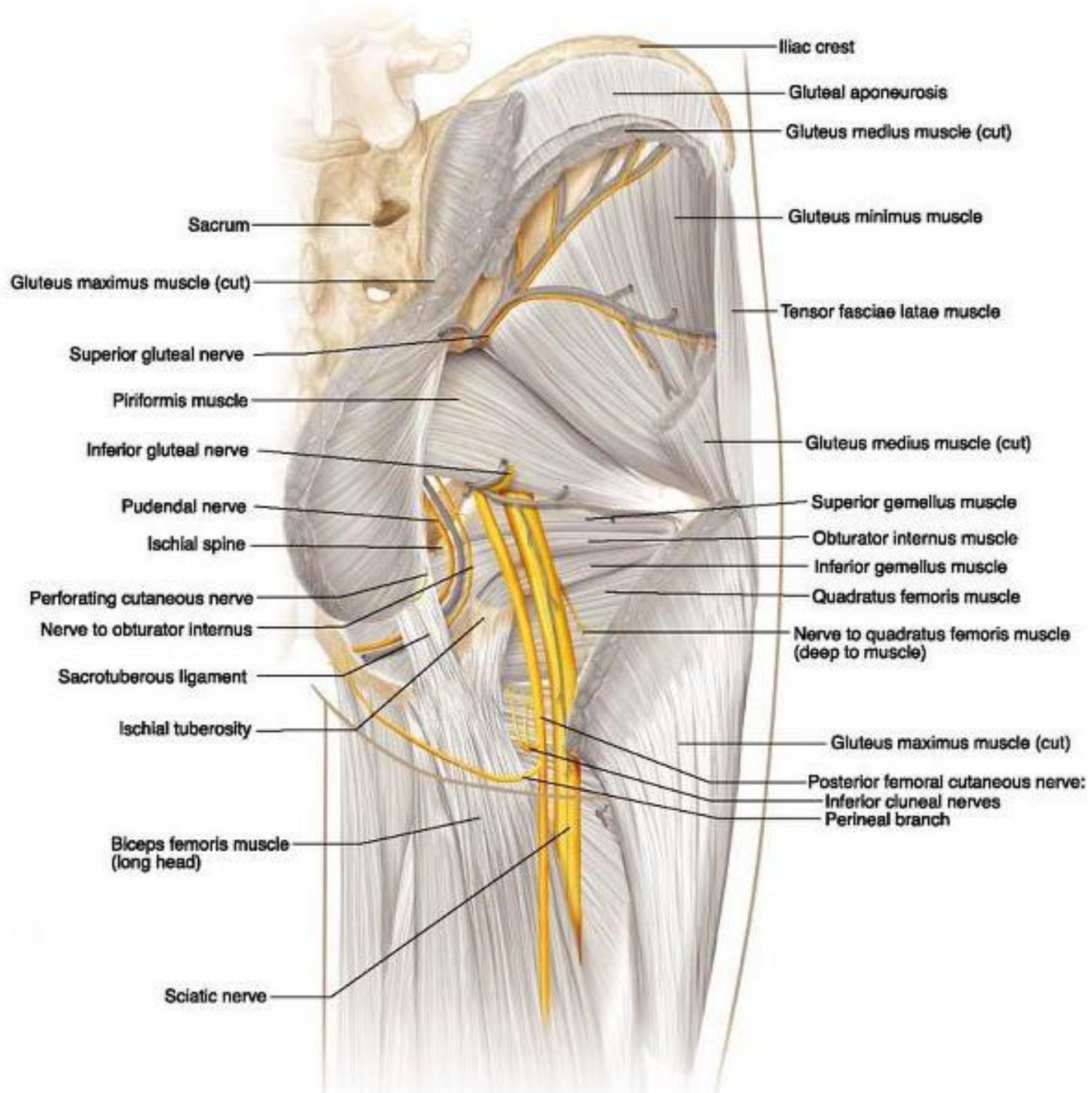


Sciatic nerve (L4-S3)

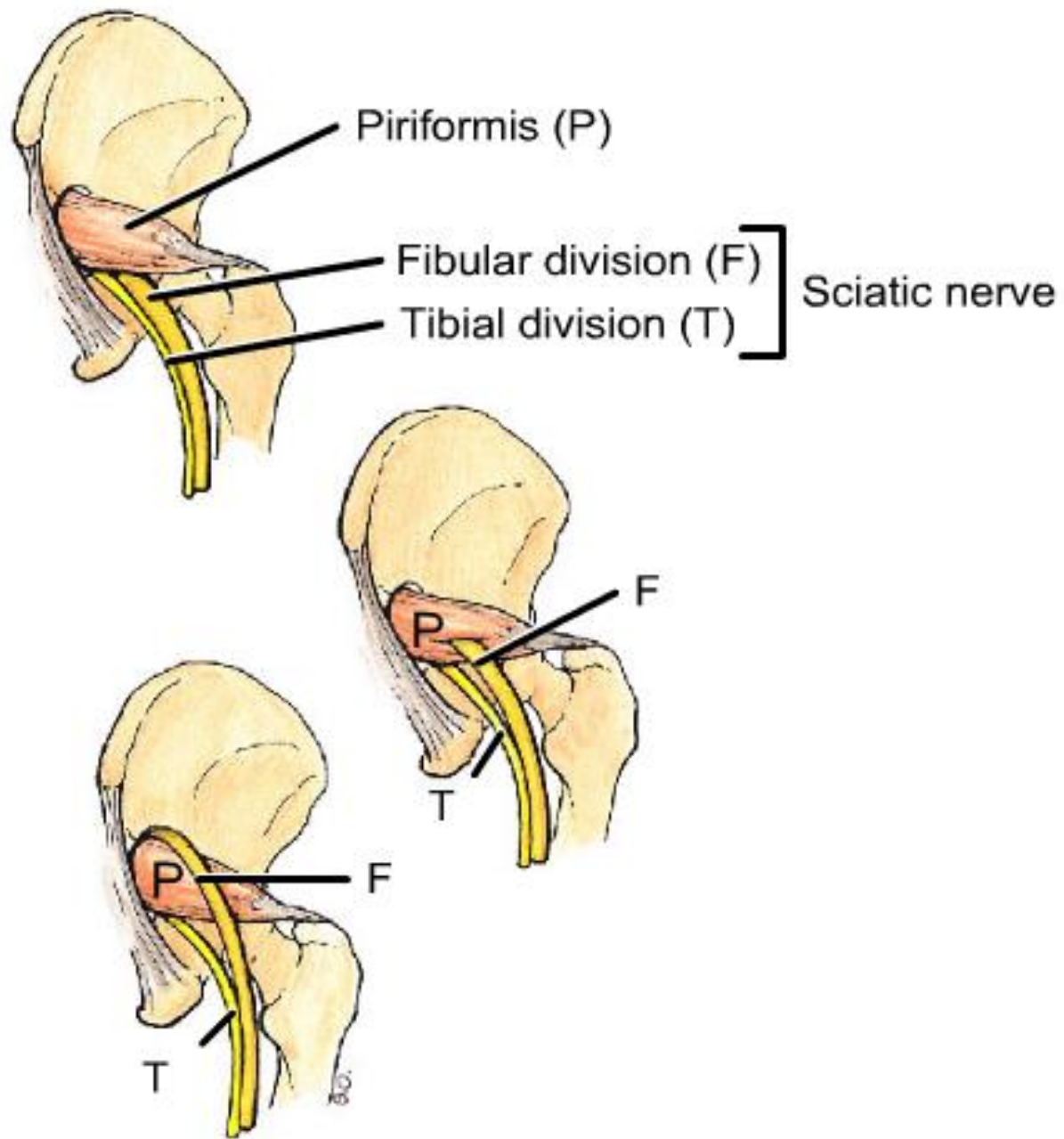
- The largest nerve in the body
 - Thick flattened band approximately 2 cm wide
- The continuation of the main part of the sacral plexus
 - The rami converge at the inferior border of the piriformis to form the sciatic nerve
- Terminate in the middle of the thigh by dividing into tibial and common fibular nerves

- Course
 - Leaves pelvis through greater sciatic foramen below piriformis
 - The sciatic nerve is the most lateral structure emerging through the greater sciatic foramen inferior to the piriformis
 - Medial to it are the inferior gluteal nerve and vessels, the internal pudendal vessels, and the pudendal nerve
 - Enter gluteal region
 - Runs inferolaterally under cover of the gluteus maximus, midway between the greater trochanter and ischial tuberosity

- The nerve rests on the ischium and then passes posterior to the obturator internus, quadratus femoris, and adductor magnus muscles
- Descends deep to biceps femoris
- Bifurcates into tibial and common peroneal nerves at apex of popliteal fossa
- The sciatic nerve is so large that it receives a named branch of the inferior gluteal artery, the artery to the sciatic nerve (L. arteria comitans nervi ischiadici)

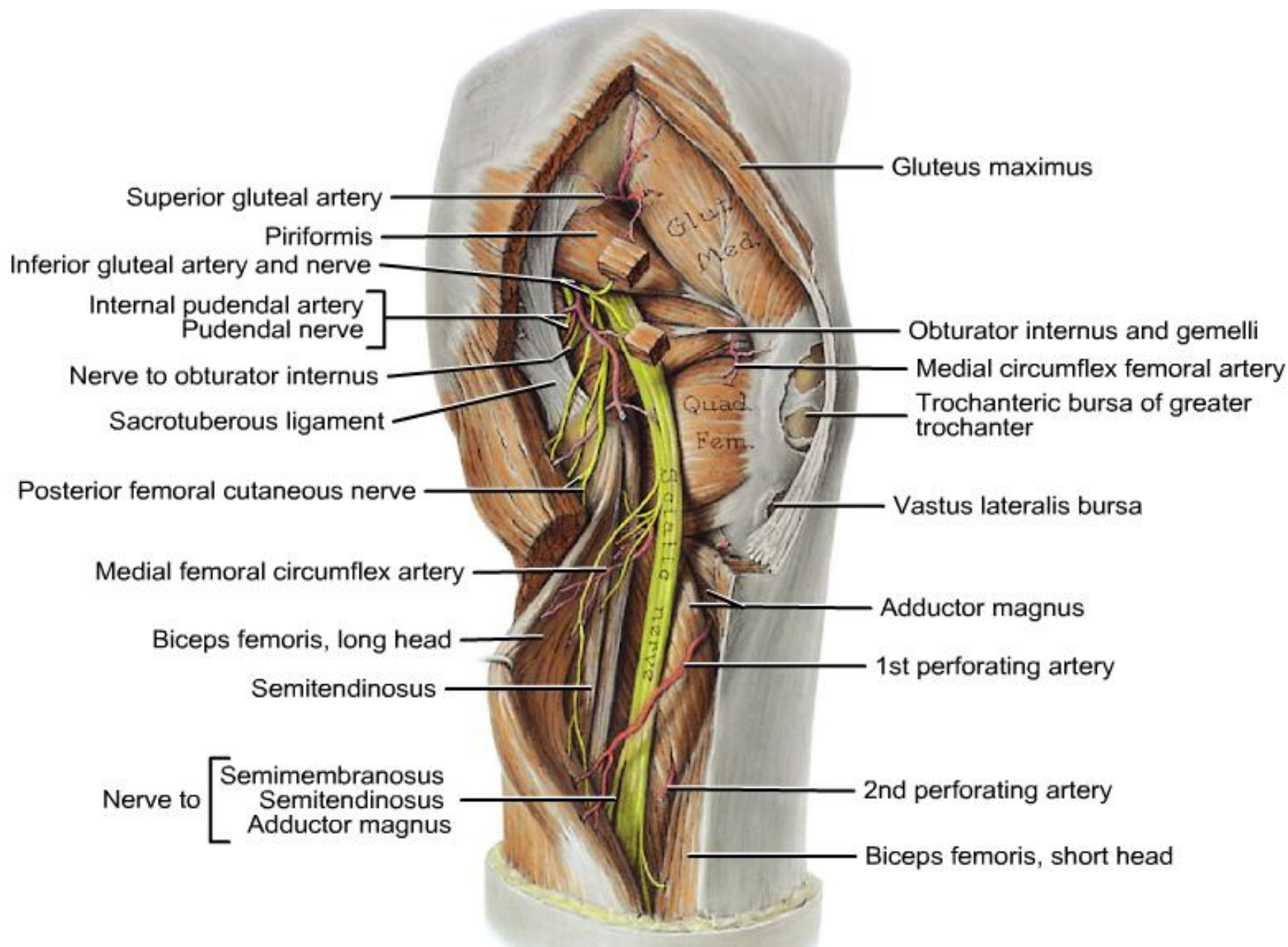


- The sciatic nerve is really two nerves which are loosely bound together in the same connective tissue sheath
 - tibial nerve - derived from anterior (preaxial) divisions of the anterior rami
 - common fibular nerve - derived from posterior (postaxial) divisions of the anterior rami
- The tibial and common fibular nerves usually separate approximately halfway or more down the thigh; however, in approximately 12% of people, the nerves separate as they leave the pelvis
 - In these cases, the tibial nerve passes inferior to the piriformis, and the common fibular nerve pierces this muscle or passes superior to it



- Distribution

- The sciatic nerve supplies no structures in the gluteal region
- It supplies the posterior thigh muscles, all leg and foot muscles, and the skin of posterior thigh, most of the leg and the foot
- Motor for hamstring muscles
 - The nerve to the short head of the biceps femoris comes from the lateral (peroneal) division whereas branches to other hamstrings arise from the medial (tibial) division
- It also supplies the articular branches to all joints of the lower limb



Anesthetic block of the sciatic nerve

- Sensation conveyed by the sciatic nerve can be blocked by the injection of an anesthetic agent a few centimeters inferior to the midpoint of the line joining the PSIS and the superior border of the greater trochanter
- Paresthesia radiates to the foot because of anesthesia of the plantar nerves, which are terminal branches of the tibial nerve derived from the sciatic nerve

Injury to sciatic nerve

- Piriformis syndrome
 - A pain in the buttock may result from compression of the sciatic nerve by the piriformis muscle
 - Individuals involved in sports that require excessive use of the gluteal muscles (e.g., ice skaters, cyclists, and rock climbers) are more likely to develop this syndrome

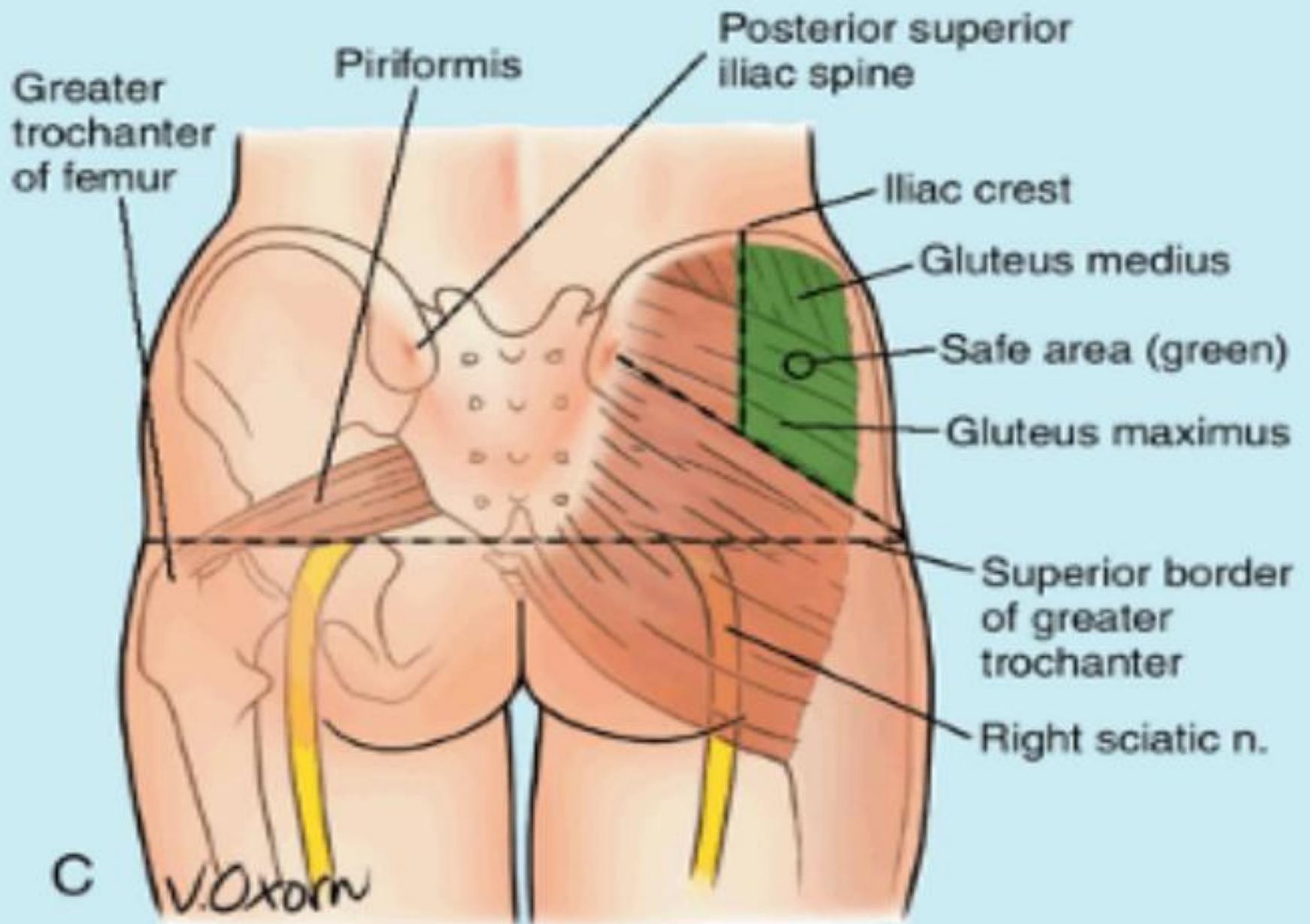
- Incomplete section of the sciatic nerve (e.g., from a stab wound) may also involve the inferior gluteal and/or the posterior femoral cutaneous nerves
- Complete section of the sciatic nerve is uncommon
 - When this occurs, the leg is useless because extension of the hip is impaired, as is flexion of the leg
 - All ankle and foot movements are also lost

- Recovery from a sciatic lesion is slow and usually incomplete
- With respect to the sciatic nerve, the buttock has a side of safety (its lateral side) and a side of danger (its medial side)
- Wounds or surgery on the medial side of the buttock are liable to injure the sciatic nerve and its branches to the hamstrings on the posterior aspect of the thigh
 - Paralysis of these muscles results in impairment of thigh extension and leg flexion

Intragluteal injections

- The gluteal region is a common site for intramuscular injection of drugs
- The gluteal region is a favorable injection site because the muscles are thick and large; consequently, they provide a substantial volume for absorption of injected substances by intramuscular veins
- Complications of improper technique include nerve injury, hematoma, and abscess formation

- Injections into the buttock are safe only in the **superolateral quadrant** of the buttock or superior to a line extending from the PSIS to the superior border of the greater trochanter (approximating the superior border of the gluteus maximus)
- Intramuscular injections can also be given safely into the **anterolateral part of the thigh**, where the needle enters the tensor of the fascia lata as it extends distally from the iliac crest and ASIS



Superior gluteal nerve (L4, 5, S1)

- Runs along with the superior gluteal vessels, through the greater sciatic foramen (superior to the piriformis)
- Runs laterally across the lateral surface of the ilium between the gluteus medius and minimus, along with the deep branch of the superior gluteal artery
- It divides into a superior branch that supplies the gluteus medius and an inferior branch that continues to pass between the gluteus medius and the gluteus minimus to supply both muscles and the tensor of the fascia lata

Injury to superior gluteal nerve

- Injury to the superior gluteal nerve results in a characteristic motor loss, resulting in a **disabling gluteus medius limp**, to compensate for weakened abduction of the thigh by the gluteus medius and minimus, and/or a gluteal gait, a compensatory list of the body to the weakened gluteal side
- As a result of this compensation, the center of gravity is placed over the supporting lower limb
- Medial rotation of the thigh is also severely impaired

- When a person with **paralysis of the superior gluteal nerve** is asked to stand on one leg, the pelvis descends on the unsupported side, indicating that the gluteus medius on the contralateral (supported) side is weak or non-functional
- This observation is referred to clinically as a **positive Trendelenburg test**



A. The pelvis is shown during normal gait

B. A positive Trendelenburg test indicates paralysis of superior gluteal nerve

Inferior gluteal nerve

- Leaves the pelvis through the greater sciatic foramen, inferior to the piriformis and superficial to the sciatic nerve, accompanied by multiple branches of the inferior gluteal artery and vein
- It also divides into several branches, which provide motor innervation to the overlying gluteus maximus

Nerve to quadratus femoris

- Leaves the pelvis **anterior to the sciatic nerve** and obturator internus and passes over the posterior surface of the hip joint
- It supplies an articular branch to this joint and innervates the inferior gemellus and quadratus femoris

Posterior cutaneous nerve of the thigh

- Supplies more skin than any other cutaneous nerve
- Its fibers from the anterior divisions of **S2 and S3** supply the skin of the perineum
- Some of the fibers from the **posterior** divisions of the anterior rami of **S1 and S2** supply the skin of the inferior part of the buttock (via the inferior clunial nerves); others continue inferiorly in branches that supply the skin of the posterior thigh and proximal part of the leg
- Unlike most nerves bearing the name cutaneous, the main part of this nerve lies deep to the deep fascia (fascia lata), with only its terminal branches penetrating the subcutaneous tissue for distribution to the skin

Pudendal nerve

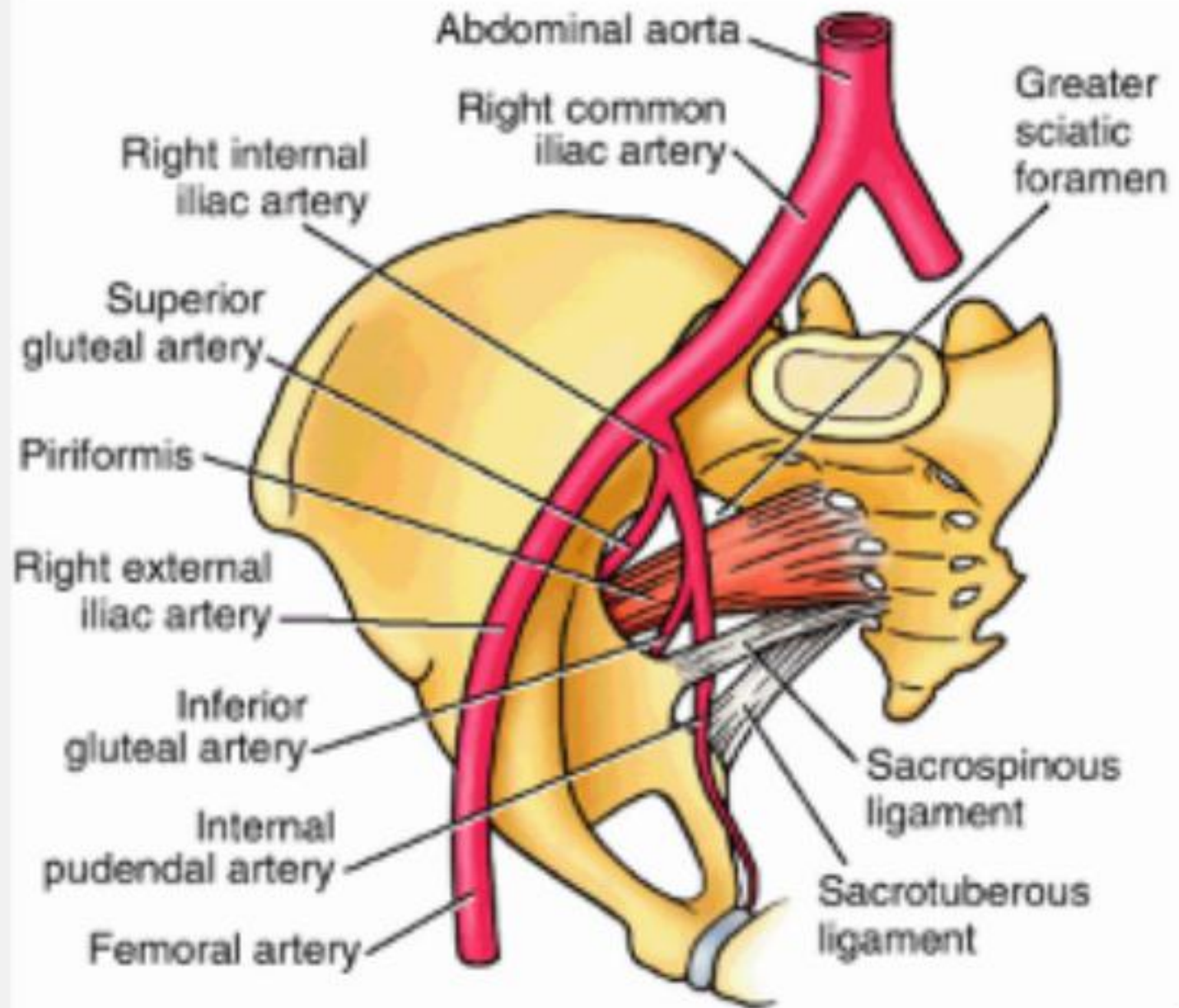
- The pudendal nerve is the most **medial structure** to exit the pelvis through the **greater sciatic foramen** inferior to the piriformis muscle
- It descends posterolateral to the sacrospinous ligament and enters the perineum through the **lesser sciatic foramen**
- Supply structures in the perineum; it supplies no structures in the gluteal region or posterior thigh

Nerve to obturator internus

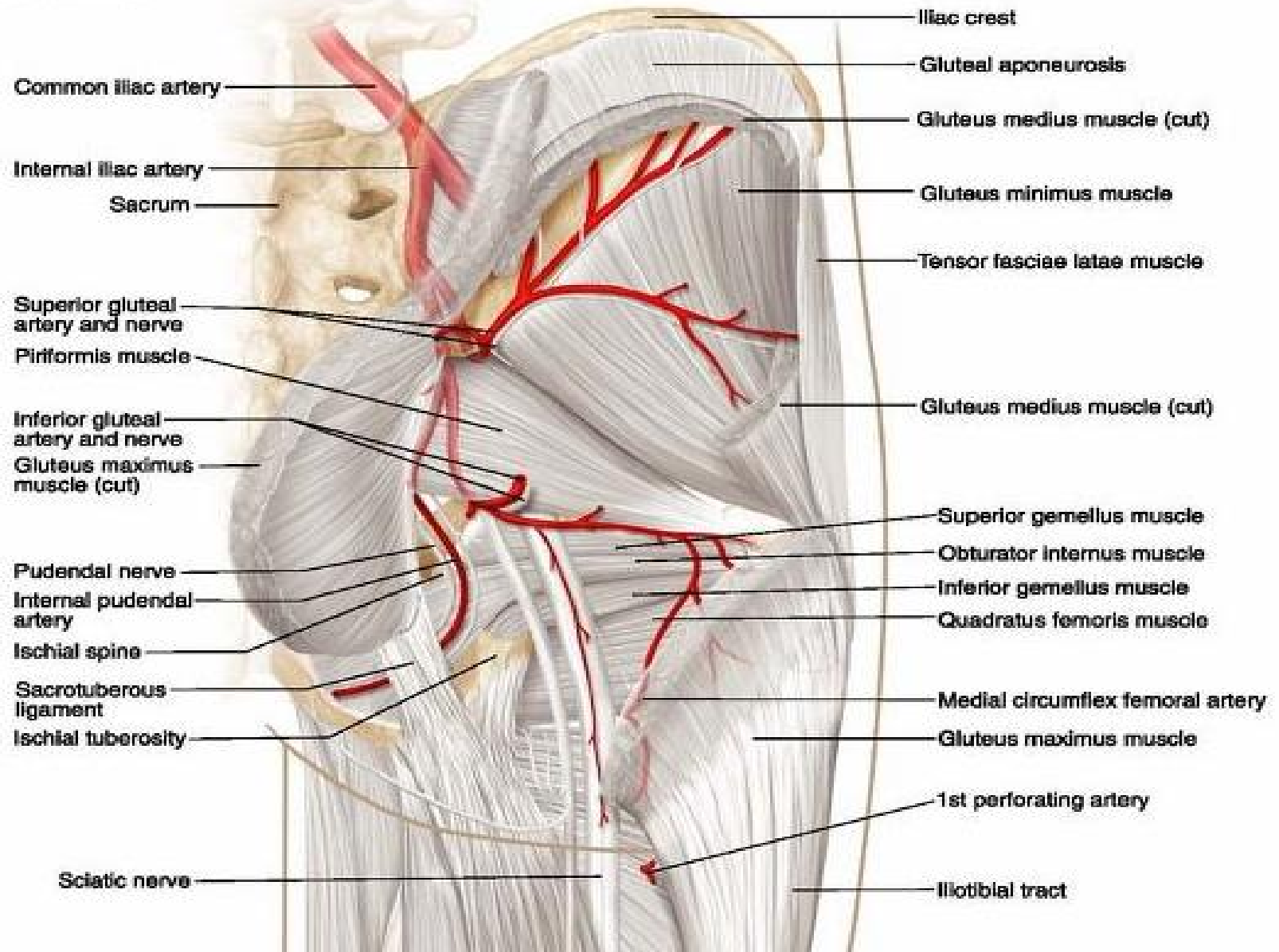
- Arises from the anterior divisions of the anterior rami of the L5 - S2 nerves and parallels the course of the pudendal nerve
- As it passes around the base of the ischial spine, it supplies the superior gemellus
- After entering the perineum via the lesser sciatic foramen, it supplies the obturator internus muscle

Arteries of the gluteal region and posterior thigh

- The arteries of the gluteal region arise, directly or indirectly, from the **internal iliac arteries**, but the patterns of origin of the arteries are variable
- The major branches of the internal iliac artery that supply or traverse the gluteal region are
 - **superior gluteal artery**
 - **inferior gluteal artery**
 - **internal pudendal artery**
- After birth, the posterior compartment of the thigh has no major artery exclusive to the compartment; it receives blood from multiple sources: **inferior gluteal, medial circumflex femoral, perforating, and popliteal arteries**



A. Arteries



Superior gluteal artery

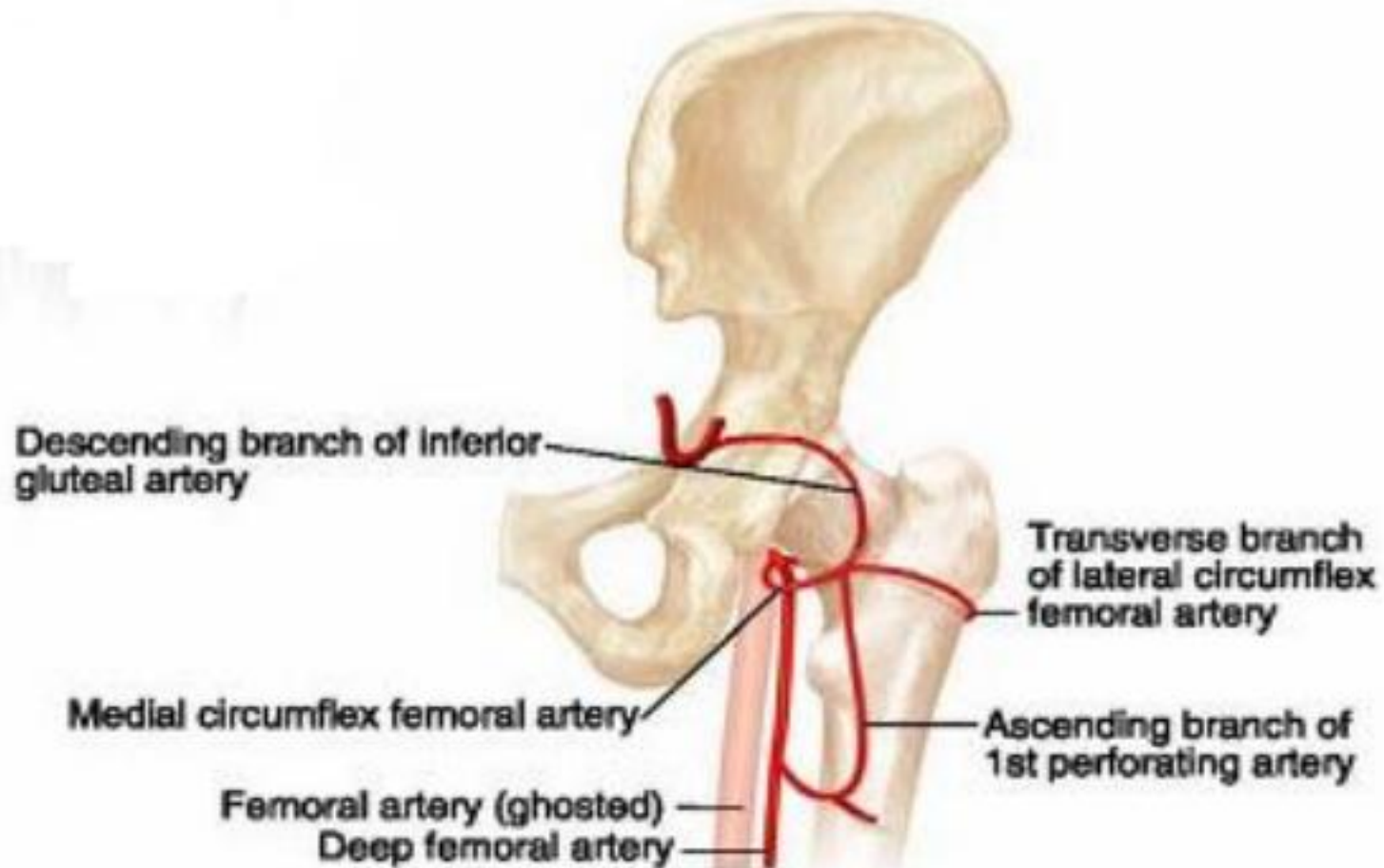
- The **largest branch** of the internal iliac artery
- Passes posteriorly between the lumbosacral trunk and the S1 nerve
- Leaves the pelvis through the greater sciatic foramen, superior to the piriformis, and divides immediately into superficial and deep branches
- The superficial branch supplies the gluteus maximus and skin over the proximal attachment of this muscle
- The deep branch supplies the gluteus medius, gluteus minimus, and tensor of the fascia lata
- The superior gluteal artery anastomoses with the inferior gluteal and medial circumflex femoral arteries

Inferior gluteal artery

- Arises from the internal iliac artery
- Passes posteriorly through the parietal pelvic fascia, between S1 and S2 (or S2 and S3) nerves
- Leaves the pelvis through the greater sciatic foramen, inferior to the piriformis
- Enters the gluteal region deep to the gluteus maximus
- Descends medial to the sciatic nerve
- Supplies **gluteus maximus, obturator internus, quadratus femoris, and superior parts of the hamstrings**

- It anastomoses with the **superior gluteal artery** and frequently participates in the **cruciate anastomosis of the thigh**, involving the first perforating arteries of the deep artery of the thigh and the medial and lateral circumflex femoral arteries
- Whether the cruciate anastomosis is formed or not, these vessels all participate in supplying the structures of the proximal posterior thigh
- Developmentally, the inferior gluteal artery is the major artery of the **posterior compartment**, traversing its length and becoming continuous with the popliteal artery
- This part of the artery diminishes, however, persisting postnatally as the artery to the sciatic nerve

B. Cruciate anastomosis



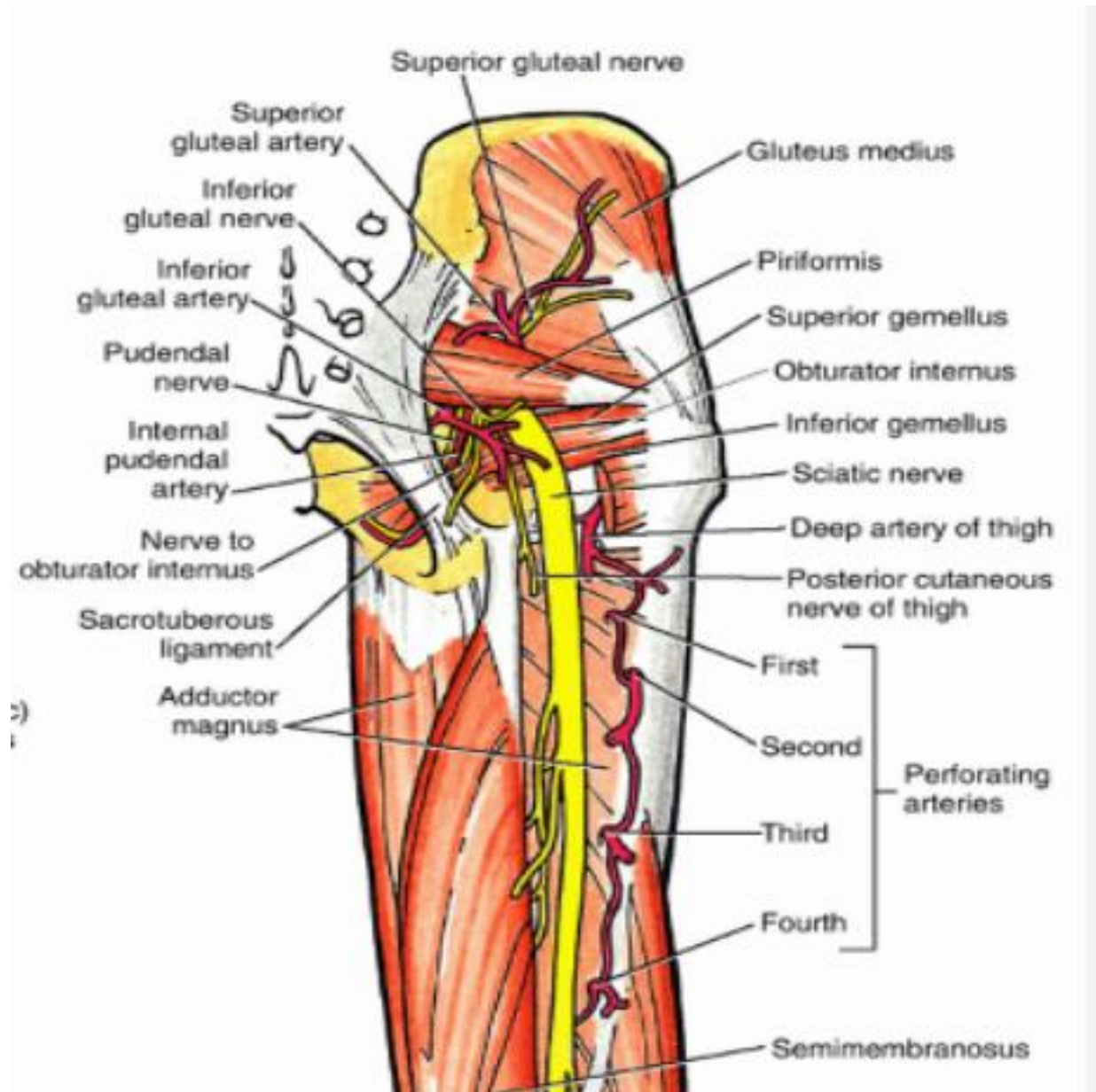
Internal pudendal artery

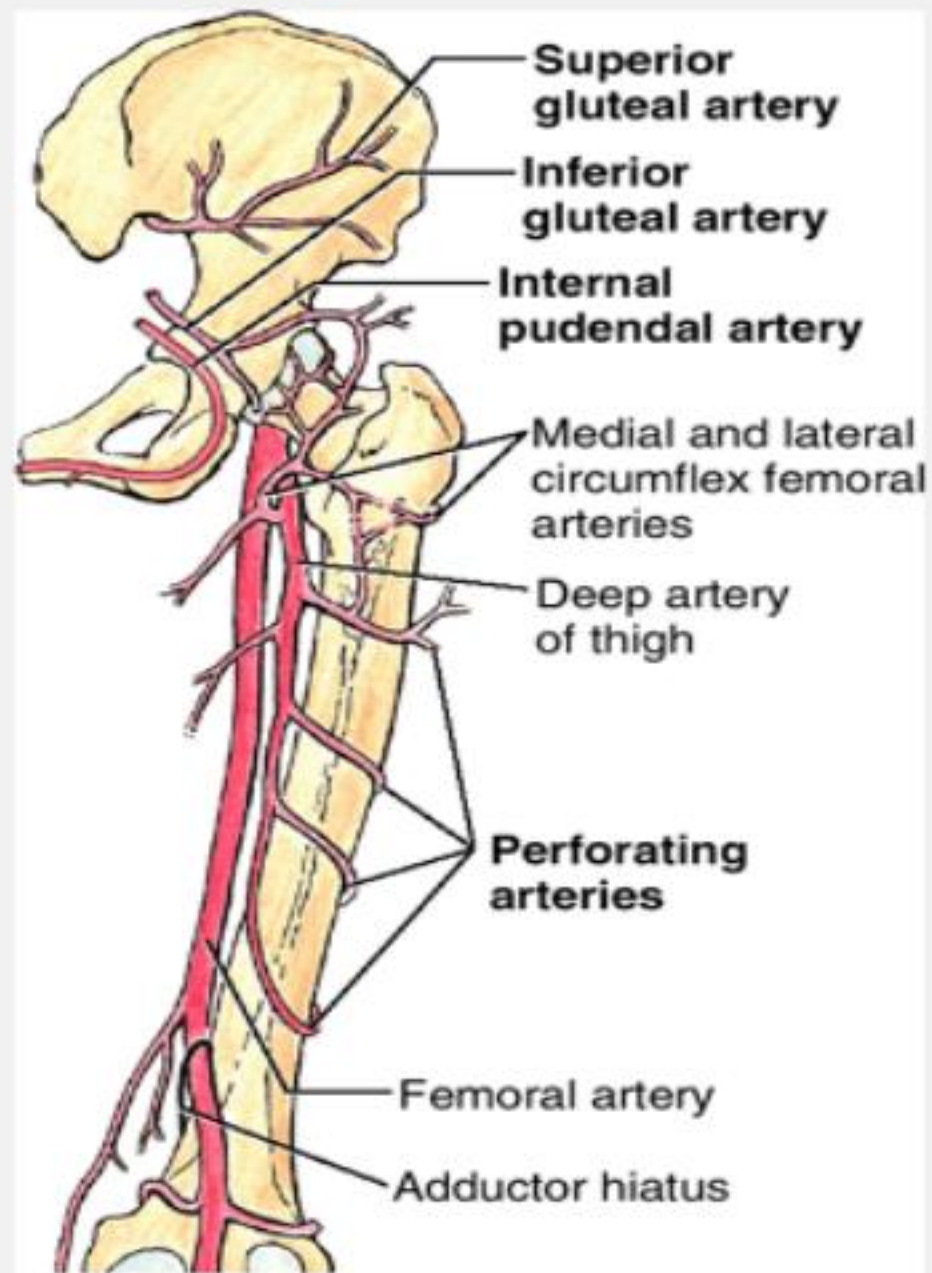
- Arises from the internal iliac artery and lies anterior to the inferior gluteal artery
- Its course parallels that of the pudendal nerve, entering the gluteal region through the greater sciatic foramen inferior to the piriformis
- Leaves the gluteal region immediately by crossing the ischial spine/sacrospinous ligament and enters the perineum through the lesser sciatic foramen
- Like the pudendal nerve, it supplies the skin, external genitalia, and muscles in the perineal region
- It does not supply any structures in the gluteal region or posterior thigh

Perforating arteries

- There are usually four perforating arteries of the deep artery of the thigh, three arising in the anterior compartment and the fourth being the terminal branch of the deep artery itself
- The perforating arteries are large vessels, unusual in the limbs for their transverse, intercompartmental course
- Surgeons operating in the posterior compartment are careful to identify them to avoid inadvertent injury
- They perforate the aponeurotic portion of the distal attachment of the adductor magnus to enter the posterior compartment

- Within the posterior compartment, they typically give rise to muscular branches to the hamstrings and anastomotic branches that ascend or descend to unite with those arising superiorly or inferiorly from the other perforating arteries or the inferior gluteal and popliteal artery
- A continuous anastomotic chain thus extends from the gluteal to popliteal regions, which gives rise to additional branches to muscles and to the sciatic nerve
- After giving off their posterior compartment branches, the perforating arteries pierce the lateral intermuscular septum to enter the anterior compartment, where they supply the vastus lateralis muscle





Posterior view

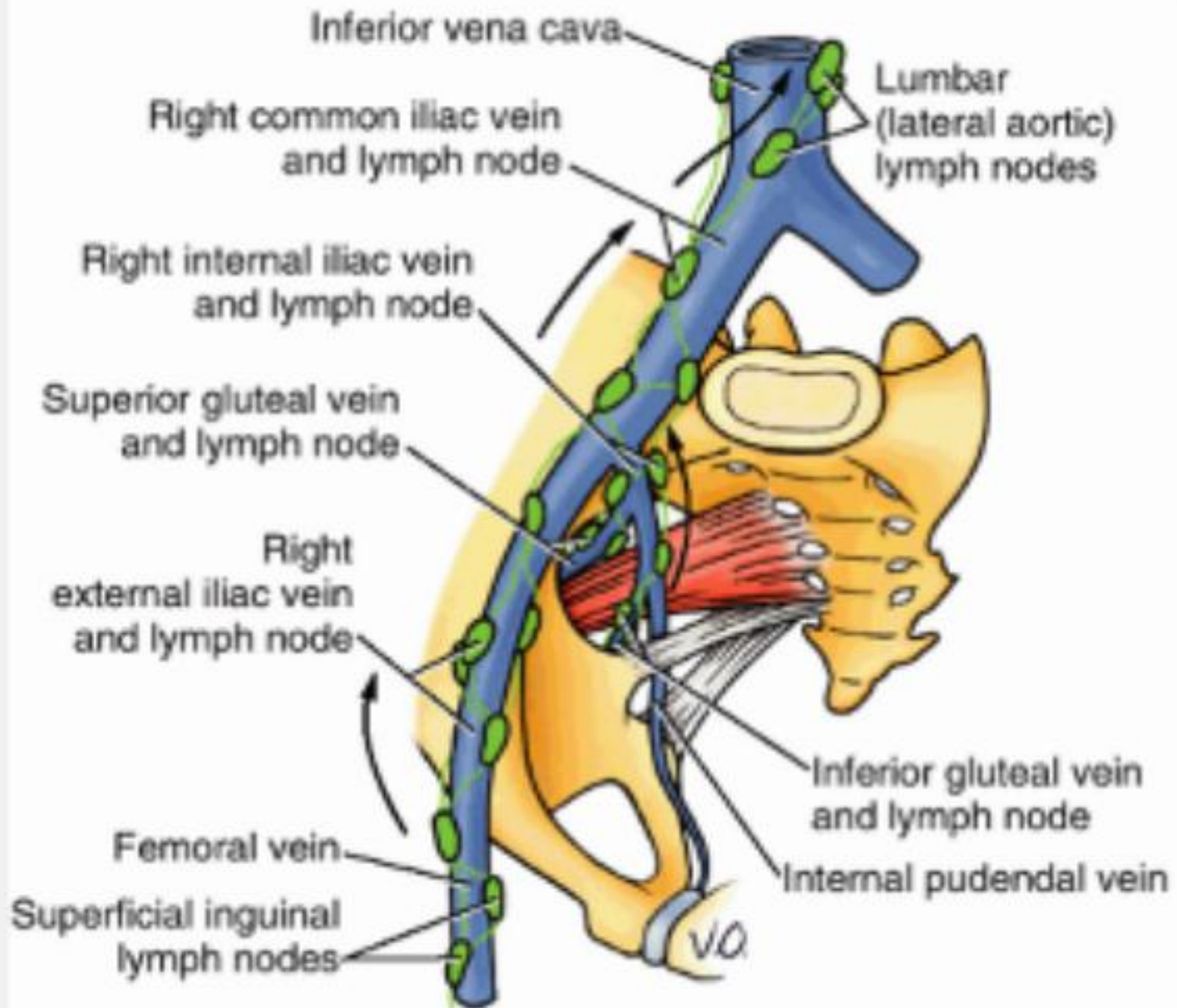
Veins of the gluteal region and posterior thigh

- The gluteal veins are tributaries of the internal iliac veins that drain blood from the gluteal region
- Superior and inferior gluteal veins
 - accompany the corresponding arteries through the greater sciatic foramen, superior and inferior to the piriformis, respectively
 - they communicate with tributaries of the femoral vein, thereby providing alternate routes for the return of blood from the lower limb if the femoral vein is occluded or has to be ligated

- Internal pudendal veins
 - accompany the internal pudendal arteries
 - join to form a single vein that enters the internal iliac vein
 - these veins drain blood from the external genitalia or pudendum
- Perforating veins
 - accompany the arteries of the same name
 - drain blood from the posterior compartment of the thigh into the deep vein of the thigh
 - the perforating veins, like the arteries, usually also communicate inferiorly with the popliteal vein and superiorly with the inferior gluteal vein

Lymphatic drainage of the gluteal region and thigh

- Lymph from the deep tissues of the buttocks
 - follows the gluteal vessels to the superior and inferior gluteal lymph nodes
 - from them to the internal, external, and common iliac lymph nodes
 - from them to the lateral aortic lumbar (caval) lymph nodes
- Lymph from the superficial tissues of the gluteal region
 - enters the superficial inguinal lymph nodes, which also receive lymph from the thigh
 - all the superficial inguinal nodes send efferent lymphatic vessels to the external iliac lymph nodes



Thank you!